

FlowCHAOS

Comprehensive **H**ydraulic **A**nalysis & **O**ptimization **S**ystem

User Manual

Version 1.0.0, for Windows 10 or later

Compiled by Hobby Bwanali, initiator and project lead

Copyright 2026 The FlowCHAOS contributors. Freeware, licensed for commercial and non-commercial use under the FlowCHAOS Freeware Licence.

Contributors

Hobby Bwanali, initiator and project lead

Contributors (L.Gravam Consulting Services Ltd), Lilongwe, Malawi

Ideas, guidance, field experience and support from the irrigation engineers of L.Gravam Consulting Services Ltd:

Eng. Stanley Phiri, Director of Hydraulic Engineering

Lenard Kumwenda, Supervisor

Sangwani Mfune, Supervisor

Hezekiah Kaunda, Supervisor

Joseph Chawinga

Sarah Phiri

Patrick Tembo

FlowCHAOS is a shared project: everyone who reports an issue, shares a network, tests a release or writes a plugin is part of it.

About this manual

FlowCHAOS designs, analyses and presents pressurised water networks: town supplies, rural gravity schemes, bulk transfers, building plumbing and irrigation. It brings together the drawing tools of a design office, the layers and coordinate systems of a GIS, a hydraulic and water-quality engine, terrain and surface maps, quantities, construction planning and printable map sheets, in one program that works offline.

This manual follows the order in which a project is usually built. Part I introduces FlowCHAOS, installs it and works through a complete example, a village supply from a spring. Part II explains each part of the work: projects, drawing and editing, layers and data, labels, the network model and the physics behind it, running an analysis and its options, controls and rules, water quality, reading the results, map views and animation, terrain, the design tools, quantities and planning, and map sheets. Part III is the reference: shortcuts, file formats and the FlowCHAOS network file, units, coordinates, errors and warnings, how a network is solved, a glossary, questions and answers, and troubleshooting. The appendix is for those who extend FlowCHAOS with plugins.

Menu commands are written in bold with their menu, for example Analysis > Water Quality Analysis. Buttons and fields are written in bold as they appear on the screen. Values are in SI units unless a table says otherwise; pressures and heads are in metres of water.

The same chapters are in the program (Help > Help Contents, F1, also offline) and on the FlowCHAOS website, which always has the latest version of this manual. Corrections and questions are welcome through Help > Report an Issue and Help > Support and Contact.

Contents

Part I. Getting started **1**

1. Introduction 2
2. Install 5
3. Tutorial: a village supply 7
4. The window 10

Part II. User guide **13**

5. Working with projects 15
6. Drawing and editing 18
7. Layers and data 22
8. Labels 36
9. The network model 39
10. Running an analysis 46
11. Analysis options 50
12. Controls and rules 55
13. Water quality 59
14. Reading the results 65
15. Map views, legends and animation 69
16. Terrain and surface maps 74
17. Design tools 78
18. Quantities and planning 86
19. Map sheets and reports 91

Part III. Reference **95**

20. Keyboard shortcuts 97
21. File formats 101
22. The network file 104
23. Units of measurement 109
24. Coordinates and regional settings 111
25. Errors and warnings 113
26. How a network is solved 117
27. Glossary 120
28. Questions and answers 122

29. Troubleshooting 124

Appendix. Extending FlowCHAOS 126

30. Writing plugins 128

31. Tutorial: build a plugin 131

32. Plugin API 135

33. Project model 138

34. Plugin examples 141

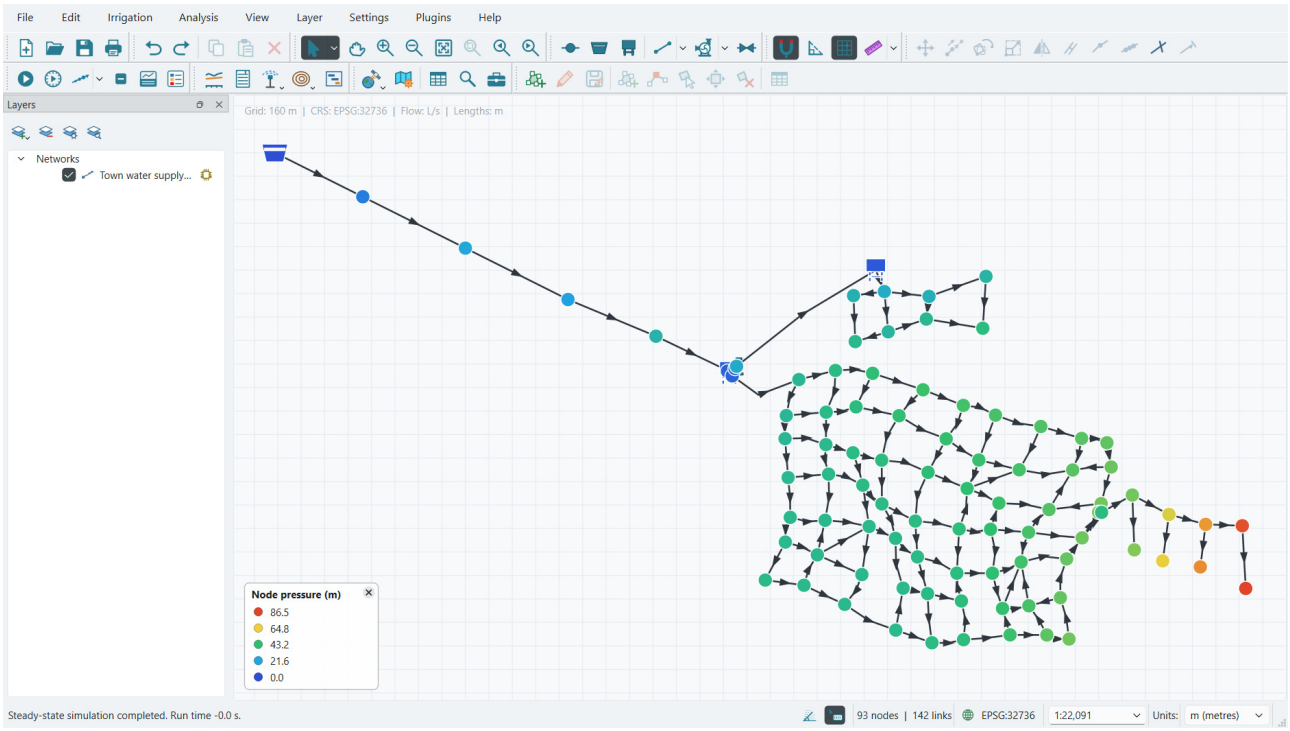
Part I. Getting started

Introduction, Install, Tutorial: a village supply, The window

1

Introduction

What FlowCHAOS is, what it models, and how a study runs from the first line to the last report.



A network drawn, analysed and coloured by its results

What FlowCHAOS is

FlowCHAOS (Comprehensive Hydraulic Analysis & Optimization System) is a Windows program for the design and analysis of pressurised water networks: town and city supplies, rural gravity schemes, bulk transfer mains, building plumbing and irrigation systems. It brings into one model the steps an engineer usually spreads over several programs:

- drawing the network on real maps, with survey points, scanned drawings, satellite imagery and ground levels;
- the hydraulic and water quality analysis of the network, at one moment or through days of use;
- the design tools that size pipes, check fire flow, find critical mains and lay out irrigation;
- the quantities, costs and construction programme that follow from the design;
- map sheets, profiles, reports and animations that present it.

Because every step works on the same model, the network that is analysed is the network that is priced, drawn and built.

What a network model is

A network model describes a water system as **nodes** joined by **links**:

- nodes are the points where water enters, leaves or is stored: **junctions** (where pipes meet and water is taken),

reservoirs (sources with a fixed water level, such as a lake, a river intake or a bulk supply) and **tanks** (storage whose level rises and falls);

- links carry water between nodes: **pipes**, **pumps** and **valves**.

Each element carries the data the physics needs: ground and hydraulic levels and withdrawals at the junctions; lengths, diameters and roughness of the pipes; curves of the pumps; types and settings of the valves; sizes and levels of the tanks. Around them sit the data that describe time: **demand profiles** (how use varies through the day), **curves** (pump head, efficiency, tank volume, head loss), **controls** and **rules** (when pumps, pipes and valves change).

What FlowCHAOS calculates

For a moment in time (a **steady state**) or a sequence of moments (an **extended period**), the analysis finds:

- the **flow** and **velocity** in every pipe, pump and valve, and the direction the water moves;
- the **head** (the level water would rise to in a standpipe) and the **pressure** at every node;
- the water **delivered** at each junction, which can be less than the design withdrawal when the pressure is too low (pressure-driven analysis);
- the **level** of every tank, the **status** of every pump and valve, and the **energy** used by pumping and its cost;
- the **water quality**: the concentration of chlorine or another substance, the **age** of the water, or the share of water coming from a chosen source, everywhere in the network.

What it can model

Hydraulics	Water quality
Networks of any size and layout, branched or looped	Movement of a substance with the flow, mixing at junctions
Friction by Hazen-Williams, Darcy-Weisbach or Chezy-Manning	Reactions in the water (bulk) and at the pipe walls, first or higher order
Minor losses at bends and fittings	Limiting concentrations (a reaction stops at a level)
Fixed-speed and variable-speed pumps, with their curves	Four ways tanks can mix their water
Pressure reducing, sustaining and breaking valves, flow control, throttle control and general valves with their own loss curves	Sources at any node: inflow concentration, boosters, flow-paced dosing, mass injection
Check valves and pipes that can be opened or closed	Water age and source tracing
Tanks of any shape (with a volume curve)	Sources that act for part of a run (an intrusion)
Demands that vary in time, several categories at a junction	Compliance against limits, by node and by delivered water
Demand-driven or pressure-driven withdrawal, emitters for leaks and sprinklers	
Controls on time, clock time, tank level and pressure, and rules with AND, OR, ELSE and priorities	

How a study runs

A typical study follows these steps; each has a chapter in this manual.

1. **Set up the project** with its coordinate system and units (Working with projects).
2. **Bring in the ground and the existing data**: basemaps, a DEM, survey points, drawings, older models (Layers and data).
3. **Draw or import the network** and give its elements their properties (Drawing and editing, The network model).
4. **Describe time**: demand profiles, pump and tank curves, controls and rules (Controls and rules).
5. **Choose the analysis options** (Analysis options) and **run** (Running an analysis).
6. **Read the results** on the map, in tables and graphs, and fix what does not meet the design criteria (Reading the results, Map views).
7. **Design**: size the pipes, check fire flow and critical mains, plan irrigation (Design tools).
8. **Check the water quality** (Water quality).
9. **Deliver**: quantities, construction programme, map sheets and reports (Quantities and planning, Map sheets and reports).

Who it is for

FlowCHAOS is made for water and irrigation engineers, consultants, water utilities and boards, contractors, plumbers, lecturers and students. A network of a few pipes and one of many thousands are handled the same way; large networks draw quickly (labels and symbols are simplified on their own when there are many elements) and long work, such as runs, contours and DEMs, happens in the background with its progress in the status bar, while the window stays usable.

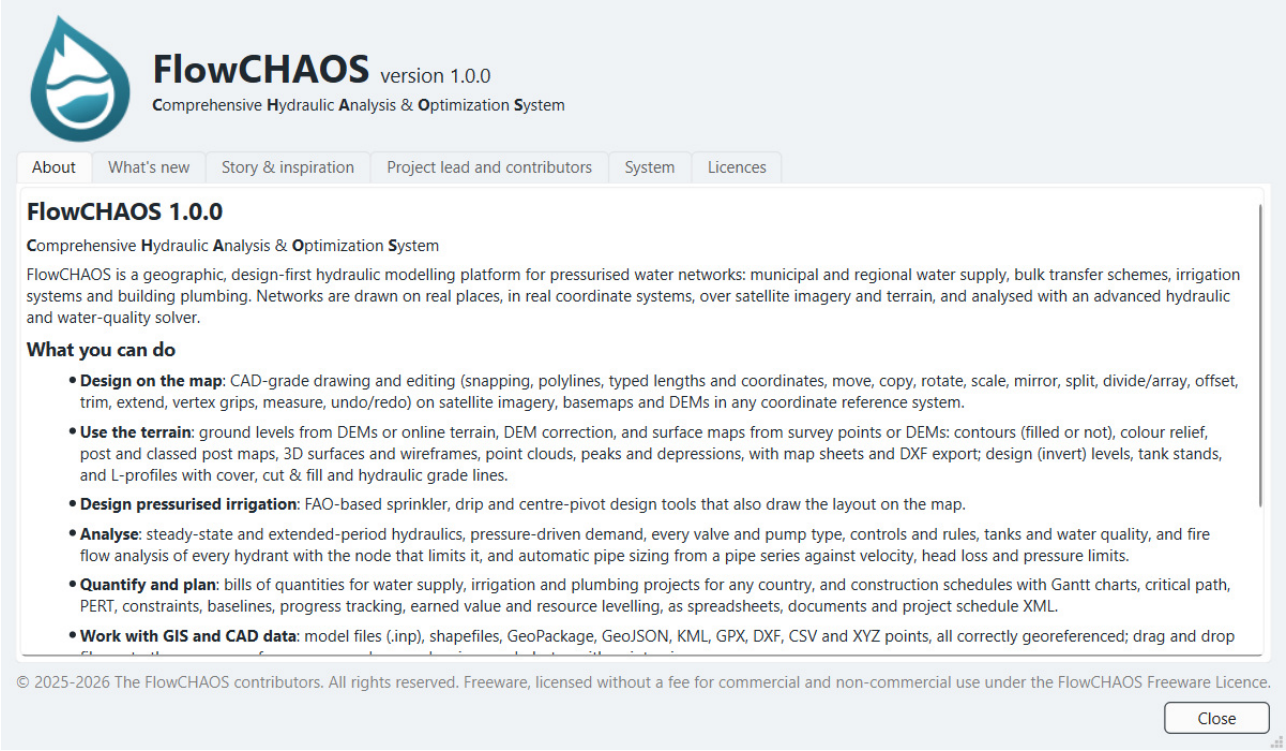
Conventions in this manual

- Menu commands are written as **Menu > Command**, for example **Analysis > Water Quality Analysis**.
- Buttons, fields and tabs are written in bold, as they appear on the screen.
- Lengths and levels are in metres, pressures and heads in metres of water, flows in litres per second unless the project uses other units (Units of measurement).
- A worked example is set out as numbered steps; tips and cautions follow the text they belong to.

2

Install

Put FlowCHAOS on your computer with its installer.



The About window: version, licence and the people behind FlowCHAOS

What you need

System	Windows 10 or later, 64-bit (Windows 10 or 11 recommended)
Memory	4 GB (8 GB for large networks and terrain)
Disk	1 GB
Internet	only for basemaps, satellite imagery, place search and online terrain

From an installer

Download the installer from the download page. Each file is listed with its SHA-256 checksum so you can verify it: certutil -hashfile FlowCHAOS-Setup.exe SHA256 must give the same value.

There are two kinds:

- **Setup (EXE)** for one computer: a guided setup that adds FlowCHAOS to the Start menu and opens imported hydraulic model input files (.inp) with it.
- **Installer (MSI)** for organisations: installs for every user and can be deployed silently: msiexec /i FlowCHAOS-x64.msi /qn.

Updating and removing

Install a newer version over the old one; your settings and plugins are kept. Remove FlowCHAOS from **Settings > Apps** in Windows. The setup program then asks whether to remove your settings, the map tile cache, logs, reference data and plugins too: choose Yes for a clean start, No to keep them for a reinstall. A silent uninstall keeps them unless /PURGE=yes is given. Your projects are ordinary files and stay where you saved them.

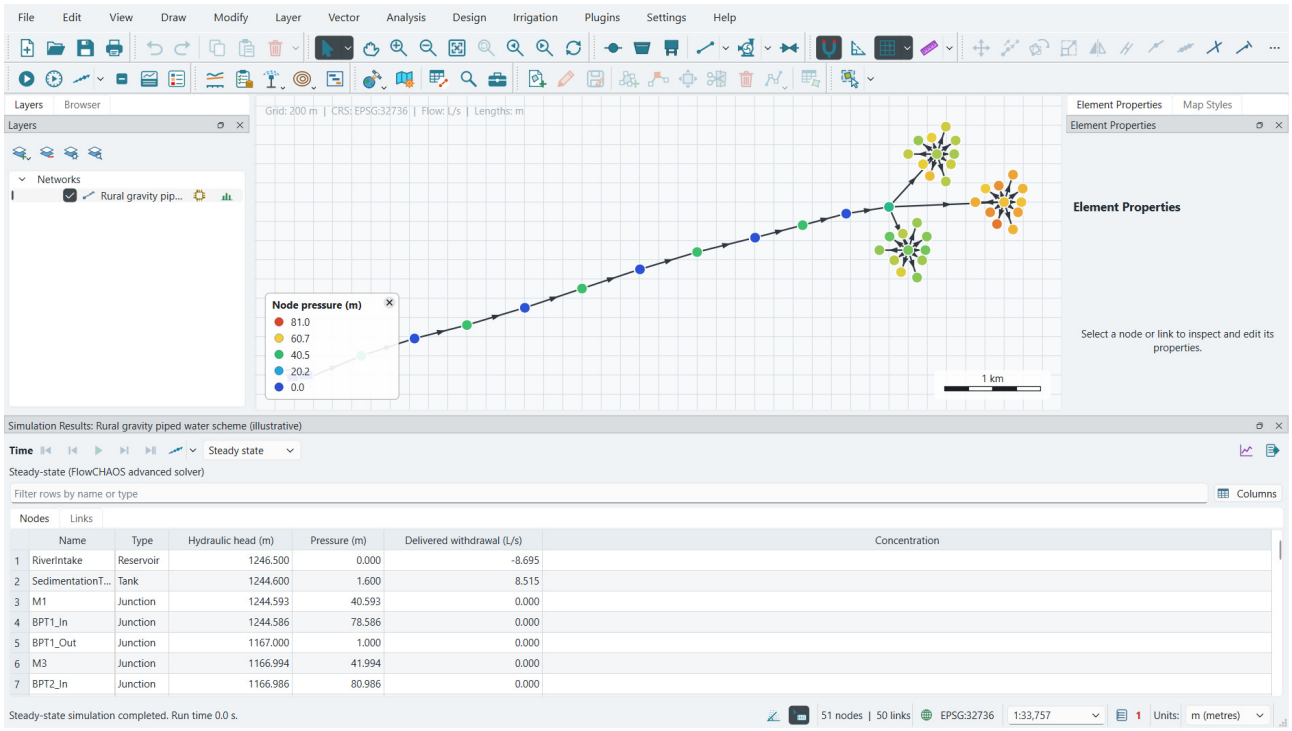
Where FlowCHAOS keeps things

What	Where
Your plugins	`%USERPROFILE%\.flowchaos\plugins`
Your own reference data	`%APPDATA%\FlowCHAOS\data`
The log (for bug reports)	`%LOCALAPPDATA%\FlowCHAOS\FlowCHAOS\logs\flowchaos.log`

3

Tutorial: a village supply

A worked example: a village supply from a spring, drawn, analysed, improved and reported, step by step.



The rural gravity scheme after a run: the kind of network the tutorial builds

This tutorial builds a small gravity scheme: a protected spring feeds a storage tank, which supplies four communal taps. It takes about forty minutes and touches every main part of FlowCHAOS. The values are typical of a rural scheme in Malawi; use your own when you repeat it for a real site.

1. Start a project

1. **File > New**. The project uses UTM zone 36 South (EPSG:32736); for a site in another zone, click the coordinate system in the status bar and choose yours.
2. **File > Project Properties > General**: name the project "Village supply tutorial".
3. Press the satellite button on the toolbar and choose a satellite basemap to see the ground. Go to your site with **View > Go to Coordinates**, or with the place search on the toolbar.
4. **Analysis > Analysis Options > Hydraulics**: flow units **L/s**, head loss formula **Hazen-Williams**. Leave the rest as it is.

2. Draw the network

The network is drawn in a **network layer**; the first drawing tool creates one and starts editing it.

1. Choose **Reservoir** on the drawing toolbar and click where the spring is. In **Element Properties** set its **Total**

Head to the level of the spring's water surface, for example **1148 m**. Name it **Spring**.

2. Choose **Tank** and click about 850 m below the spring. Set its ground elevation (from the DEM, or type **1132 m**), **diameter 4.0 m**, **minimum level 0.3 m**, **maximum level 2.5 m** and **starting level 1.5 m**. Name it **Tank**.
3. Choose **Pipe**, click the spring, then the tank: the pipe follows your clicks; each click on empty ground places a bend. Double-click, Enter or right-click to finish.
4. From the tank draw the distribution main to a junction **J1**, 420 m away, then branches to four taps, **Tap1** to **Tap4**, 200 to 350 m from J1. Type a length (for example 260) and press Enter to place a point exactly that far.
5. Select the four taps (Ctrl+click) and in Element Properties give them a **design withdrawal** of **0.15 L/s** each (about 13 m3 a day per tap: some 500 people at 25 litres a day).
6. Ground levels come from a DEM or online terrain automatically (**Layer > Ground Elevation Source**); check them in Element Properties, and type corrections where you surveyed.

3. Give the pipes their properties

1. Select the spring-to-tank pipe: material **HDPE PE100**, internal diameter **79 mm** (the 90 mm PN10 pipe), roughness **C 150**.
2. Select the distribution pipes with a box (drag across them): main **63 mm** pipe (internal about **55 mm**), branches **50 mm** (internal about **44 mm**). The material also feeds the bill of quantities.
3. **Select by Attribute** (on the toolbar and the map's right-click menu) selects elements by any property, to change many at once.

4. Run a steady state

1. Press **Run Steady-State** (F5). The Simulation Results panel opens and the nodes are coloured by pressure, with a legend on the map.
2. Click **Tap4**: Element Properties shows its head, pressure and delivered withdrawal.
3. In **Map Styles**, under Links, choose **Velocity**: the pipes are coloured too.
4. Open **Analysis > Solver Warnings**: it should be empty. If not, see Errors and warnings.

Check the pressures: at a communal tap, 5 to 10 m is comfortable; much more wastes water and wears taps, much less gives a weak flow.

5. Make it move through the day

Communal taps are busy at dawn and late afternoon and closed at night.

1. **Analysis > Demand Profiles**. Press **New Profile**, name it **TAP**, then from **Preset** choose the communal water point shape. The chart shows the multiplier for each hour.
2. Select the taps and set their **demand profile** to **TAP**.
3. **Analysis > Analysis Options > Times**: duration **48:00**, hydraulic timestep **1:00**.
4. **Analysis > Run Extended Period**. The time bar of the Simulation Results panel now steps through 48 hours: press play to watch the tank level and the tap pressures change.
5. **Analysis > Advanced Graphs**, **Time Series** of **Tank**, **Level**. A healthy tank fills at night and drains in the day without emptying.

6. Add a control

If the tank overflows at night, close the inlet when it is full:

1. **Analysis > Scheduled Pipe Controls**, **Start from a Sample**, **Close the tank inlet before it overflows**. The sample is written with your tank and its inlet pipe; adjust the levels if needed.
2. Run the extended period again and look at the tank's level graph.

7. Check the chlorine

1. **Analysis > Water Quality Analysis**, choose **Chlorine residual compliance**. The spring is the source at 1 mg/L; set the bulk rate from a bottle test if you have one.
2. Press **Run Water Quality Analysis**. The summary says whether every tap keeps at least 0.2 mg/L.
3. **Show on Map** and play the time bar to watch the chlorine travel.

8. Size the pipes

1. **Analysis > Automatic Pipe Sizing**: pipe series **HDPE PE100 PN10 (SDR 17)**, maximum velocity **1.5 m/s**, minimum pressure **5 m**, peak factor **1.0** (the profile already holds the peaks).
2. **Size Pipes**. The table lists the size chosen for each pipe; **Apply Sizes** writes them into the network (Undo takes them back).

9. Present and price it

1. Select the pipes from the spring to Tap4 and choose **Edit > Longitudinal Profile**: the ground, the pipe and the hydraulic grade line along them.
2. **File > Export > Bill of Quantities**: pipes by size and material, fittings, tanks and earthworks, priced from your rates.
3. **File > Export > Print Layout**: a map sheet with the network, the legend, a scale bar, a north arrow and a title block, exported to PDF.

10. Save

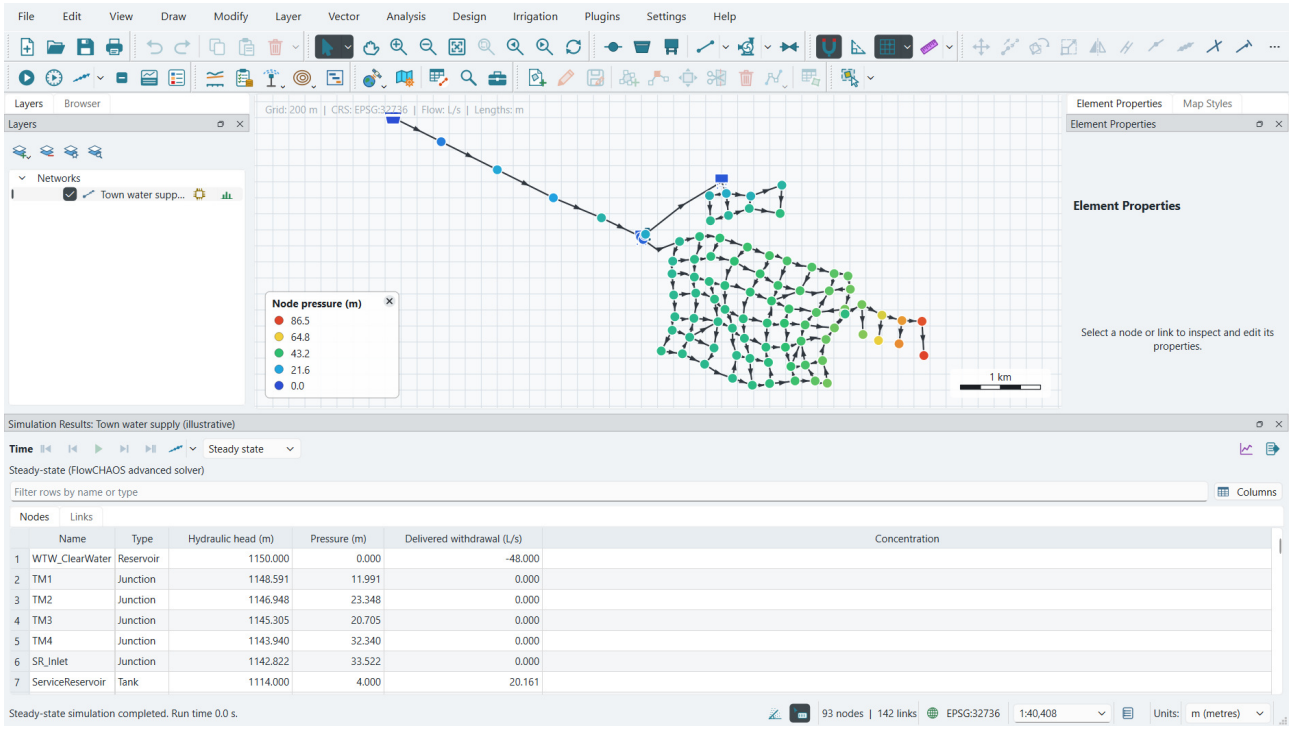
File > Save writes the project with everything: the network, results, styles, layouts and plans. A green bar confirms it. Every step can still be undone with **Ctrl+Z**.

Next: the window, drawing and editing and the network model.

4

The window

A tour of the window: where everything is and how it fits together.



The FlowCHAOS window with a town network after a steady-state run

The map

The map in the middle shows everything you draw and import, over imagery, terrain or a plain background. Scroll to zoom, drag with the middle button (or the pan tool) to move, and use **Ctrl+0** to see the whole network. **Ctrl+Left** and **Ctrl+Right** step back and forward through the views you have had.

Toolbars

- **File and edit**: new, open, save, print, undo and redo.
- **Navigate**: select, pan and the zoom tools.
- **Draw**: junction, reservoir, tank, pipe, pump and valve. The small arrow beside pipe and pump tools chooses how they draw.
- **Drafting**: snap, ortho, the grid and measure.
- **Modify**: move, copy, rotate, scale, mirror, offset, split, divide, trim and extend.
- **Run** (first on the second row): steady-state and extended-period analysis, Animate Map, Stop / Clear Results, the Simulation Results panel and the Map Styles panel.
- **Digitizing**: editing of vector layers (add, select, move and delete features).

- **Analysis and design**: profiles, quantities, planning, surface maps and the design tools.
- **Map and data**: basemaps and satellite imagery, map options and layers.

A command is available only when it can act: Export Results needs results, Rotate needs a selection. Point at a greyed command to see what it needs.

Panels

- **Layers** (left): Networks, Vector layers, reference images, rasters, basemaps and surface maps, each group with its layers. The toolbar above the list adds every kind of layer, removes, opens properties and zooms. Double-click a layer for its properties (a 3D surface opens its 3D view).
- **Element Properties** (right): the values of what you select, ready to edit. The fields fit the panel and widen with it.
- **Map Styles** (right, at any time): what nodes and links are coloured by, each with its colour band and legend. A newly opened project shows each layer in its own colour until you choose a view.
- **Simulation Results** (bottom, after a run): the time bar, playback and the result tables.
- **Value Relay** (right): the source and destinations of Value Relay and its history.
- **Log Messages** (bottom): what the program reports, in tabs: all, messages, analysis, tasks, plugins, online and program. The button at the right of the status bar opens it and counts new warnings.
- **Measure** (under Element Properties, while measuring): the values of the measurement. Esc clears the measurement and leaves the tool.

Panels on the same side share it as tabs (Element Properties, Map Styles and the Processing Toolbox on the right; Layers and Plugins on the left), so each keeps its full height. Drag a tab to move a panel, or out of the window to float it; **View** brings closed panels back. Each start shows the Layers, Element Properties and Map Styles panels and every toolbar again, in the places you left them.



Log Messages: what the program reports, in tabs

The status bar

From left to right: the cursor coordinates, the length and angle being drawn, two drawing aids (polar tracking and dynamic input), the network size, the coordinate system (click it for the project's coordinates), the map scale (pick or type 1:N) and the units of lengths.

Toolbar buttons show their name in bold, the keyboard shortcut beside it and a short description below; a button that cannot be used yet adds what it needs.

Every save and export confirms itself in a bar over the bottom of the window: green when the file was written, red when it could not be, with the reason and a **Details** button. The bar closes by itself; point at it to keep it open.

Tool windows

Big tools open in their own windows beside the map so you can keep working: print layouts, the planner, graphs, the attribute table, profiles, quantities, the georeferencer, 3D views, pipe sizing and fire flow. Every window fits the screen and keeps its buttons reachable.

Light and dark

FlowCHAOS starts in the light theme. **Settings > Theme** switches between light, dark and the system setting.

Quick tour

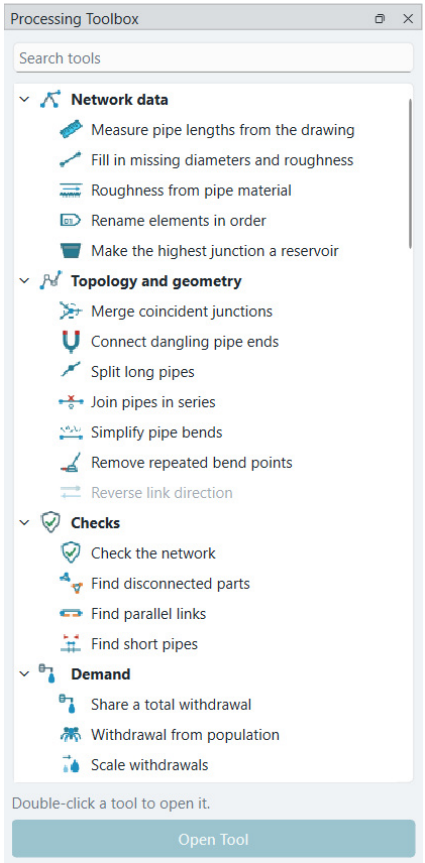
The first time FlowCHAOS starts, a short tour points out the map, the layers, the drawing and run buttons, map styles, results and the design tools. Use Next and Back (or the arrow keys), and Esc to close it. Start it again at any time from **Help > Quick Tour**.

Help

Help > User Manual opens the full FlowCHAOS User Manual (PDF) in your PDF reader; it is installed with the program, so it works offline. **Help > Help Contents** (F1) opens this documentation inside FlowCHAOS, with search, also offline. The Help menu also has Getting Started, Keyboard Shortcuts, What's New, Report an Issue, Get Involved, Support and Contact, the FlowCHAOS home page and Check for Updates.

Processing Toolbox

View > Processing Toolbox lists ready-made network tools in groups, each with its icon: network data (lengths from the drawing, missing sizes, roughness by material, renaming in order), topology and geometry (merging and snapping nodes, splitting and joining pipes, simplifying bends), checks (the whole network, disconnected parts, parallel and short pipes), demand (sharing a total, from population, scaling), analysis and design windows, terrain, irrigation, and reports and export, and five groups for vector layers (analysis, geoprocessing, geometry, selection and research, data management; see Vector layers under Data). Double-click a tool: its window has the parameters, a help page, and after **Run** a log and a table you can copy; elements it finds are selected on the map, and a change it makes is one step you can undo. Tools that cannot run yet are greyed with the reason in their tooltip. Plugins can add tools.



The Processing Toolbox: network tools and tools for vector layers

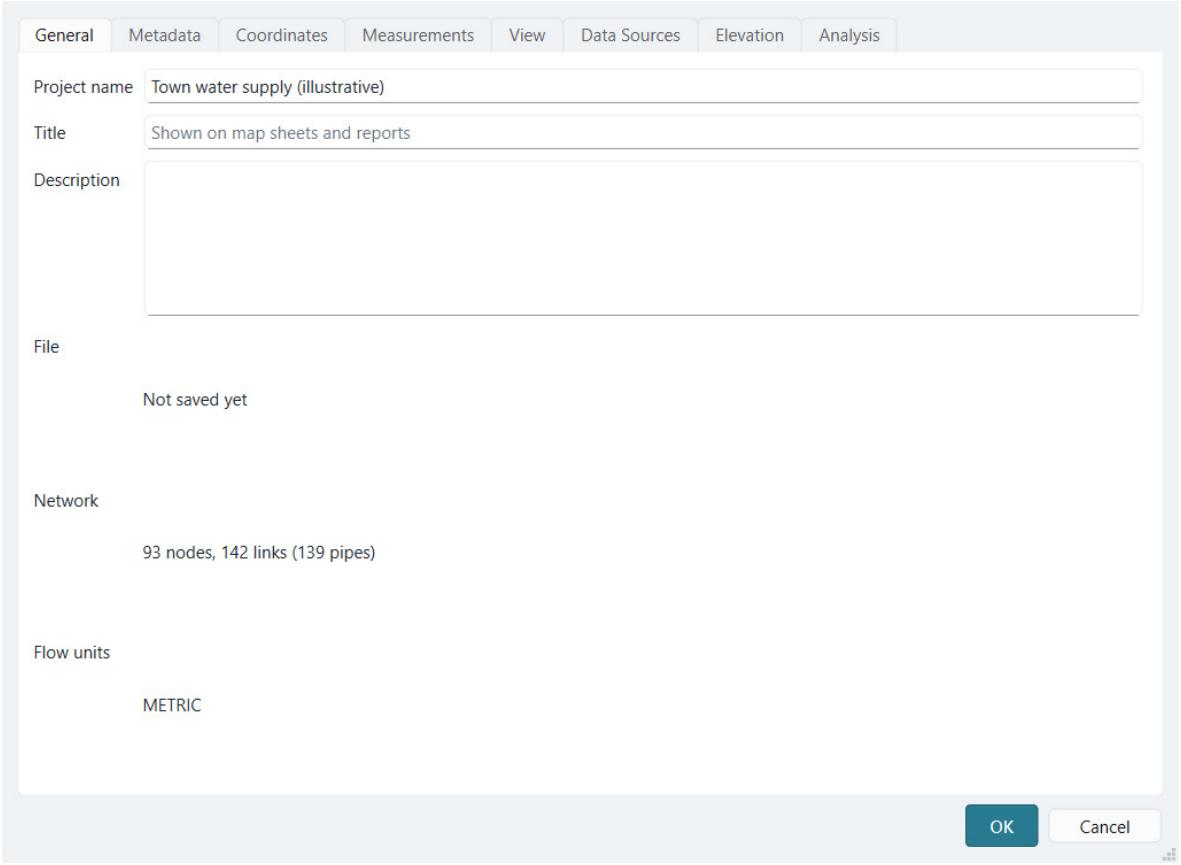
Part II. User guide

Working with projects, Drawing and editing, Layers and data, Labels, The network model, Running an analysis, Analysis options, Controls and rules, Water quality, Reading the results, Map views, legends and animation, Terrain and surface maps, Design tools, Quantities and planning, Map sheets and reports

5

Working with projects

Projects and files: starting, opening, saving and setting up a project, and the defaults it uses.



Project Properties: general details, coordinates, units and data sources

What a project holds

A FlowCHAOS **project** (a .fchaos file) holds the whole work: every network layer, vector layer, raster and basemap; surface maps and contours; the analysis options, demand profiles, curves, controls and rules; the last results; labels, legends and map styles; print layouts; bills of quantities and construction plans; and the project's own settings. One file is everything another engineer needs to open the work where you left it.

A **network file** (.fcnet) holds a single network, to share it or to add it to another project as a layer. FlowCHAOS also **imports and exports** hydraulic model input files (.inp), the format of the widely used public-domain network modelling program, so models can move between programs. It is not a FlowCHAOS format: FlowCHAOS's own files are the project (.fchaos) and the network file (.fcnet) (File formats).

Starting a project

- **File > New** starts an empty project in UTM zone 36 South (EPSG:32736). Change the coordinate system in

Project Properties > Coordinates, or click it in the status bar.

- **File > Open** opens a project, or starts a new project holding a network file as its first layer (.fcnet, .inp, .net, CSV, shapefile, KML, GPX, DXF).
- **File > Open Recent** lists the last projects opened.
- **File > Open Sample Projects** opens ready networks by discipline: town supply, rural gravity scheme, bulk transfer, chlorine study, building plumbing, georeferencing practice, and irrigation schemes from a 1.5 ha smallholder drip system to a 10 ha banana farm and a 50 ha centre pivot.
- Files can also be dragged onto the map: a project opens, other files are added as layers.

Saving

File > Save (Ctrl+S) writes the project; **File > Save As** writes it under another name. The title bar shows an asterisk while there are unsaved changes, and FlowCHAOS asks before closing or opening another project. Every save shows a green bar over the bottom of the window, or a red one with the reason if the file could not be written. Results are saved with the project, so a project reopens with its results on the map.

Auto-save and recovery. While you work, FlowCHAOS writes a recovery copy of the project with its unsaved changes every few minutes (**Settings > Auto-save**: off, or every 1 to 30 minutes; 5 by default). The copy is kept in FlowCHAOS's own folder, never over your file, and is removed when the project is saved or FlowCHAOS closes normally. If FlowCHAOS stops unexpectedly, the next start offers to **Recover** the work: it opens as an unsaved project with its own file name, so Save puts it back.

Undo and redo

Every change, from moving a node to applying pipe sizes, is one step of **Edit > Undo** (Ctrl+Z) and **Edit > Redo** (Ctrl+Y). The history covers the session; saving does not clear it.

Project Properties

File > Project Properties sets what applies to the whole project:

Tab	What it sets
General	the project name, title and description, the file, a summary of the network and its flow units
Metadata	author, organisation, keywords and an abstract, kept with the project and used in reports
Coordinates	the project's coordinate system (CRS), with its details
Measurements	how coordinates are shown (metres, degrees and minutes) and their decimals; units of lengths, areas and angles
View	the predefined map scales and how the map draws
Data Sources	the files the project's layers come from
Elevation	where ground levels come from: a DEM, online terrain, or typed values
Analysis	the head loss formula, demand model, duration and time step of runs, and the design standard

The **design standard** sets the starting values of pipe sizing, fire flow, criticality and the withdrawal tools: water use per person, peak factor, losses, minimum and maximum pressure, velocity limits and fire flow. The presets describe typical schemes; check them against the standard that applies to your project, and add your own to design/standards.json

in the reference data.

Coordinates

Every project has a coordinate system, and every layer keeps its own: a shapefile in geographic coordinates and a DXF in a local grid are each shown in the project's system without changing the files. Lengths of pipes are true ground lengths in that system. See Coordinates and units.

Defaults for new elements

Settings > ID Labels and Element Defaults sets the names of new elements and the properties they are drawn with (see Analysis options). Draw quickly with sensible defaults, then correct what differs with **Select by Attribute** and the attribute table.

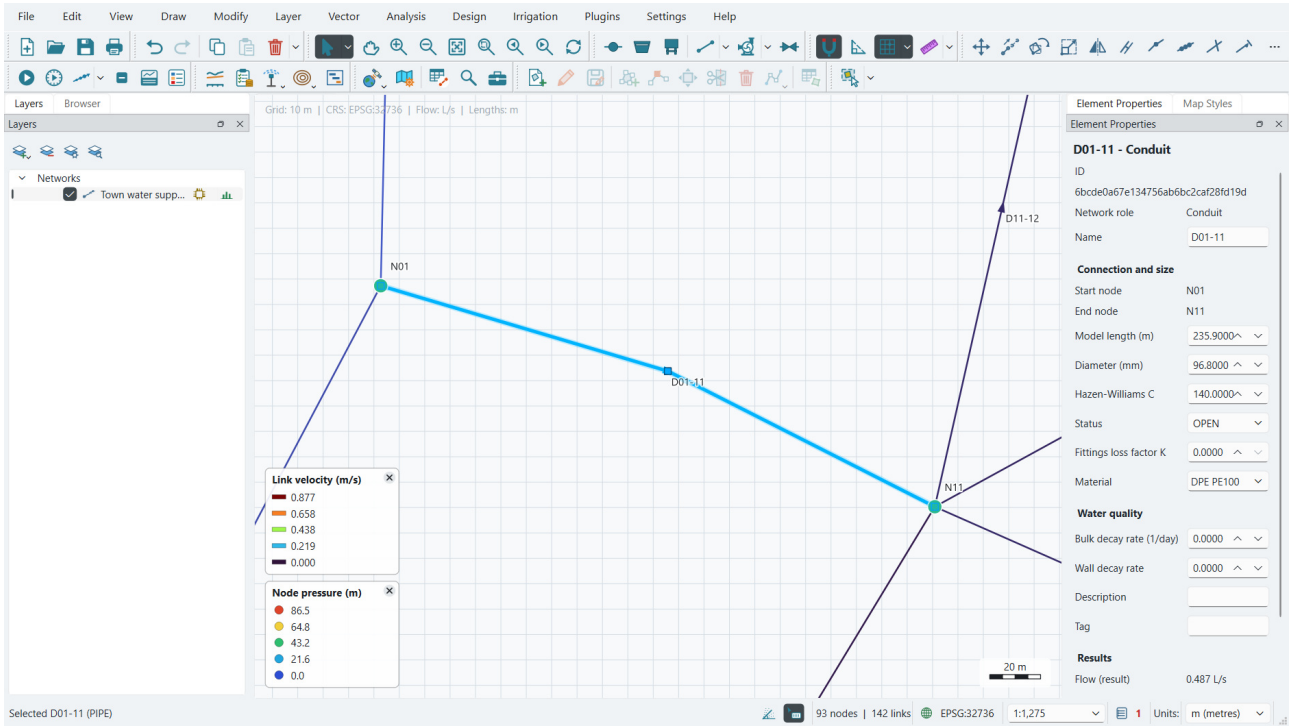
Where FlowCHAOS keeps your settings

Window layouts, recent files, colour ramps, the language and similar preferences are kept for your Windows account, not in the project. **Settings > Language and Region** chooses the language of the menus and the number formats. Reference data you add (pipe series, standards, rates, curves) goes in %APPDATA%\FlowCHAOS\data and overrides the data installed with the program.

6

Drawing and editing

Draw pipes, nodes, pumps and valves precisely, then move, copy, rotate, scale and mirror them.



A pipe selected: its bend grips on the map, its properties beside it

Drawing pipes

Choose the pipe tool and click to place nodes; every click adds a bend or a node depending on the pipe mode (the arrow beside the tool). While drawing, type to enter values exactly:

Type	Meaning
`25`	25 m towards the cursor
`25<30`	25 m at 30 degrees
`@10,-5`	10 m east and 5 m south of the last point
`500200,8501300`	an absolute coordinate

Press **Enter** to accept and **Esc** to stop. A typed length is kept exactly: a pipe drawn as 1000 is 1000 m long, also when it is finished with a double-click.

Drawing aids

On the drafting toolbar:

- **Snap**: to nodes, pipes and the grid. Snapping to a pipe splits it when you place a node. With snapping off, a point goes exactly where you click and the grid is ignored, but a click right on a node or a pipe still lands on it, so a pipe never starts beside a node or doubles it.
- **Ortho**: horizontal and vertical segments only.
- **Grid**: the drawing grid (hidden under basemaps).

In the status bar:

- **Polar tracking**: segments lock to every 15 degrees when you are near one.
- **Dynamic input**: length and angle in a small box beside the cursor while you draw or move.

Moving nodes

With **Select**, drag a node to move it; its pipes follow, and a bend that sat on the node's old place goes with it. Drop the node on another node and the two become one, with all their pipes; drop it on a pipe (snapping on) and the pipe is split there and joined to the node. A click without a drag only selects. Click an empty place on the map to clear the selection.

Changes go into the network layer being edited. If no layer is being edited, the first change starts editing the layer it is in. If another layer is being edited, nothing changes, the map shows the network as it is, and the status bar says which layer to edit (right-click it, **Toggle Editing**).

Modify

Select elements, then use the Modify toolbar or right-click on the map. The button of the tool in use stays pressed; **Esc** stops it and returns to **Select**.

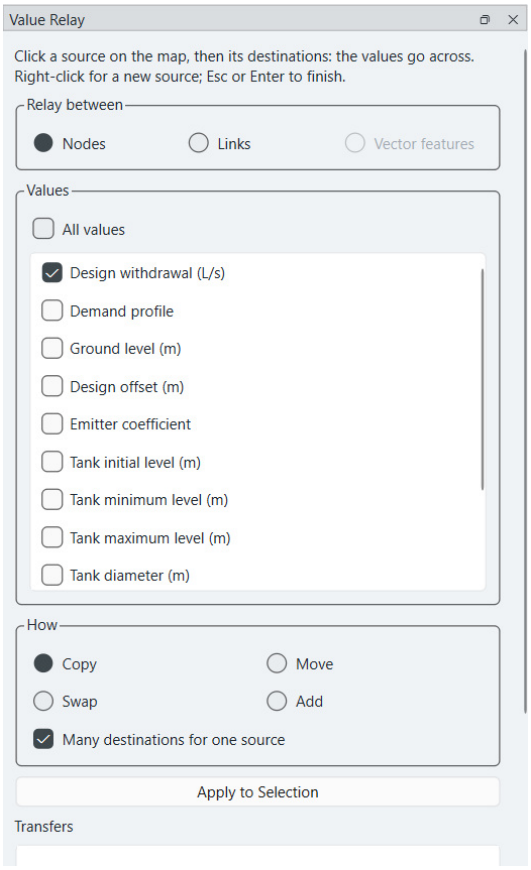
- **Move**: pick a base point, then the destination (or type a distance). Clicking an element that is not selected moves that element alone.
- **Copy with Base Point**: as many copies as you click; Esc ends.
- **Rotate**: pick the centre, then the angle. Press **R** for a reference angle, **C** to rotate a copy.
- **Scale**: pick the fixed point, then the factor. **R** makes a length exactly so long; **C** scales a copy.
- **Mirror**: two points of the mirror line; the original stays unless you press **E**.
- **Offset**, **Split** and **Divide** for the selected pipe.
- **Trim**: hover over a pipe and the piece between bends, crossings and nodes turns red; click to remove it.
- **Extend**: hover near the end of a pipe and a green line shows how far it goes straight on to the next pipe; click to extend it. That pipe is split where they meet and joined to it. A free end moves there; an end that has other pipes stays, and a new pipe continues from it.
- **Edit > Connect Link End to Node**: with one link and one node selected, the end of the link nearer the node moves onto it.
- **Edit > Dissolve Nodes**: junctions that join two pipes are removed and the pipes become one line, so four nodes joined by three pipes become one pipe between the two outer nodes. It works on the selected nodes, on the nodes inside the selected pipes, or on every junction of the network. By default only pipes of the same diameter, roughness and material are joined and junctions with a withdrawal stay; you can also share a withdrawal between the two ends. The removed nodes become bends unless you ask for a straight line. Also on the right-click menu.
- **Edit > Pipe Shape**: **Reverse Direction** swaps the ends of the selected links; **Simplify Shape** removes bends that hardly change the line (closer than the distance you give to a straighter line); **Fillet Bends** turns every bend into a smooth curve of the radius you give (polyethylene is laid to about 25 times its diameter); **Set Exact Length** moves the free end of the selected pipe along its last straight piece until the pipe is exactly so long.

- **Select by Region** and **Delete by Region** (Edit menu, or the arrow beside the Select button): click the corners of an area, double-click or press Enter to close it, and everything inside is selected or deleted. Right-click removes the last corner, Esc stops.

Every step shows a preview before you click and can be undone.

Value Relay

Edit > Value Relay (Shift+V) passes a property from one element to others. Choose the property (diameter, roughness, withdrawal, ground level, pressure class, material and others) and whether to **copy** it (the source keeps its value) or **move** it (the source is set to zero or emptied). Click the source: it is marked SOURCE in red. Then click a destination of the same kind: it takes the value at once and is marked DESTINATION in green. Pick the next pair the same way; Enter or Esc asks whether to leave the tool. Every transfer is one step to undo.



The Value Relay panel: values, how they pass and the transfers made

Converting elements

Edit > Convert To (or right-click the selection) changes the type of the selected elements without drawing them again:

- junctions, reservoirs and tanks become one another; the place, ground level, layer, labels and your own fields are kept, and what only the new type has takes the element defaults;
- a junction between two pipes becomes a pump or a valve: it is split in two and the pump or valve is put between the halves;
- pipes, pumps and valves become one another.

A name made from the old type's prefix (J12) takes the next name of the new type (R1), and controls and rules follow the new name. New projects show the IDs of nodes and pipes from the start.

ID labels and element defaults

Settings > ID Labels and Element Defaults sets how new elements are named and what they are given:

- **ID Labels**: a prefix for junctions, reservoirs, tanks, pipes, pumps and valves (J, R, T, P, PU and V by default), the first number and the increment (10 gives J10, J20, J30). The page shows the next names.
- **Element Defaults**: ground elevation (used where no elevation model or survey gives one), the design offset and withdrawal of junctions, their demand profile and emitter coefficient; tank stand height, diameter and levels; pipe diameter, roughness and minor loss; and the type of new valves.

The settings are kept with the project.

Changing many elements at once

The Edit menu has tools that save hours on a large design (also on the map's right-click menu when pipes are selected):

- **Insert Components on Pipes**: one component on every selected pipe: an isolation valve, a pressure reducing or sustaining valve, a flow control or pressure breaker valve, a check valve, a meter or a pump. Choose where: at the start or the end of each pipe as drawn, at its upstream or downstream end (by the flows of the last run), or in its middle. The component takes a short piece of the pipe (0.5 m by default) next to a new junction, so the pipe keeps its name, layer and properties. For example: select every tertiary pipe with **Select by Attribute** (diameter below 63 mm), then insert isolation valves at their upstream ends, named IV1, IV2 and so on.
- **Mark High and Low Points**: along the selected pipes, an air valve at every junction higher than its neighbours and a washout at every one lower than them, ready for the bill of quantities.
- **Connect Points to Nearest Pipe**: a service pipe from each selected unconnected node (a house, a tap, a hydrant) to the nearest pipe, which is split at the tee.

Each is one step of Undo.

Selecting

Drag left to right for a window selection (whole elements), right to left for a crossing selection. **Select Similar**, **Select Connected Network** and **Select by Attribute** (on the toolbar and the map's right-click menu) select by rule.

Selected elements are drawn in the selection colour with a halo that stays the same size at every zoom. **View > Map Options > Selection** sets the colour, turns the halo off, or makes it pulse so a selected node is easy to find on a busy map.

7

Layers and data

Bring in networks, survey points, drawings, rasters and basemaps, each in its own coordinate system.

Adding layers

Use the **Add Layer** button above the layers list (or drag files onto the map):

- **Network layer**: FlowCHAOS network files (.fcnet), hydraulic model files (.inp, .net), network CSV, shapefiles and KML.
- **Survey points**: XYZ, CSV, TXT and spreadsheet files. Points become unconnected nodes with their ground level and every column of the file, ready to label and contour.
- **Raster or DEM**: GeoTIFF and other rasters. A DEM can supply ground levels. A PDF drawing or map comes in as an image: choose its page (see Georeferencing below).
- **Vector layer from file** (Ctrl+Shift+V): shapefiles, GeoPackage, GeoJSON, KML and DXF drawings as vector layers, every layer and kind of geometry of the file.
- **CAD drawing (DXF)**: lines, polylines, arcs and splines become a network (see below), or the whole drawing comes in as vector layers (see DXF drawings as vector layers).
- **Surface map**: contours and other maps from points or a DEM.
- **Satellite and basemaps**: online imagery, street and topographic maps, vector basemaps, your own tile services and offline tile packages.

Networks are layers

Every network is a layer, listed under **Networks** in the Layers panel: the network of a new project, networks added with **Add Layer** or **File > Import Network > Hydraulic Model**, a file dropped on the map, a file opened with **File > Open** (a new project holding it as its first layer), survey points, and empty layers made with **Layer > Add Empty Network Layer**. A layer brings everything it needs to run: curves, demand profiles, controls and rules come with it, and names already used in the project are renamed (T1 becomes T1_2) together with every reference to them. The first layer of an empty project also sets the analysis options, times, energy and report settings.

A network layer is changed only while it is being edited, as in a GIS: right-click it and choose **Toggle Editing** (its name turns bold, with a pencil). What you draw goes into that layer, a new pipe connects only to its nodes, and the other layers stay as they are; an attempt to change them says which layer to edit. Choosing a drawing tool starts editing by itself when it is clear which layer: the only one, or a new one in an empty project; with several layers FlowCHAOS asks.

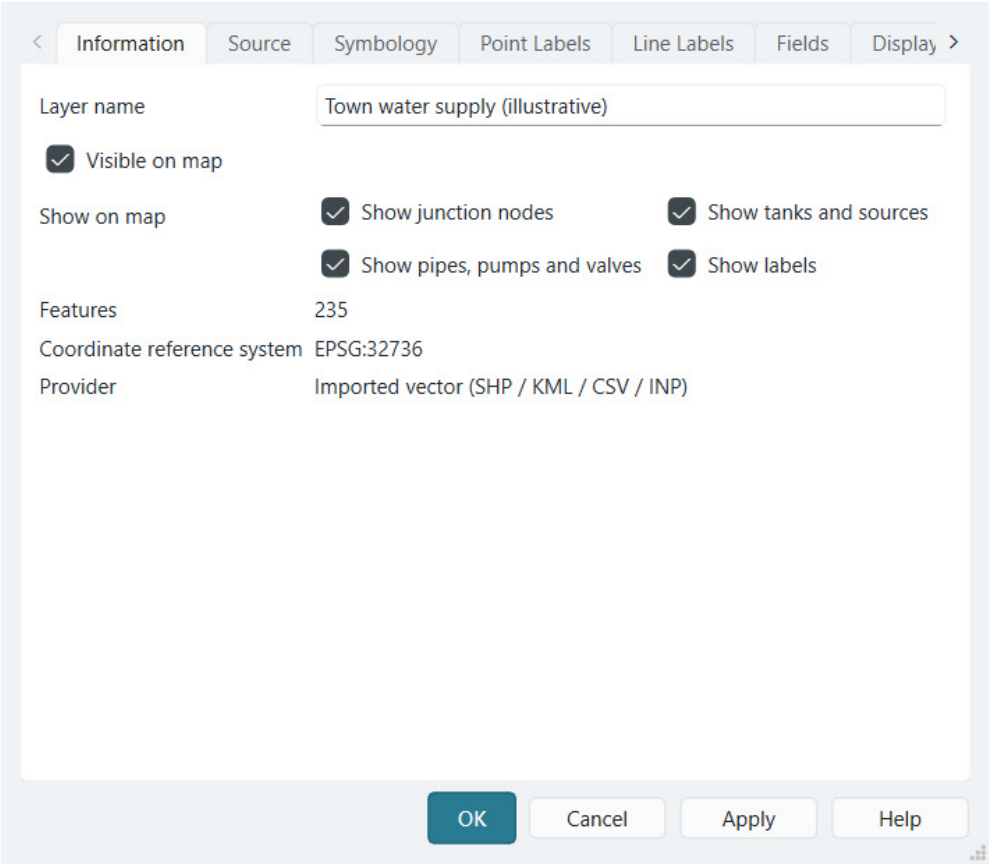
Right-click a network layer for its menu: Toggle Editing, Zoom to Layer, Attribute Table, labels, **Analyse This Network**, **Export Layer As** (FlowCHAOS network, hydraulic model, CSV, GeoJSON, shapefile, GeoPackage, KML or DXF of that layer alone), Rename, Properties and Remove. Right-clicking an element on the map shows the same commands for its layer, under **Layer**.

A **FlowCHAOS network file** (.fcnet) holds one network with every property, the profiles, curves, controls and rules it uses, its coordinate system and its look (symbol and labels). Use it to share a network or to add it to another project as a layer; a project file (.fchaos) holds the whole work.

Properties of a network layer has, on its Information page, **Show on map** switches: junction nodes, tanks and sources, pipes, pumps and valves, and labels. Hiding the junctions shows the pipes alone on a dense network; the hidden elements stay in the network and in the analysis. Hide a layer and its elements leave the map and the analysis; its labels, colours and attribute table are its own (labels follow Map Options until the layer is given its own). After a run the map shows the network that was analysed.

A progress window shows each stage of a long import; what was imported, and any notes, go to the status bar and the

log. A message box opens only when something needs your attention.



Properties of a network layer, with the Show on map switches

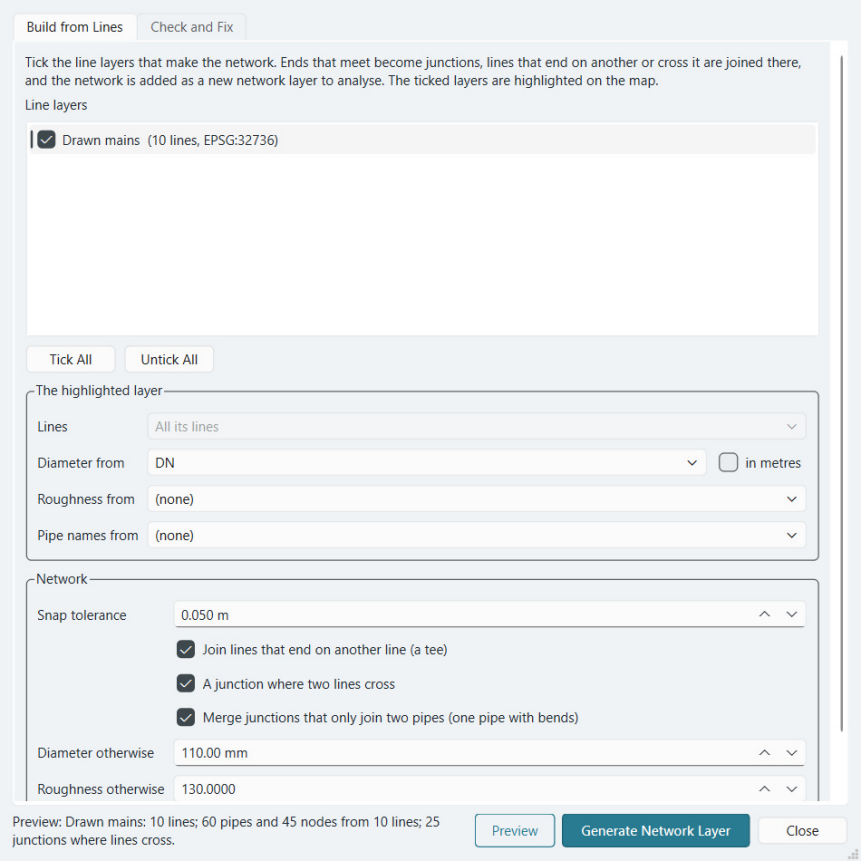
Networks from lines (DXF, shapefiles, KML, GPX)

Lines become pipes and their ends become junctions. A junction is placed only where pipes meet: at the ends of lines, where lines join end to end, and at a tee, where the end of one line lies on another, which is then split there. Bends stay as vertices of a pipe, so a curved main is one pipe with its true length, not a chain of short pipes. Two pipes that meet end to end with the same diameter, roughness and drawing layer are joined into one. Point features become junctions with their elevation and withdrawal when the file has them.

From a DXF, lines, polylines, 3D polylines, arcs, splines and ellipses are read. The drawing layer of each line is kept as a field; a diameter in the layer name (DN150, D110, 160mm) becomes the pipe diameter, and the heights of 3D polylines become ground levels.

Build Network from Lines

Layer > Build Network from Lines (or right-click a line layer, **Build Network from This Layer**) makes a new network layer from line layers already in the project. Tick the layers to use (each is highlighted on the map as you choose), and for each one the fields that hold the diameter, roughness, material, name and ground level, with a value to use where a field is empty. Set the snapping tolerance that joins line ends that nearly meet, whether lines are split where another ends on them (tees), and whether a junction is put where two lines cross (untick it when pipes cross at different depths without being connected), and whether pipes that meet end to end with the same properties are joined. **Preview** counts the pipes, junctions and joins it will make; **Generate** adds the network as a layer of its own, ready to run. Dropping line files on the Networks group in the Layers panel does the same.



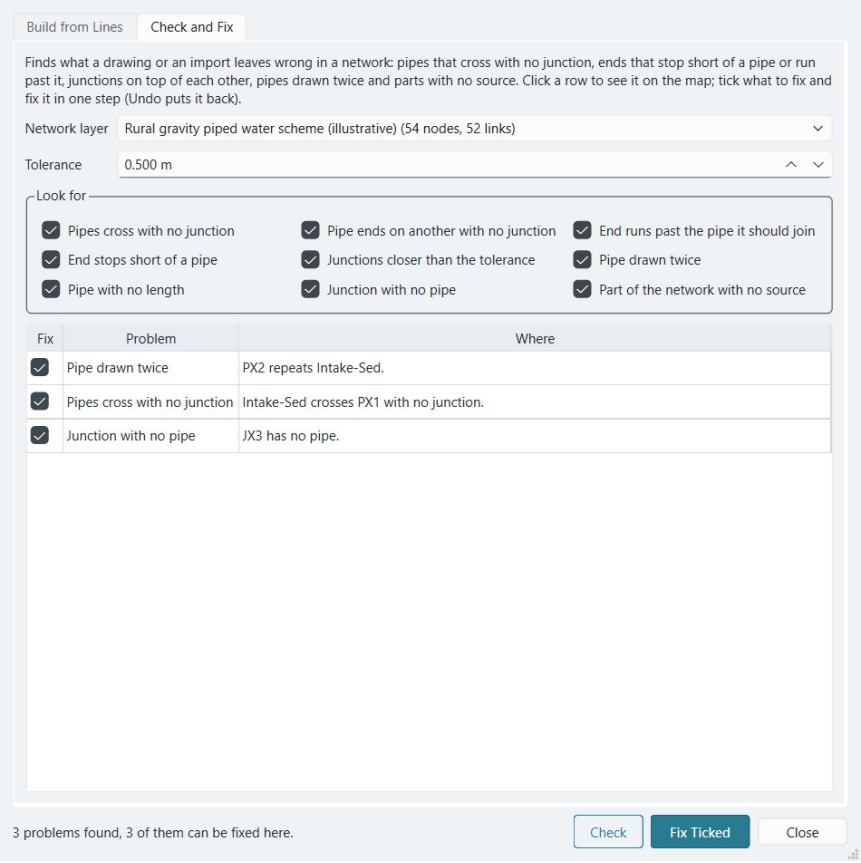
Build Network from Lines: layers, field mapping and the preview

Check and Fix Network

Layer > Check and Fix Network (the second page of the Network Builder) finds what a drawing or an import leaves wrong in a network layer, within a tolerance in metres:

- pipes that cross with no junction, and pipe ends that lie on another pipe with no junction;
- ends that run just past the pipe they should join (overshoots) or stop just short of it (undershoots);
- junctions closer together than the tolerance, pipes drawn twice, pipes with no length and junctions with no pipe;
- parts of the network with no reservoir or tank (reported for you to decide).

Check lists them; click a row to select the elements and zoom to the place. **Fix Ticked** fixes the ticked ones together: junctions are put where pipes cross or meet, overshooting tails are cut off, short ends are joined to the pipe, doubled pipes and loose junctions are removed, and levels of new junctions are taken along their pipe. The whole fix is one step that Undo puts back.

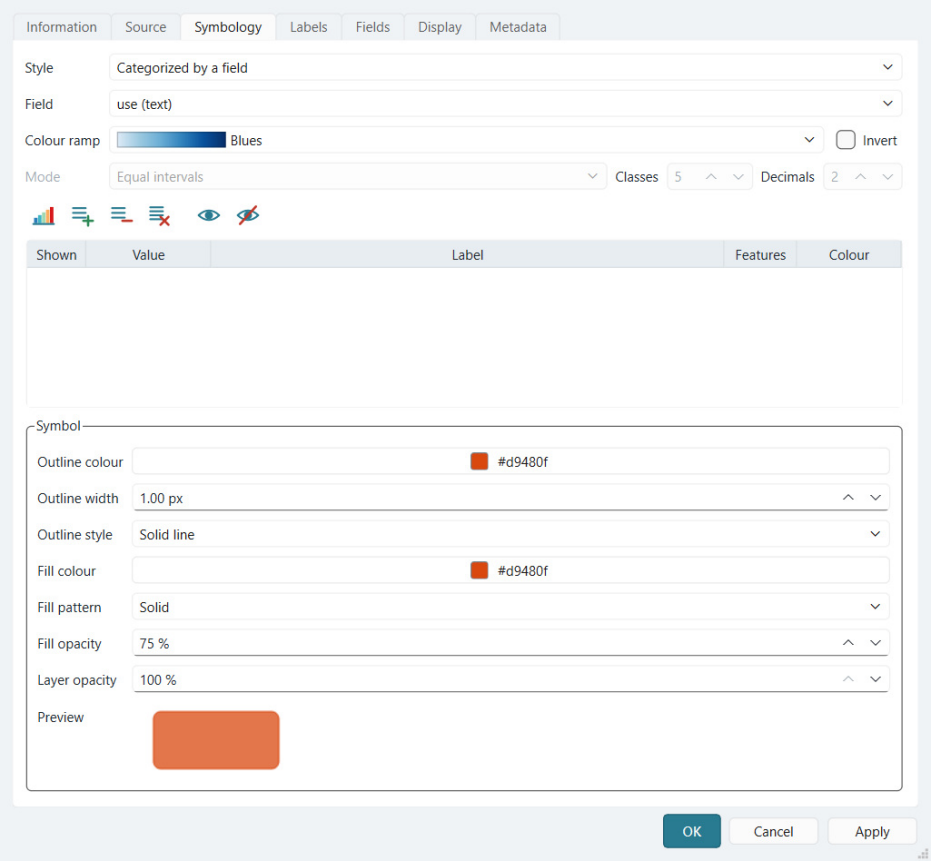


Check and Fix: the problems of a network, each shown on the map

Vector layers

Vector layers hold points, lines and polygons that are not a network: plots and parcels, roads, rivers, service areas, valve chambers, survey lines.

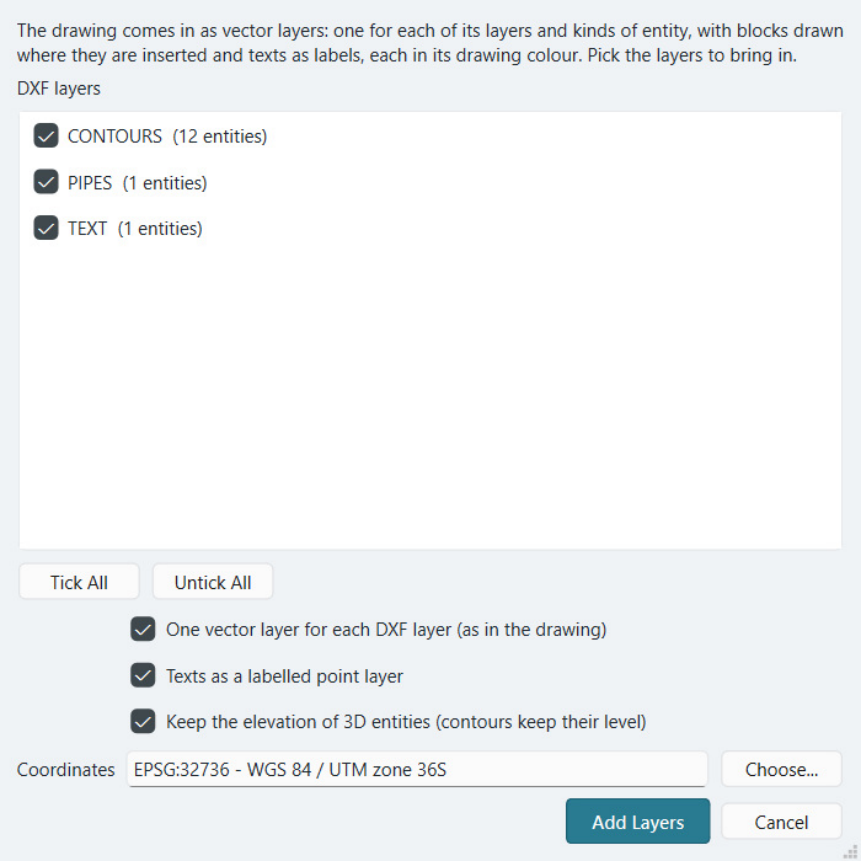
New Shapefile Layer (Layer menu, or the first button of the Digitizing toolbar) creates a layer of points, multipoints, lines, multilines, polygons or multipolygons, as a new shapefile is created in a GIS: give it a name, its geometry type, its coordinate system and its fields (text, whole number, decimal number, date or yes/no). A new layer starts with an id field that numbers each new feature by itself (1, 2, 3) and a name field; with **Number new features automatically** ticked, adding a point does not stop to ask for its values: fill them in later in the attribute table. The layer is written as a shapefile where you choose, or kept in the project only. **Layer > Add Vector Layer** reads shapefiles, GeoPackage, GeoJSON, KML and DXF files as editable layers (not as a network): a file with several layers, or with points, lines and polygons together, gives one vector layer for each. Vector layers are listed under **Vector layers** and saved with the project (the layers read from one file, such as the drawing layers of a DXF or the tables of a GeoPackage, sit under a group named after the file, which shows or hides them together); right-click one to reproject it to another coordinate system.



Vector layer properties: a categorized symbology

DXF drawings as vector layers

A DXF added as a vector layer asks which of its drawing layers to bring in, and comes in as the drawing is organised: one vector layer for each drawing layer and kind of geometry (points, lines, polygons), or one for each kind when the drawing layers are not split. Blocks are drawn where they are inserted, texts become a labelled point layer at their height and angle, and every feature keeps its drawing layer, line type, colour and handle, with the colour of the drawing on the map. 3D entities keep their elevation in a field, so contour lines of a survey keep their level and can make a DEM. A DXF does not say its coordinate system: choose the one its coordinates are in.



A DXF drawing added as vector layers: its drawing layers to bring in

Selecting features

The **Select Features** button (on the Selection toolbar, and in **Vector > Selection**) selects the features of any vector layer, whether it is being edited or not. A click on it selects in the way last chosen; the arrow beside it opens every way to select:

- **Select by Click or Box**: click a feature, or drag a box over several.
- **Select by Drawn Polygon**: click the corners of a polygon; a double-click, Enter or a right-click selects what it touches.
- With either, **Shift** adds to the selection, **Ctrl** takes from it (Ctrl and a click on a feature that is not selected adds it), and **Esc** clears the selection.
- **Select by Expression** (area_ha > 2 and crop == 'maize'), **Select by Spatial Relation** (the features that intersect, lie within, contain, touch, cross, overlap, are apart from or are near the features of another layer, or only its selected ones), **Select All**, **Invert Selection** and **Clear Selection**.

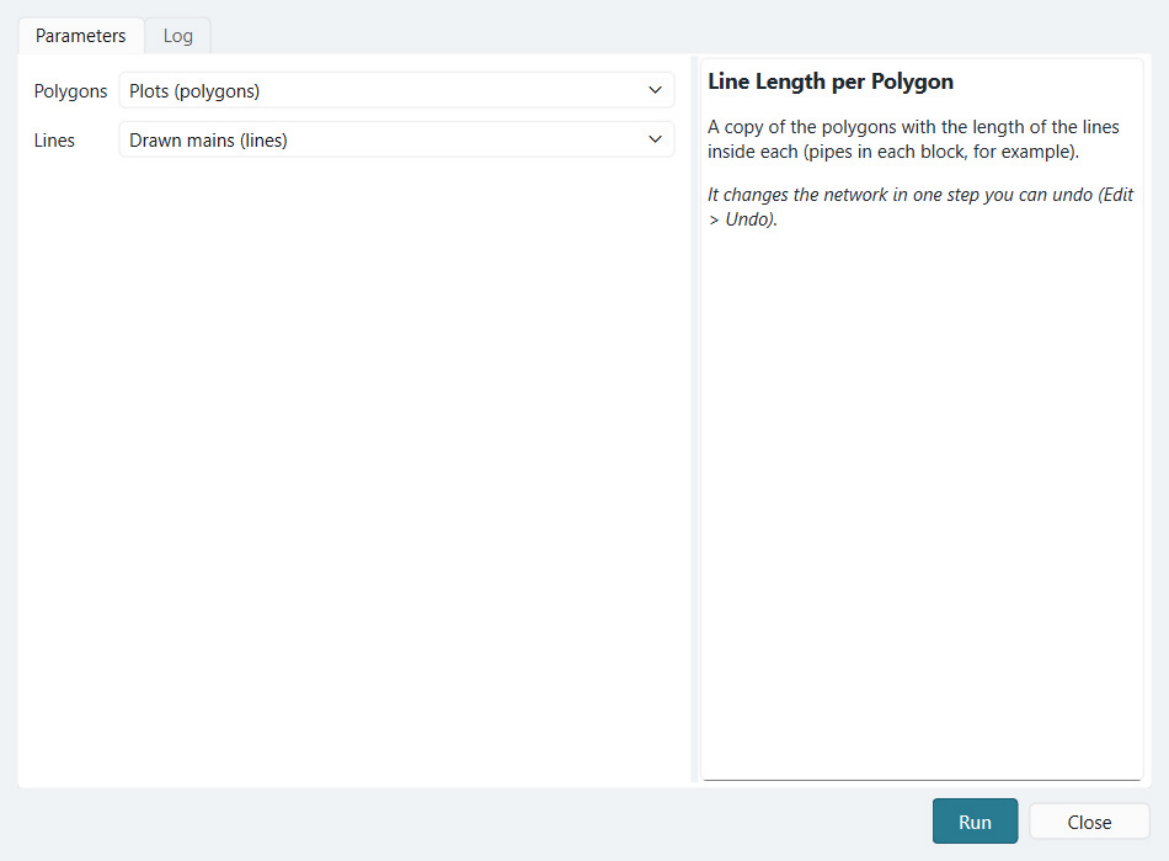
There is one selection tool for vector layers: it selects in any layer, and in the layer being edited the selected features are the ones Delete, Move, Split and the Advanced Editing commands work on. The network has its own, **Select Network Elements**, on the Navigate toolbar.

Features are picked from the vector layer chosen in the Layers panel, or from every visible vector layer when none is chosen. Selected features are drawn in the selection colour, and the attribute table and the tools that work on "only the selected features" use them.

Processing tools for vector layers

The **Vector** menu (and the Processing Toolbox) has five groups of tools for vector layers. Each reads a layer (all its features, or only those selected on the map), works in metres for any coordinate system, and gives its result as a new vector layer, a selection or a table:

- **Measures and Summaries**: Field Summary, Totals by Class, Points per Polygon, Line Length per Polygon (pipes in each block), Distance to Closest Feature, Crossing Points, Centre of Features.
- **Overlays**: Offset Zone (a band at a distance around features), Cut to Outline, Common Parts, Cut Away, Combine Polygons, Parts Not Shared, Fuse Features, Enclosing Outline, Cut along Lines.
- **Shapes and Conversions**: Centre Points, Outlines of Polygons, Fill Closed Lines, Break Lines into Segments, Reduce Vertices, Round Corners, Add Vertices, Points at Intervals (with chainage), Vertices as Points, Size and Position Fields, Bounding Rectangles, Split Multipart Features, Repair Shapes, Supply Zones around Points (the area each junction or well serves).
- **Selection and Sampling**: Select by Spatial Relation, Filter by Spatial Relation, Random Sample, Filter by Expression, Copy Selection to New Layer, Layer Footprint, Square Grid, Scatter Points in Polygons.
- **Tables and Layers**: Append Layers, One Layer per Value, Link Tables by Field, Link Attributes by Position, Change Coordinate System, Remove Repeated Shapes, Number Features, Drop Field, Field Calculator.



A processing tool for vector layers: line lengths in polygons

Editing

Select a vector layer in the Layers panel: **Toggle Editing** and **Open Attribute Table** on the toolbar act on it (they are greyed until a vector layer is selected). Choosing a **Digitizing** tool with a vector layer selected starts editing it. While a layer is edited the Digitizing tools work on it:

- **Add Feature**: click the vertices; a point needs one click, and a line or polygon is finished with a double-click, Enter or a right-click. Backspace removes the last vertex, Esc drops the sketch, and with Shift held the new part

is added to the selected feature of a multi layer. A new feature is numbered automatically and nothing is asked: a layer edited for the first time without a numbering field gets an **id** field (1, 2, 3 in the order of its features), and each new feature takes the next number. A field all the layer's features share one value of (a drawing layer's name) is filled in too; the rest are filled in the attribute table. An id typed by hand must be free: one used by another feature is refused, with the next free number. A layer can ask for the values with a form instead (its **Properties > Fields** page).

- Selecting: the **Select Features** button of the Selection toolbar; Delete removes the selected features of the layer being edited.
- **Move Feature**: drag a feature; it follows the pointer and lands where you release it.
- **Vertex Tool**: drag a vertex to move it (the shape follows it as it moves), double-click a segment to add a vertex, right-click a vertex to remove it.
- **Split Features**: select the polygons or lines to cut (or none, to cut every one the line crosses), choose Split Features, then click a cutting line across them, starting and ending outside them, and finish it with a double-click, Enter or a right-click. Each piece keeps the attributes; Undo puts the feature back whole. With snapping on, the clicks jump to nearby nodes and grid points.

Advanced Editing (the dropdown beside Delete) has **Merge Selected Features** (one feature from the selected ones, keeping the attributes of the one you choose), **Separate Parts** (each part of a multipart feature as a feature of its own), **Simplify Selected Features** (fewer vertices, within a tolerance in metres), and **Copy Features** and **Paste Features**, which bring features from one layer into the layer being edited, with fields matched by name and the coordinates moved into its system. Dissolve, union, clip and the other overlays are in the Processing Toolbox.

Save Layer Edits writes the layer back to the file it came from when that file can be written (a shapefile, GeoPackage or GeoJSON). A layer drawn in the project, or read from a drawing or KML, is a temporary layer: its edits are saved in the project and in a temporary file, without asking for a file name; **Save Layer As** makes a file of it. **Toggle Editing** to stop asks to **Save** the edits, **Discard** them (the layer goes back to what it was when editing started) or **Cancel** (keep editing). Every edit is one step of Undo.

Pieces made by **Split Features**, **Separate Parts** and pasting are numbered: the layer's numbering field (or a whole-number field called id, fid or objectid) takes the next free numbers, so nothing is asked and no number repeats. A cutting line drawn a little short of a polygon's edge is drawn on to it.

Writing expressions

Expressions (the Update bar, the Field Calculator, Select by Expression, Find and Query) name fields as they read, in any case: area_ha * 1.3, "Design withdrawal (l/s)" * 4 / 2 (double quotes for a name with spaces or brackets), design withdrawal (l/s) * 4 / 2 or design_withdrawal_(l/s) * 4 / 2. 'Single quotes' are text. Type two letters, or a double quote, for a list of the fields and functions that match; while typing in the Update bar, a click on a column heading or a cell puts that column's field into the expression, as a spreadsheet takes a cell into a formula.

Every expression is checked as it is written: a green tick (and the line under the Expression Builder's box) when it can be worked out, with what the first row gives; a red cross with what is wrong, such as an unknown field or a bracket left open. In the Expression Builder the functions show in bold teal, fields in orange, text in green and numbers in purple. The Update list names every column: those that cannot be set (measured from the shape, or worked out by the analysis) are greyed, with the reason, and can still be used in the expression.

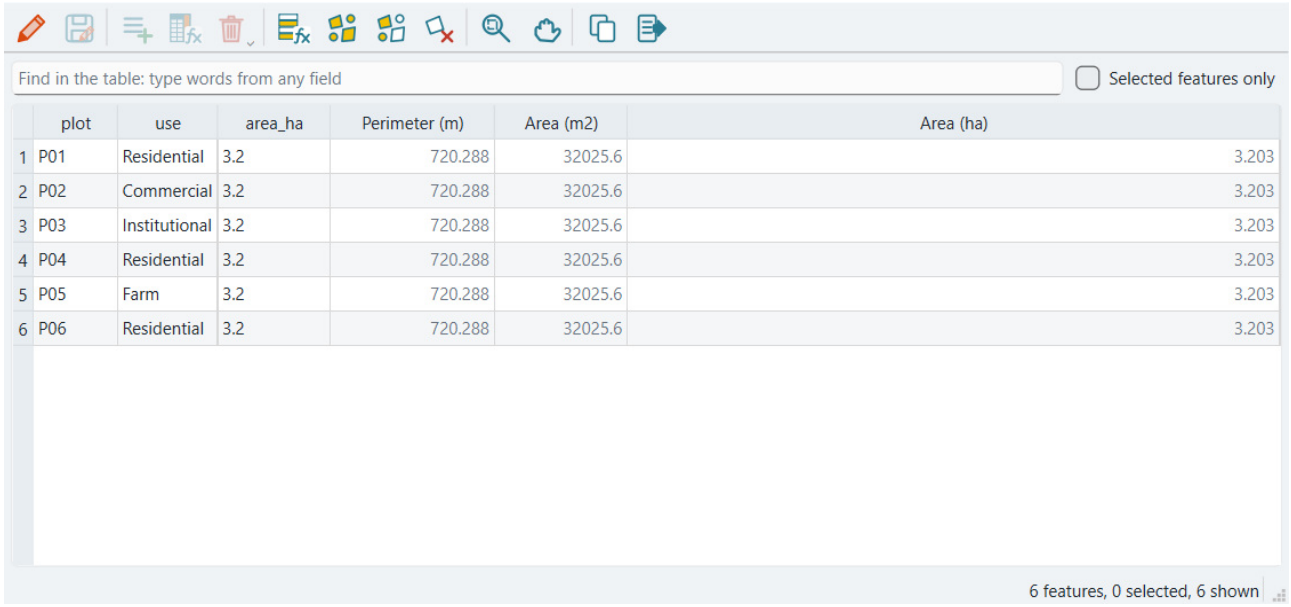
Attribute table

The attribute table lists every feature with its values and its calculated fields: length_m for lines; perimeter_m, area_m2 and area_ha for polygons, holes left out. They are measured on the ellipsoid, the true ground values a GIS gives, or on the map grid when **Project Properties > Measurements > Measure areas** says so (a UTM grid is up to 0.1 % off the ground near its central meridian, more towards the edges of its zone). They are worked out from each shape, always current (shown in grey), and can be used like any other field to colour the layer, label it and in expressions. Large layers open at once: rows are drawn as they come into view.

- **Selecting**: click rows, with Ctrl or Shift to add more; **Select by Expression** (with the Expression Builder, as a new selection, added to it, taken from it or within it), **Select All**, **Invert Selection** and **Deselect All**. The table and the map share one selection.
- **Finding**: type words in the find box, or tick **Selected features only**; the status line counts the features, the selected ones and those shown.

- **Columns**: click a heading to sort (numbers as numbers); right-click a heading to sort, fit, hide or show columns. Double-click a row number to zoom the map to that feature.
- **Editing**: **Toggle Editing** opens the cells for typing and keeps the table in front. The **Update** bar sets a field from a value or an expression (area_ha * 1.3, if(crop == 'maize', 2, 1)) for the selected features or all of them. **Delete Fields** removes several fields at once (Tick All for a fresh start); the id field stays, so new features keep their numbers, and **Add Field** adds your own. **Add Field**, the **Field Calculator** and the **Delete** button (its menu has Delete Selected Features and Delete Field) complete it; every change is one step of Undo, and **Save Layer Edits** writes the layer to its file.
- **Copy Selected Rows** (to paste into a spreadsheet) and **Save Selected Features As** a new file.

The **Field Calculator** fills a new or an existing field from an expression of other fields, for example area / 10000 for hectares or concat(name, ' ', zone). It knows + - * /, brackets and round, abs, min, max, int, float, str, len, sqrt, upper, lower and concat, and can update the selected features only.



	plot	use	area_ha	Perimeter (m)	Area (m2)	Area (ha)
1	P01	Residential	3.2	720.288	32025.6	3.203
2	P02	Commercial	3.2	720.288	32025.6	3.203
3	P03	Institutional	3.2	720.288	32025.6	3.203
4	P04	Residential	3.2	720.288	32025.6	3.203
5	P05	Farm	3.2	720.288	32025.6	3.203
6	P06	Residential	3.2	720.288	32025.6	3.203

The attribute table of a vector layer, with its toolbar

Properties

Double-click a vector layer (or right-click, Properties) for its properties:

- **Information** and **Source**: the geometry, the number of features, the extent, the file.
- **Symbology**: a **single symbol**, **categorized** (one colour for each value of a field) or **graduated** (classes of a number field). The symbol has a stroke colour, width and style (solid, dash, dot, dash-dot), a fill colour, opacity and style (solid, hatched, cross-hatched, diagonal, dotted or none), and for points a marker (circle, square, triangle, diamond, star, pentagon, hexagon, cross or X) and a size.

  Classes take their colours from a **colour ramp**: ramps for values (from light to dark, or two colours around a middle), sets of distinct colours for categories, or **random colours**; **Invert** turns a ramp round, and changing the ramp recolours the classes at once. Graduated classes are made by **equal intervals**, **equal counts** (quantiles), **natural breaks** (Jenks: similar values kept together), **standard deviation** or **round numbers**, with the number of classes and the decimals of their labels.

  The buttons above the class table: **Classify** (categories add the values not listed yet; graduated classes are made again), **Add Class**, **Delete Class**, **Delete All Classes**, **Show All** and **Hide All**. In the table, untick a class to hide it, edit a value, a range or a label in its cell, and click a colour to change it; **Features** counts the features in each class. The ramps are kept in style/colour_ramps.json in the reference data: add your own there.
- **Labels**: a field, the font, size, colour, bold, a halo and the placement (over, above, below, right or left of the

feature). A polygon's label sits at the point deepest inside it; a label that would cover another is left out until you zoom in, unless **Show every label** is ticked.

- **Fields**: the fields with their types.
- **Display**: the scales the layer shows at.
- **Metadata**: author, abstract and keywords.

Apply shows the change at once; OK keeps it.

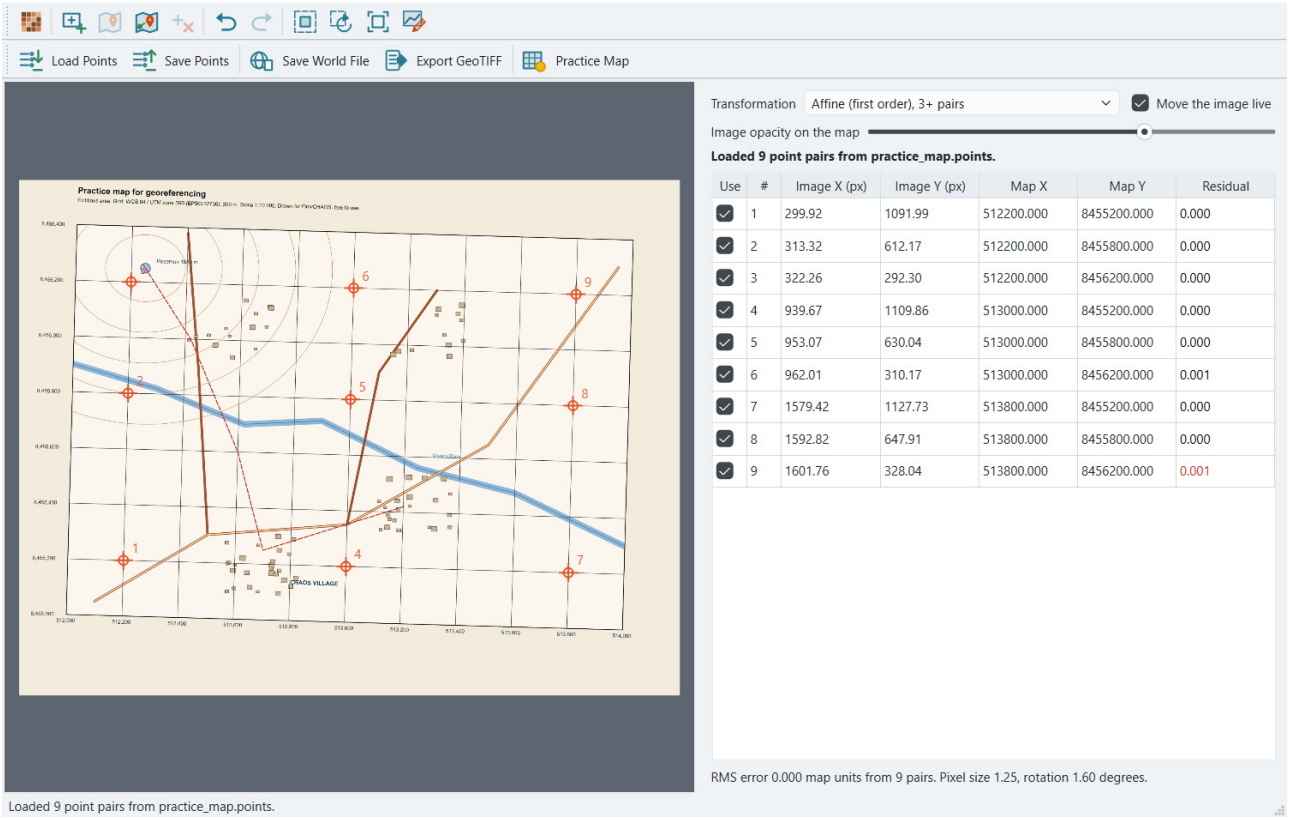
Georeferencing and tracing a map or drawing

Layer > Georeferencer places a scanned map, a drawing, a photo or a page of a PDF on the map. A PDF asks for its page first: every page is shown as a preview with its sheet size (A1 landscape, for example), and the chosen page is drawn on white at 150 to 400 dpi and kept as an image beside the PDF. There is one Georeferencer window: it opens empty, or with the image chosen in the Layers panel, and **Add Image** loads another. Opening another image (or the practice map) asks first whether to save the point pairs of the one shown.

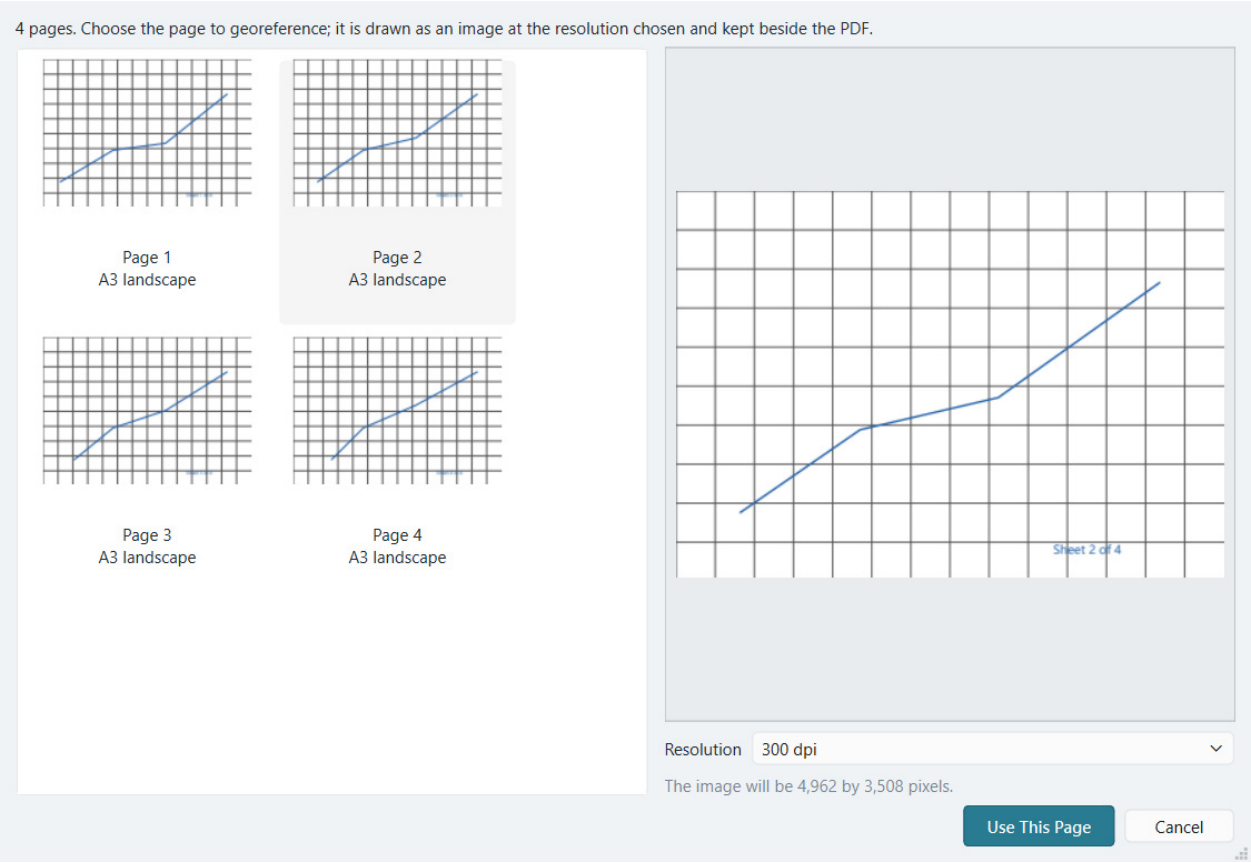
1. **Add Point**: click points on the image (grid crossings, road junctions, building corners). Each click adds a row to the table; the window stays where it is, so you can add all the image points first.
2. Select a row and click **Pick Map Point**: the main map comes forward, click the same place on it (or type its coordinates in the table), and the window comes back with the next row waiting.
3. **Pick on Map Only** works on the main map alone: click a point of the overlaid image, then where it belongs, pair after pair; the window returns when the chosen method has the pairs it needs.

The table shows each pair with its residual and the fit its RMS error. **Delete Point** (or the Delete key) removes the selected pair. **Undo** and **Redo** (Ctrl+Z, Ctrl+Y) step back and forward through the pairs and the placement; **Place in View** puts the whole image in the middle of the map as a starting position, and **Reset Placement** returns it to where it was when opened. The second row of buttons keeps the result: **Load Points**, **Save Points**, **Save World File** and **Export GeoTIFF** (a georeferenced copy of the image, added as a layer). When it fits, **Trace Over Image** keeps the image faintly under the network and starts the pipe tool, so the network is drawn along the lines of the image. The image's Layer Properties have a **Georeferencing** page with its state.

To learn it, open **File > Open Sample Projects > Georeferencing practice map** (or the Practice Map button): a scanned-style sheet with a coordinate grid in UTM zone 36 South. Load Points with practice_map.points beside it for nine pairs at grid crossings, or add your own by typing the grid coordinates written on the sheet.



The georeferencer with the practice map and its point pairs



A page of a PDF chosen from previews, to georeference

Basemaps

Layer > Basemaps and Satellite Imagery (or the satellite button) adds online maps under the network:

Group	Basemaps
Satellite imagery	Esri World Imagery, Esri Clarity, Esri Firefly, Google Satellite, Google Satellite Hybrid, Bing Virtual Earth
Street maps	OpenStreetMap, Esri World Street Map, Carto Positron, Dark Matter and Voyager, Google Maps
Topographic and terrain	Google Terrain, Google Terrain Hybrid, OpenTopoMap, Esri World Topographic, Hillshade, National Geographic
Vector basemaps	ESRI Colored Pencil, Dark, Modern Antique, Nova, Night, Topo, Charted Territory, OSM Standard, OSM Street, OSM Light Grey, OSM Dark Grey, OSM Blueprint
Labels and overlays	place names, boundaries and roads to put over imagery

Vector basemaps are drawn from map data with their own style (patterns, line styles and labels), so they stay sharp at every zoom. Tiles are kept on disk, so maps already seen also work offline. Respect each provider's terms of use; any other service can be added as a custom tile address with {z}, {x} and {y} (or {q} for quadkey services). Once a day FlowCHAOS checks in the background that every catalogue service still answers; one that has been retired is greyed out in the menus.

Measuring

The **Measure** button measures distances, areas and perimeters, and angles and bearings. The **Measure** panel, under Element Properties, shows the values without covering the map, with the units to use: metres, kilometres, feet, yards, miles, nautical miles, centimetres, millimetres, inches, degrees or map units for lengths; square metres, square kilometres, hectares, acres, square feet, yards, miles, nautical miles, centimetres, millimetres, inches, square degrees or map units for areas; degrees, radians, gradians, mils or degrees, minutes and seconds for angles. Its buttons start again, copy the values and close (clearing the drawing and returning to Select); Esc clears it too. The default units of the project are set in **Project Properties > Measurements**, with **Measure areas**: on the ellipsoid (the default, as the areas of vector features) or on the map grid.

Merging networks

Layer > Merge Network Layers (or right-click a network layer > Merge Other Networks Into This) joins other network layers into one. Nodes closer than the join distance become one node; a node of one network lying on a pipe of the other splits that pipe and connects there; ends within the connection distance can be joined by a new pipe. Pipes, pumps and valves that are the same in both are kept once. The report lists what was joined, connected and left apart.

Coordinate systems

The project has one coordinate system (status bar, or **Project Properties > Coordinates**). A file opened into an empty project keeps its own system: the project takes it, and nothing is reprojected until you choose another system yourself. A network added to a project in another system is transformed into the project's system, and its original system is kept with the layer. When you import points, choose the system they were surveyed in.

Vector and raster layers in other systems are drawn in the project's system as they are. Right-click a vector layer > **Reproject Layer** to add a copy in another system (save it with Save As), or a raster layer > **Save Reprojected Copy**. **Properties > Coordinate system** of a vector layer only changes the system it is read in, for a file that was labelled wrongly.

Layer properties

Double-click a layer (or use the properties button above the list):

- **Symbology**: for points (junctions, tanks, reservoirs) the colour, size and opacity; for lines (pipes, pumps, valves) the colour, thickness and pattern (solid, dashed, dotted, dash and dot). A map view, when one is chosen, colours the layer instead.
- **Labels**: see *Map views, legends and labels*.
- **Fields**: every field with its type, how many features fill it and an example value.
- **Display**: show the layer only between two map scales.
- **Metadata**: author and abstract, saved with the project.

Project Properties > Data Sources lists every file the project uses; missing rasters can be pointed at their new location there.

Ground levels

New and moved nodes take their ground level automatically. Choose the source in **Project Properties > Elevation**: a DEM, online terrain, or none (type levels by hand).

Find and Query

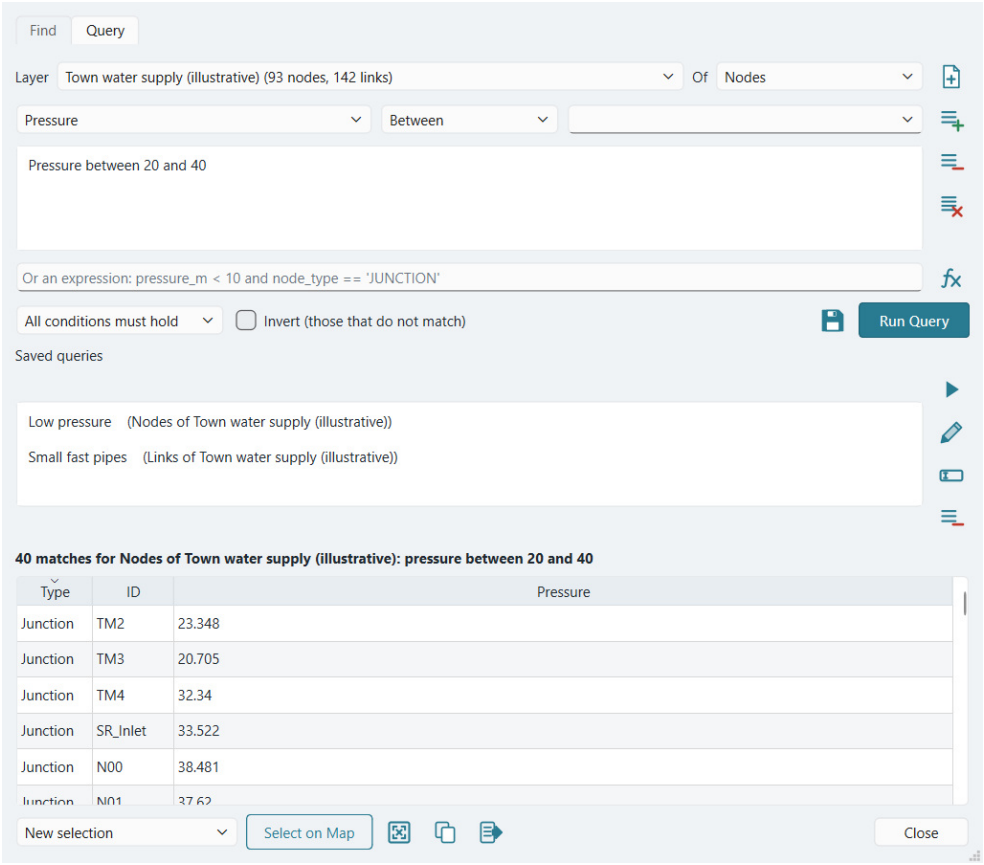
Find and Query (Edit menu, Ctrl+F, or the toolbar) is a tool window beside the map, for network elements and vector features alike.

Find matches names as you type, in all elements, one type (junctions, reservoirs, tanks, pipes, pumps, valves) or one vector layer (any text value of its features), by contains, starts with, exact or ends with, optionally with the links of the nodes found. When the project has several network layers, choose the one to search in, or all of them.

Query runs on the layer you choose, named in full: the nodes or links of one network layer, of all network layers, or a vector layer. A query is made of conditions (a field, a comparison and a value) that must all hold, or any of them, or of an expression such as pressure_m < 10 and node_type == 'JUNCTION' built with the **Expression Builder**. The comparisons are below, at most, equal to, not equal to, at least, above, between (10 and 20), one of (110, 160, 200 or PVC, HDPE), contains, does not contain, starts with, ends with, is empty and is not empty. **Invert** finds the ones that do not match. **New Query** starts again; double-click a condition to change it, and Delete removes it. The label above the matches says exactly what was run and on which layer, and with several networks each match shows the network it belongs to.

Save Query keeps a query by name with the project. Each saved query runs on its own, on its own layer, with **Run** (or a double-click); **Load into the Editor** brings it back to change, and saved queries can be renamed and removed.

The matches are listed with the value that matched. **Select on Map** makes them a new selection, adds them to it, takes them from it or keeps only those already selected; **Zoom To**, **Copy IDs** and **Export CSV** act on the selected rows, or on all of them.



Find and Query on a chosen layer, with saved queries

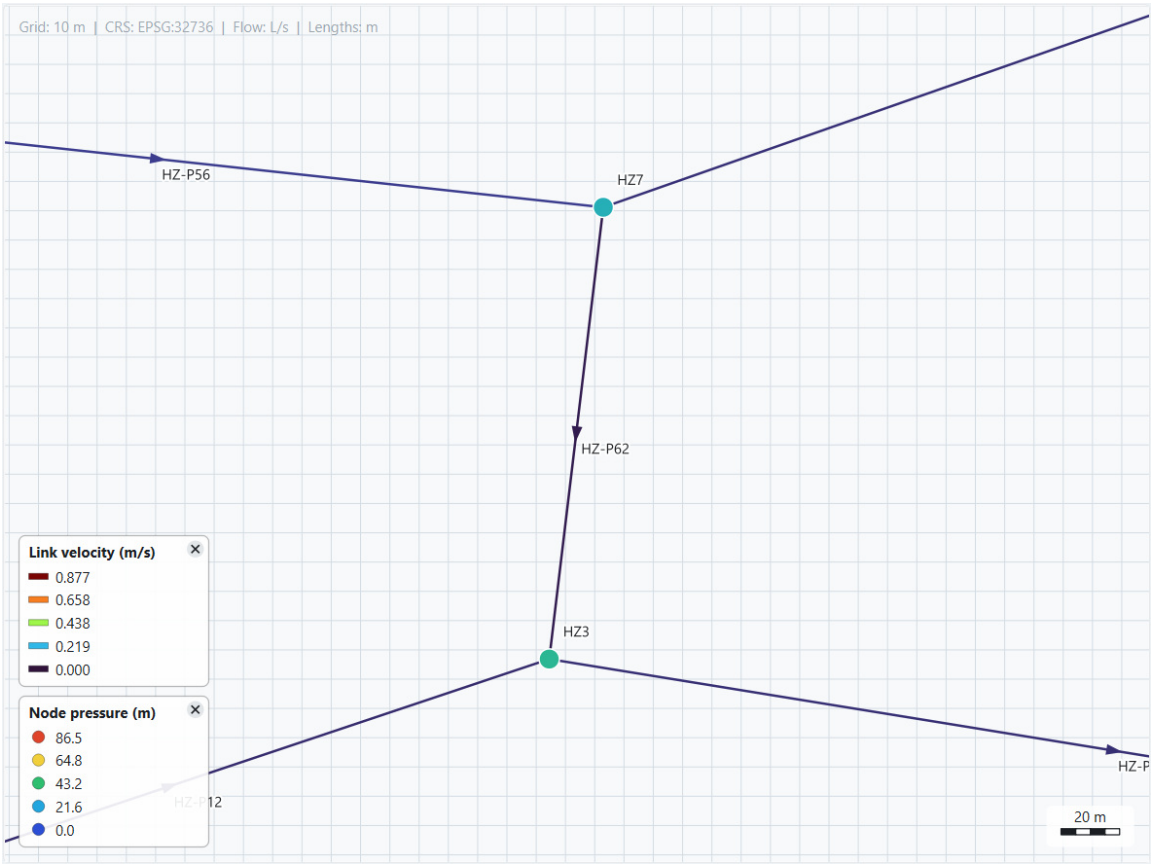
Attribute tables

Attribute Table opens the table of the layer selected in the Layers panel: each imported network layer has its own table, and an open table is brought forward rather than opened twice. Selection is shared with the map both ways: selecting rows selects those elements on the map, and selecting on the map highlights their rows. Edits in the table can be undone like any other edit.

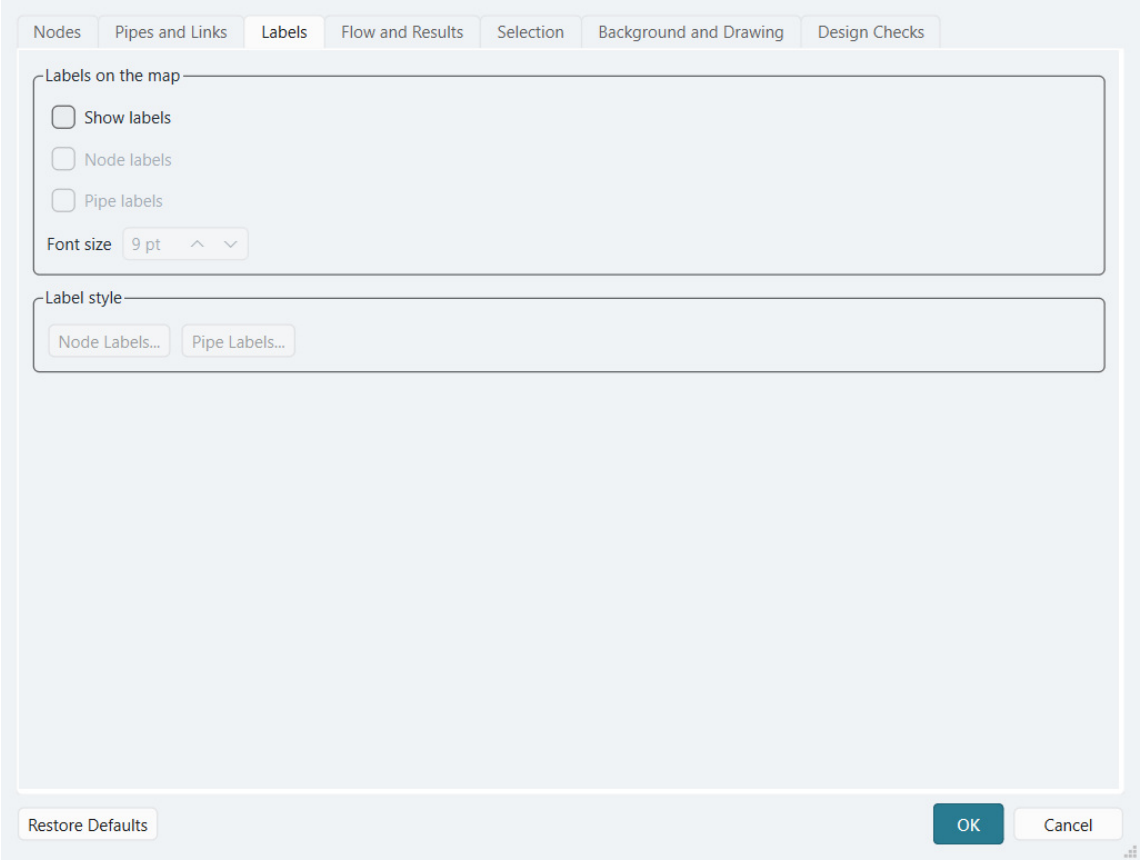
8

Labels

Label any layer by any field: one value, several values with units, or your own expression.



Labels of nodes and links on the map



Map Options: the labels of nodes and links

One place for labels

Every network layer is labelled in its **Layer Properties > Labels** tab, its nodes and its pipes apart. A layer without labels of its own follows **Map Options > Labels**, which sets the labels of the whole project; new projects show the IDs of nodes and pipes. The label switches and the right-click menu change the same settings, so they always agree.

What a label shows

Mode	Example
One field	`110` (the diameter)
Several fields with units	`P12  D=110 mm  v=1.20 m/s`
Expression	`{name} D={diameter_mm} mm`

Fields include everything an element has: its name and type, levels, withdrawal, diameter, length, roughness and material, every column of the file it came from, and its results after a run.

How it looks

Font, size, bold and italic, colour, a buffer (halo) and a background box, placement around points, offset, the scales between which labels show, priority and whether labels may overlap. A live example shows the first feature's label as you change the settings.

Rules

Only label where limits labels to the features that meet a condition, for example `diameter_mm > 150` or `name starts with PS`.

Quick changes from the map

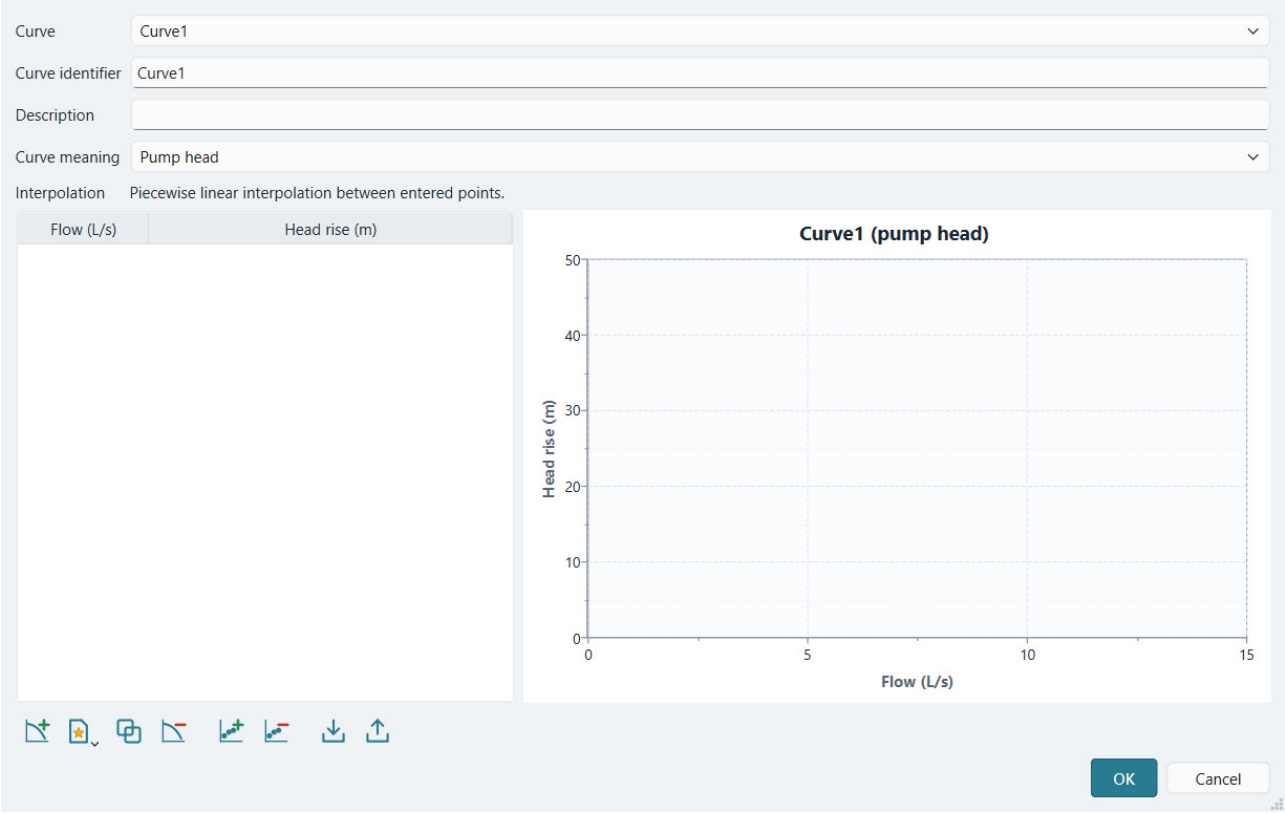
Right-click a feature:

- **Label Layer By** labels its whole layer by one field in one step;
- **Layer Labels...** opens the full settings;
- **Label** changes only this feature: hide or show its label, give it its own text, move it, or change its size and colour. **Reset to the Layer's Labels** undoes that.

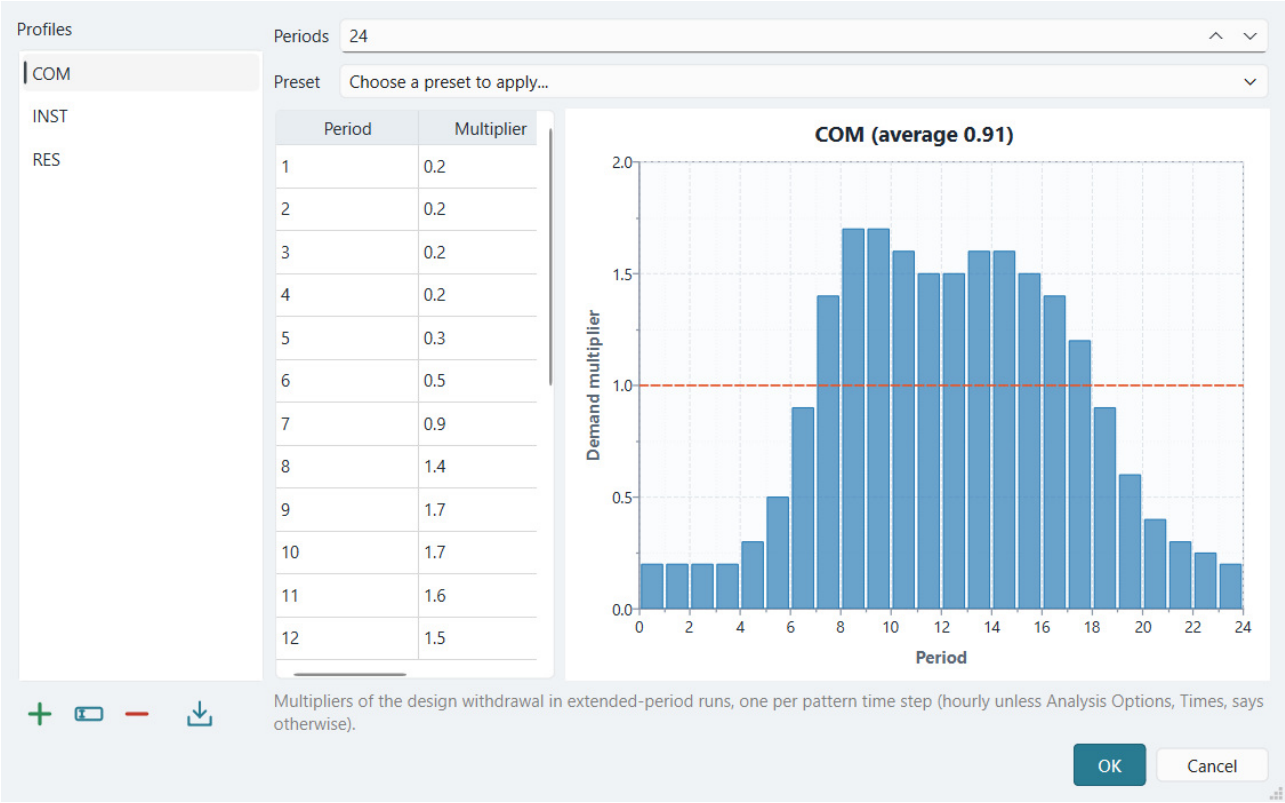
9

The network model

The elements of a network, the data each one carries, and the physics FlowCHAOS applies to them.



The curve library with a pump head-flow curve



Demand profiles: the factor of each hour of the day

Physical elements

Junctions

A junction is a point where pipes meet and where water can be taken out (or put in, as a negative withdrawal). Its data:

Property	Meaning
Ground elevation	the level of the ground at the junction, from a DEM, online terrain or typed in
Design offset	how far the pipe lies below (negative) or above the ground; the hydraulic elevation is ground plus offset
Design withdrawal	the average flow taken at the junction (L/s), the base demand
Demand profile	multipliers of the withdrawal through the day; none means the project's default profile
Demand categories	several withdrawals at one junction, each with its own profile (houses, a school, a leak), as imported models give them; FlowCHAOS keeps and analyses them
Outflow coefficient	an emitter: an outflow that grows with pressure, q = C p^n, for leaks, sprinklers and hydrants

Starting concentration	the water quality at the start of a run
Injection	a water quality source here (see Water quality)

The analysis gives each junction its head, its pressure (head minus hydraulic elevation), the withdrawal actually delivered, and its water quality.

Reservoirs

A reservoir is a source or sink of unlimited capacity whose water level does not change with the flow: a lake, a river, a groundwater aquifer at its pumping level, or a connection to a bulk supply.

Property	Meaning
Total head	the level of the water surface (m above datum)
Water level above ground	another way to set it: the head follows the ground level when it changes
Head profile	multipliers that make the level vary in time (a tide, a river that rises and falls)
Starting concentration, injection	the quality of the water it supplies

A borehole is modelled as a reservoir at its **pumping water level** (static level plus drawdown), with the submersible pump as a pump link from it.

Tanks

A tank stores water; its level rises when more flows in than out and falls otherwise.

Property	Meaning
Ground elevation and stand height	the bottom of the tank is the ground plus the stand (for an elevated tank)
Diameter	of a cylindrical tank
Minimum, maximum and starting levels	the water depth limits and the depth at the start of the run
Minimum volume	water below the minimum level that is never used
Shape curve	volume against depth for tanks that are not cylinders; the diameter is then ignored
Spills when full	extra inflow is lost over the top, rather than the inlet being closed

Mixing behaviour, inlet zone fraction, decay rate in tank	how water mixes and reacts inside it (see Water quality)

When a tank reaches its maximum level its inflow stops (unless it spills); when it reaches its minimum level its outflow stops.

Pipes

A pipe carries water from its start node to its end node; the flow is negative when water runs the other way.

Property	Meaning
Length	taken from the drawing, or a model length when the drawing is schematic
Diameter	internal diameter in mm
Roughness	the friction coefficient of the chosen head loss formula
Fittings loss factor K	the minor losses of bends, tees and valves, as a sum of their K values
Starting status	open, closed, or a check valve (water flows one way only)
Material	used by quantities, wall decay by material and pipe series
Bulk and wall decay rates	this pipe's own reaction rates; empty means the project's

Pumps

A pump adds head to the water flowing through it, from its start (suction) to its end (delivery).

Property	Meaning
Head-flow curve	the pump's curve from the Performance Curve Library
Power	a constant-power pump when no curve is given
Speed	relative speed (1.0 is the speed of the curve); a variable speed drive changes it
Speed profile	multipliers of the speed through the day
Efficiency curve, energy tariff, tariff profile	for the energy report
Starting status	open or closed at the start of the run

A pump that cannot deliver the head the system needs closes for that time step, and the solver warns about it. A pump

never lets water flow backwards.

Valves

A valve controls pressure or flow at a point. Each type has its own setting:

Type	Setting	What it does
Pressure reducing valve (PRV)	pressure (m)	keeps the pressure after it at the setting; closes if that would reverse the flow, opens fully if the pressure before it is lower
Pressure sustaining valve (PSV)	pressure (m)	keeps the pressure before it at the setting, for example to protect high ground upstream
Pressure breaker valve (PBV)	pressure drop (m)	takes away a fixed head, like a break-pressure tank
Flow control valve (FCV)	flow (L/s)	limits the flow to the setting
Throttle control valve (TCV)	loss coefficient K	a partly closed valve; 0 is fully open, as an isolation valve
General purpose valve (GPV)	a head loss curve	any device with its own head loss against flow: filters, meters, backflow preventers

A valve can also be forced open or closed. Valves should not be joined directly to a reservoir or tank, nor two pressure valves set against each other in series: give them a short pipe between.

Data that describe time and behaviour

- **Demand profiles** multiply the design withdrawals, period by period (usually hourly over a day). The profile library has typical shapes to start from.
- **Curves**: pump head and efficiency against flow, tank volume against depth, head loss against flow for general purpose valves. The library has samples for each type. In an element's properties each curve field lists only the curves of its kind (a tank's **Volume curve**, a pump's **Head-flow curve** and **Efficiency curve**); the button beside it opens the library at that curve, or starts a new one of the right kind, and assigns it when you close the library. A pump's **Speed profile** and **Tariff profile** are chosen from the demand profiles.
- **Controls** change one pump, pipe or valve at a time or when a level or pressure crosses a value; **rules** combine several conditions (Controls and rules).
- **Analysis options** set the units, the friction formula, the demand model, the solver's limits and the times of an extended period (Analysis options).

The hydraulic model

For each moment the analysis solves two sets of equations together:

- **conservation of mass** at every junction: the flows in equal the flows out plus the withdrawal;
- **conservation of energy** along every link: the difference of head between its ends equals the head lost in a pipe or valve, or minus the head added by a pump.

Friction

Head lost to friction along a pipe of length L (m), diameter D (m) and flow Q (m3/s):

Formula	Head loss	Roughness	Use it for
Hazen-Williams	h = 10.67 L Q^1.852 / (C^1.852 D^4.87)	C, larger is smoother	water supply in the usual range of velocities
Darcy-Weisbach	h = f L V^2 / (2 g D)	roughness height e (mm); f from the Colebrook-White law or the laminar law	any fluid and flow, the most exact
Chezy-Manning	h = 10.29 n^2 L Q^2 / D^5.33	Manning n, smaller is smoother	larger mains and open-channel habits

Typical roughness of new pipe:

Material	Hazen-Williams C	Darcy-Weisbach e (mm)	Manning n
PVC, HDPE, PE, PPR	145 to 150	0.0015 to 0.007	0.009
GRP	150	0.01	0.009
Copper	140	0.0015	0.010
Ductile iron, cement lined	130 to 140	0.03 to 0.1	0.011 to 0.012
Steel, cement lined	130 to 140	0.05	0.012
Galvanised iron	120	0.15	0.014
Concrete	120 to 140	0.3 to 3	0.012 to 0.015
Old cast iron	60 to 100	1 to 5	0.015 to 0.017

Roughness grows as pipes age and scale; calibrate it against field pressures and flows.

Minor losses

Fittings add a loss h = K V^2 / (2 g), with K the sum of the loss factors of the bends, tees, valves and meters along the pipe (a gate valve fully open about 0.2, a 90 degree bend about 0.3 to 0.9, a globe valve about 10).

Pumps

A pump adds the head of its curve at the flow through it. A curve given by one point (the design flow and head) is extended to a full curve with shut-off at 133 % of the design head and the largest flow at twice the design flow. A three-point curve (shut-off, design, largest flow) is fitted by a smooth law; a curve with more points is followed point by point. At a relative speed s the curve scales as h(sQ) = s^2 h(Q).

Demand-driven and pressure-driven withdrawal

In a **demand-driven** analysis every junction receives its full withdrawal whatever the pressure; negative pressures then show where the network cannot deliver it. In a **pressure-driven** analysis the delivered withdrawal follows the pressure

p:

- nothing below the minimum pressure Pmin;
- the full withdrawal Q at or above the required pressure Preq;
- in between, q = Q ((p - Pmin) / (Preq - Pmin))^n, usually with n = 0.5.

Pressure-driven analysis is the one to use for intermittent supply, for networks that cannot meet the peak, and for any study of what customers actually receive.

Emitters

An outflow coefficient C at a junction adds an outflow q = C p^n (n is the emitter exponent of the analysis options, 0.5 for an orifice). Emitters model leaks, sprinklers, fire hydrants and open taps.

The water quality model

The quality analysis follows the substance with the water. Its steps, at each quality time step:

1. **Transport**: water moves along each pipe at its velocity, carrying its concentration in segments.
2. **Mixing at junctions**: the water leaving a junction has the flow-weighted average of the water arriving, plus any source there.
3. **Mixing in tanks**: by the tank's mixing behaviour:

Behaviour	How the tank mixes	Typical tank
Fully mixed	all water in the tank mixes at once	a tank with one pipe for inlet and outlet
Two compartments	an inlet zone mixes with the inflow, the rest only with the inlet zone	a large tank with separate inlet and outlet
Plug flow, first in first out	water leaves in the order it came	a tall tank with inlet and outlet at opposite ends
Stacked, last in first out	new water floats on old and leaves first	a tank filled and emptied from the bottom

4. **Reactions**:
   - in the bulk water, at a rate kb C^n (first order, n = 1: the concentration falls exponentially, halving every ln 2 / kb days);
   - at the pipe wall, at a rate that depends on the wall coefficient kw, the pipe size and how fast the substance can reach the wall (molecular diffusivity);
   - a **limiting potential** makes a reaction stop at a level (a substance that grows to a ceiling or decays to a floor).

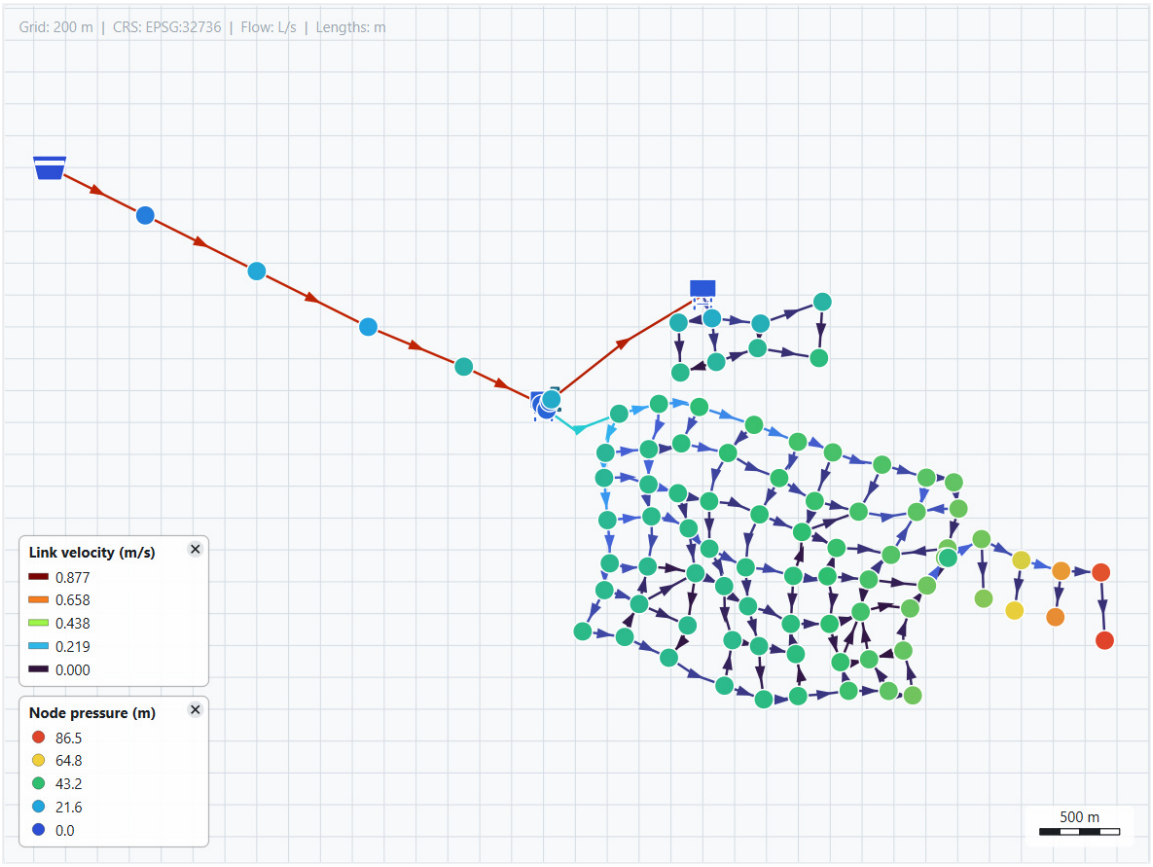
Water age is computed as a substance that grows by one hour every hour; **source tracing** as the share of water that has passed through the traced node.

Rates by material, bottle tests and the study tools are in Water quality.

10

Running an analysis

Steady-state and extended-period analysis, demand models, controls and water quality.



After a run: junctions coloured by pressure, pipes by velocity

Steady state

Run Steady State (F5) solves the network for one moment with the advanced solver. Nodes without pipes, such as survey points, are left out automatically and counted in the status bar.

Several networks in one project

A project can hold its own network and networks imported as layers. They are separate systems, so an analysis works on one of them: the one whose layer is selected in the Layers panel, or the one chosen with **Analysis > Network to Analyse**. Steady-state and extended-period runs, pipe sizing, fire flow, criticality and the solver comparison all use that network, and the status bar names it. The run commands are available while a network layer is selected; if that network is out of view, the map zooms to it when the run starts.

In the Layers panel, the third column marks the network that has results (a ticked sheet) and the one being analysed (a turning sign); the Simulation Results title names it too.

From the command line

FlowCHAOS can solve a model without opening its window, for batch studies and scripts:

```
FlowCHAOS --run model.inp report.txt --hours 24 --step-minutes 60
```

The model may be any network file or project FlowCHAOS opens. Without a report name the full report is printed in the console; without --hours the run is a steady state.

Extended period

Run Extended Period follows the network through time with demand profiles, tank levels, pump schedules and controls. Enter the duration and the time step. The run happens in the background; the window stays usable and the results arrive when it finishes. While it runs, the status bar shows the elapsed time, the simulated time reached and a progress bar that fills step by step; the run time is given when it ends. Use the time bar to step through the results or play them; the map, tables, legends and animations follow the time shown. The Simulation Results panel opens by itself the first time results arrive.

Stopping and clearing

Stop / Clear Results stops a running analysis (its results are ignored when they arrive) or, when nothing is running, clears the results so the map shows the network as drawn. Undo brings the results back.

Solver warnings

After every run the solver's warnings are kept with the results: a system that did not balance or may be unstable, negative pressures, disconnected nodes, pumps running outside their curve and valves that cannot hold their setting. The completion message says how many there are; **Analysis > Solver Warnings** lists them, and a double-click selects the pump, valve or node named.

Solution algorithms

Settings > Hydraulic Solver chooses the algorithm for steady-state runs:

Algorithm	Unknowns	Character
Advanced engine (recommended)	flows and heads	Global gradient with every valve, pump, control, rule and water-quality option
Built-in sparse Newton	flows and heads	Newton with least-squares safeguards; pressure-driven withdrawal and control valves
Global gradient (Todini and Pilati, 1988)	flows and heads	Newton reduced to a symmetric system in the heads; continuity holds after each iteration
Newton nodal (Martin and Peters, 1963)	heads	Newton on the junction heads alone, damped by a line search
Linear theory (Wood and Charles, 1972)	flows and heads	Linearised elements solved exactly, flows averaged between iterations
Hardy Cross (1936)	loop flows	One loop corrected at a time, with pseudo-loops between sources

The four comparison algorithms solve the same equations with the same head loss formula, minor losses, pump curves (one-point and three-point curves are fitted as the standard method does) and emitters. Pressure and flow control

valves, controls and rules are modelled by the advanced engine and the built-in solver; extended-period runs always use the advanced engine.

Demand-driven and pressure-driven analysis

Analysis Options > Hydraulics > Demand chooses the demand model. **Demand-driven (DDA)** delivers every withdrawal whatever the pressure. **Pressure-driven (PDA)** delivers what the pressure allows: nothing below the minimum pressure, the full withdrawal at the required pressure, and in between q = Q ((p - Pmin) / (Preq - Pmin))^n, with n = 0.5 for orifices such as hydrants and sprinklers.

PDA works in every analysis: steady and extended-period runs, pipe sizing, fire flow, criticality, water quality and the solver comparison. The global gradient, Newton nodal and linear theory algorithms solve it as well as the engine; Hardy Cross balances loops around fixed withdrawals and says so. Set the required pressure from what the outlet serves: an open canal needs little, a hose or a sprinkler much more.

Compare Solvers

Analysis > Compare Solvers runs the network with the algorithms you tick and lists, for each: whether it converged, iterations, time, the largest continuity and energy errors, and the largest and RMS differences in head and flow from a reference algorithm, with the node and link where they occur. A chart shows the convergence history on a logarithmic scale, and the window recommends an algorithm for the network. **Use for Runs** makes the selected algorithm the one steady-state runs use; **Export CSV** saves the table and every iteration for a report or a paper.

Options

Analysis > Analysis Options has a tab for each subject: Hydraulics (units, head loss formula, demand model, pressure-driven settings, convergence), Water Quality, Reactions, Times, Energy, and the **ID Labels** and **Element Defaults** of new elements (see Drawing).

Controls

- **Scheduled Pipe Controls**: a pump, pipe or valve opened, closed, slowed or reset at a time, at a time of day, or when a tank level or a junction pressure crosses a value.
- **Rule-Based Controls**: IF / AND / OR / THEN / ELSE rules on pressures, levels, flows, demand and time, with priorities.

Both start from samples written with the network's own elements; see **Controls and rules**.

- **Demand Profiles** and the **Performance Curve Library** for profiles, pump and valve curves. The curve library starts a curve of any kind from a sample (borehole, solar, surface, booster and transfer pumps; efficiency curves; horizontal, spherical and conical tanks; strainers, meters, filters and backflow preventers), and **Delete Curves** lists every curve with what uses it. The profile editor has presets from the literature to start from: urban, suburban, weekend, peri-urban and rural water points, intermittent supply, offices, schools, hospitals, hotels, industry in one or two shifts or continuous, irrigation by day, by night, in two sets or in drip pulses, livestock, night-time leakage, and monthly and weekly profiles. Each is scaled to an average of 1.0 and shows where its shape comes from; edit the values to measured use.

Water quality

Analysis > Water Quality Analysis follows chlorine, a chemical, the age of the water or the share of water from one source, with ready studies for chlorine compliance, boosters, water age, source tracing and contamination events, compliance by delivered water and a PDF report. See **Water quality**.

Result tables

The node table lists head, pressure, delivered withdrawal and quality; the link table lists start and end nodes, diameter, length, flow, velocity, velocity head, heads, total energy, head loss and residual head. Choose the columns with **Columns**, filter by name, and sort by any column.

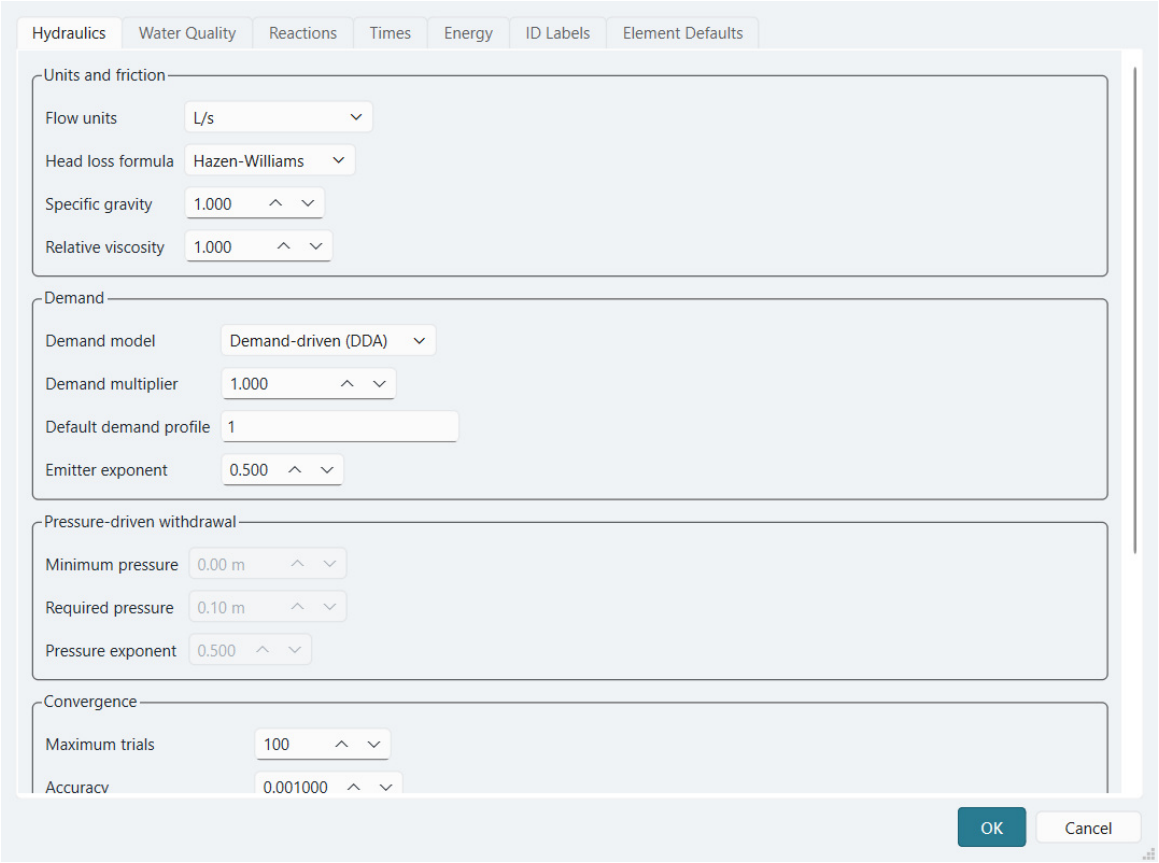
Select cells (or whole rows and columns) and right-click to **Copy** or **Copy with Headers**: the cells paste as a table into spreadsheets, documents and text editors. Ctrl+C works too, in every table of the program.

Analysis > Export Results writes a spreadsheet with a sheet for nodes and one for links.

11

Analysis options

Every option of Analysis > Analysis Options: what it does, its usual value and when to change it.



Analysis Options: hydraulics, demand model, times and more

Analysis > Analysis Options has a tab for each subject. The options are kept with the project and written into the hydraulic model file when the network is exported. Point at any field for a short explanation.

Hydraulics

Units and friction

Option	Choices or range	Default	Meaning
Flow units	L/s, L/min, ML/day, m3/hour, m3/day, US gal/min, ft3/s, US Mgal/day, Imperial Mgal/day,	L/s	units of flow in the project and its exchange files; metric flow units keep the other quantities metric

	acre-ft/day		
Head loss formula	Hazen-Williams, Darcy-Weisbach, Chezy-Manning	Hazen-Williams	the friction law; each pipe keeps a roughness for each formula, so switching back and forth loses nothing
Specific gravity	0.01 to 100	1.0	density of the fluid relative to water at 4 degrees C
Relative viscosity	0.001 to 1000	1.0	kinematic viscosity relative to water at 20 degrees C; used by Darcy-Weisbach

Demand

Option	Default	Meaning
Demand model	Demand-driven	**Demand-driven** delivers every withdrawal whatever the pressure; **pressure-driven** delivers what the pressure allows
Demand multiplier	1.0	scales every withdrawal of the run: 1.8 for a peak hour, 0.4 for night flow
Default demand profile	1	the profile of withdrawals that name none
Emitter exponent	0.5	the exponent n of emitter outflow q = C p^n (0.5 for orifices, sprinklers and most leaks; 0.5 to 1.5 for leaks in plastic pipe)

Pressure-driven withdrawal

Used when the demand model is pressure-driven.

Option	Default	Meaning
Minimum pressure	0 m	at or below this pressure nothing is delivered
Required pressure	0.1 m	at or above this pressure the full withdrawal is delivered; set it from what the outlet serves (a yard tap 5 to 10 m, a house connection 15 to 20 m, a sprinkler 25 to 35 m)
Pressure exponent	0.5	the shape of the delivery curve between the two pressures

Convergence

Option	Default	Meaning

Maximum trials	40	iterations allowed for one hydraulic state
Accuracy	0.001	the state has converged when the sum of the flow changes, divided by the total flow, is below this
Maximum head error	0 (not checked)	largest error allowed in the head loss of any link, in metres
Maximum flow change	0 (not checked)	largest flow change allowed in any link at convergence
Status check frequency	2	iterations between checks of pump, valve and check valve status
Status checks up to trial	10	after this trial, status is checked only at convergence
Damping limit	0 (none)	below this accuracy, flow corrections are halved to settle hard networks
If a state does not converge	Stop the run	**stop** the extended period, or **continue** with a warning
Status report	Changes of status	what the solver reports: status changes, nothing, or every trial (for diagnosing a network that will not balance)

Raise **Maximum trials** to 200 for large or difficult networks. A network that does not converge is usually a modelling problem (a closed loop of valves, a pump that cannot lift, disconnected nodes): **Analysis > Solver Warnings** names the elements involved.

Water Quality

Option	Default	Meaning
Parameter	None	what extended-period runs follow: chlorine, a chemical, water age or source tracing
Traced source		the node whose water is traced
Quality tolerance	0.01	the smallest difference of quality that keeps two parcels of water apart
Relative diffusivity	1	molecular diffusivity relative to chlorine in water; 0 means wall reactions are not limited by mass transfer

For a full study use **Analysis > Water Quality Analysis** (Water quality).

Reactions

Option	Default	Meaning
Bulk reaction order	1	order of the reaction in the water

Wall reaction order	1	0 or 1
Tank reaction order	1	order of the reaction in tanks
Global bulk coefficient	0	bulk rate of every pipe without its own (1/day, negative for decay)
Global wall coefficient	0	wall rate of every pipe without its own (m/day for first order)
Limiting potential	0 (none)	a concentration a reaction tends to and stops at
Roughness correlation	0 (none)	makes wall rates follow pipe roughness, so older pipes react more

Times

Durations and steps are written as H:MM or H:MM:SS.

Option	Default	Meaning
Simulation duration	0:00	the length of an extended period (0 is a steady state)
Hydraulic timestep	1:00	the step between hydraulic solutions; shorter steps follow tanks and controls more closely
Quality timestep	0:05	the step of the quality transport; shorter is more accurate
Profile timestep	1:00	the length of one period of the demand profiles
Profile start	0:00	the period of the profiles at the start of the run
Report timestep	1:00	how often results are kept
Report start	0:00	the time from which results are kept
Rule timestep	0:00:10	how often rules are checked between hydraulic steps
Start clock time	12 am	the time of day when the run starts; clock-time controls and rules count from it

Energy

Option	Default	Meaning
Global pump efficiency	75 %	the efficiency of pumps without an efficiency curve

Global energy price	0	price of a kWh for pumps without their own tariff
Energy price profile		multipliers of the price through the day (a night tariff); a pump can have its own in its properties
Demand charge	0	a charge per kW of the highest pumping power

Analysis > Energy Report gives, for each pump, the share of the time it runs, its mean flow, head and efficiency, the energy per cubic metre pumped, its mean and peak power, and its energy and cost per day, with the totals, the peak power of all pumps together and the demand charge. In an extended period the energy is summed over every time step, with the efficiency at each step's flow (from the pump's efficiency curve) and the price at that time of day.

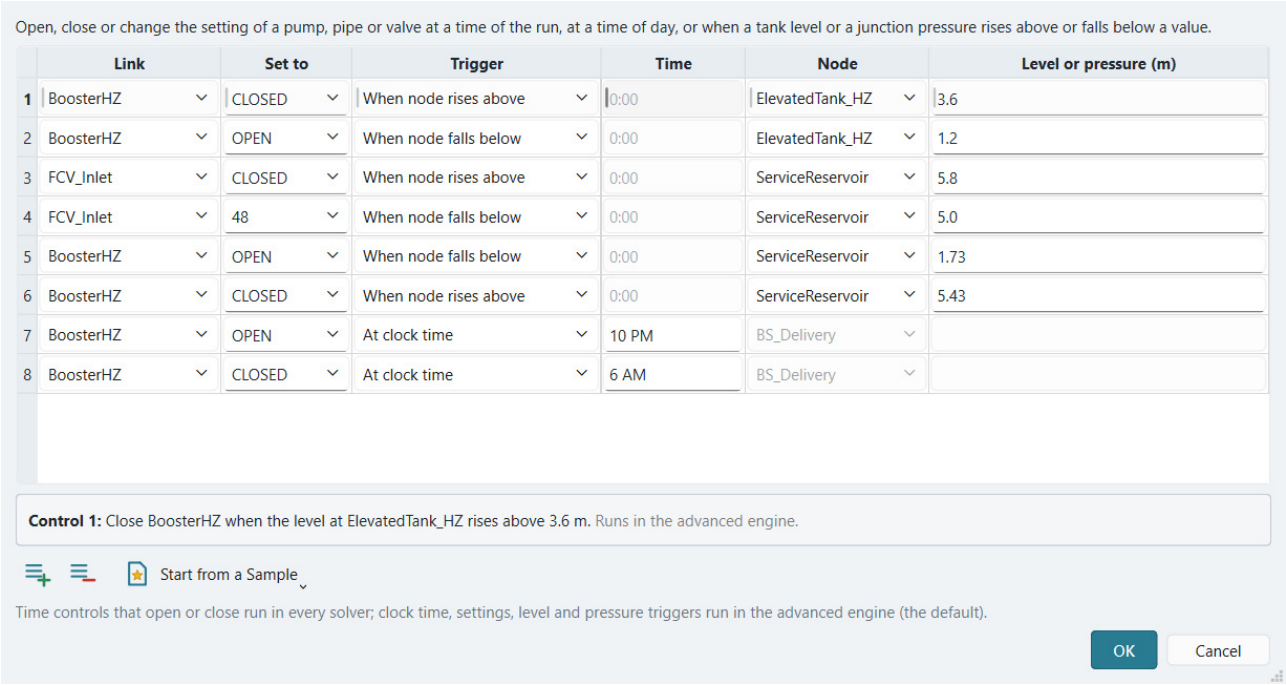
ID Labels and Element Defaults

ID Labels sets the prefix, the first number and the step of the names of new elements (J1, J2; P1, P2 and so on). **Element Defaults** gives new junctions their design offset and withdrawal, new tanks their size and levels, and new pipes and valves their diameter, roughness and material, so a network can be drawn quickly and corrected afterwards.

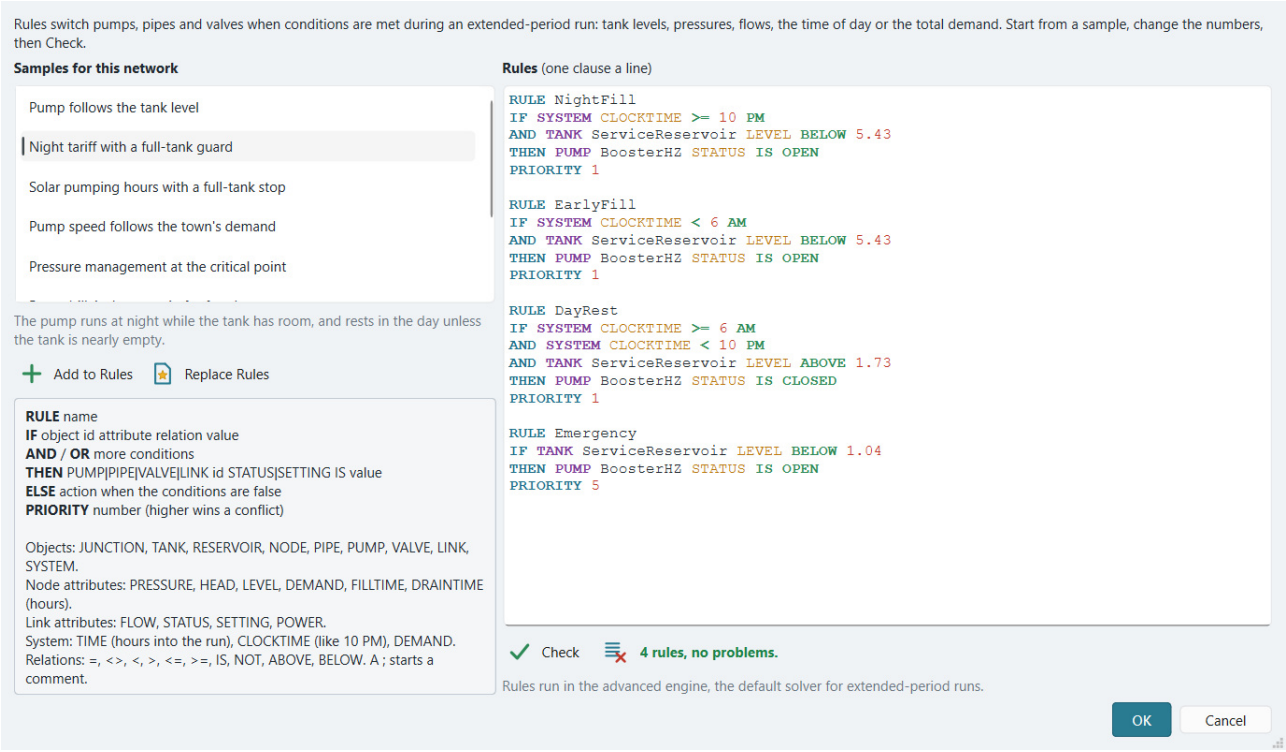
12

Controls and rules

Switch pumps, pipes and valves during a run: simple controls, rules, and ready-made samples.



Simple controls with the samples for this network



The rules editor: samples, highlighted rules and the check

Controls change a pump, a pipe or a valve while an extended-period run goes on: a pump that fills a tank and stops when it is full, a main closed for a repair, a valve that lowers night pressure. FlowCHAOS has two kinds. **Simple controls** have one trigger and one action each. **Rules** combine several conditions with AND and OR, have an ELSE, and a priority that settles conflicts.

Both editors offer samples written with your network's own pumps, tanks, valves and pipes, so a sample runs as soon as it is added and you only change the numbers.

Simple controls

Analysis > Scheduled Pipe Controls lists one control per row:

Column	Meaning
Link	The pump, pipe or valve the control changes
Set to	OPEN, CLOSED, or a number: the relative speed of a pump (1.0 is full speed) or the setting of a valve (pressure in m for a pressure valve, flow for a flow valve)
Trigger	At elapsed time, At clock time, When node rises above, When node falls below
Time	For a time trigger: hours into the run (6 or 6:30); for a clock time: 6 AM, 10:30 PM or 22:00
Node and Level or pressure	For a level trigger: a tank (its water level above its bottom) or a junction (its pressure), and

	the value in metres

The box under the table reads the selected control as a sentence, for example "Open PU1 every day at 10 PM. Runs in the advanced engine.", and turns red with the reason when a control cannot run (a pipe given a speed, a node that is not in the network, a time with 75 minutes). The row number shows the same warning. OK refuses a control that cannot run and says why.

Start from a sample offers, depending on what the network has:

- **Fill the tank: pump on when low, off when full**: the pump starts below a low level and stops above a high level, taken from the tank's own minimum and maximum levels.
- **Night-time pumping on the cheap tariff**: the pump runs from 10 PM to 6 AM.
- **Slower pump in the day, full speed at night**: 80 % speed from 6 AM, full speed from 10 PM.
- **Pipe shut for a repair from hour 8 to hour 12**: who loses water during a repair.
- **Open a bypass when pressure drops**: a pipe opens below 15 m at the highest junction and closes again above 25 m.
- **Lower the valve setting at night**: less pressure, and less leakage, from 11 PM to 5 AM.
- **Close the tank inlet before it overflows**.

Imported controls in other forms, or naming an element that is no longer in the network, are kept unchanged and listed under the table.

Elapsed-time controls that open or close run in every solver; clock times, speeds, settings and level or pressure triggers run in the advanced engine, which is the default for extended-period runs.

Rules

Analysis > Rule-Based Controls edits the rules as text, one clause a line:

RULE PumpOn

IF TANK T1 LEVEL BELOW 1.5

THEN PUMP PU1 STATUS IS OPEN

PRIORITY 2

Clause	**Form**
RULE	RULE and a name, one word
IF	object id attribute relation value
AND, OR	more conditions (AND also adds actions after THEN)
THEN	PUMP, PIPE, VALVE or LINK id STATUS or SETTING IS value
ELSE	the action when the conditions are false
PRIORITY	a number; when two rules disagree, the higher priority wins

- **Objects**: JUNCTION, TANK, RESERVOIR, NODE, PIPE, PUMP, VALVE, LINK and SYSTEM.
- **Node attributes**: PRESSURE, HEAD, LEVEL, DEMAND, FILLTIME and DRAINTIME (hours to fill or drain a tank).
- **Link attributes**: FLOW, STATUS, SETTING and POWER.
- **System**: TIME (hours into the run), CLOCKTIME (10 PM, 06:00) and DEMAND (total demand).
- **Relations**: =, <>, <, >, <=, >=, IS, NOT, ABOVE and BELOW.
- A semicolon starts a comment.

The editor colours keywords, objects, attributes, relations and numbers. The samples on the left, for this network:

- **Pump follows the tank level**: two rules keep the tank between a low and a high level.
- **Night tariff with a full-tank guard**: the pump fills the tank at night while it has room, and rests in the day unless the tank is nearly empty; an emergency rule with a higher priority starts it whenever the level is very low.
- **Pump speed follows the town's demand**: full speed above 120 % of the average demand, 75 % otherwise, as a variable speed drive does.
- **Duty and standby pumps**: the second pump starts only when the first cannot keep up.
- **Pressure management at the critical point**: a pressure valve raises its setting when the highest junction falls below 15 m.
- **Burst drill: isolate a main for four hours**.
- **Tank overflow protection**: the inlet shuts when the tank is nearly full, with the highest priority.
- **Fire reserve**: the pump starts when the level is low or the tank would drain within two hours.

Add to Rules appends the selected sample (a double-click does the same; rules that would share a name are renamed), **Replace Rules** starts again from the sample alone.

Check reads every clause against the network: unknown keywords, objects and attributes, names that are not in the network, a pump called a pipe, a pipe given a setting, a rule without a THEN, two rules with the same name. Each problem is listed with its line; click it to go to the line. OK checks too and asks before keeping rules with problems.

Rules run in the advanced engine.

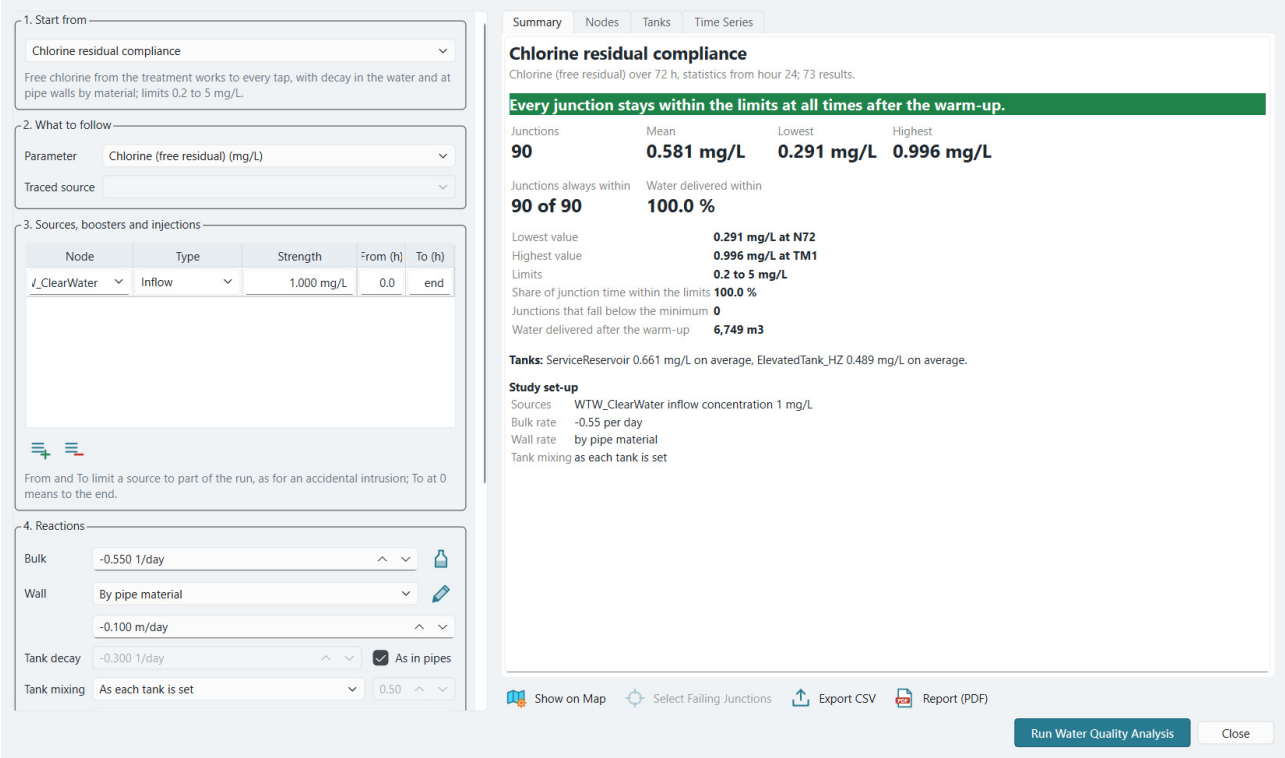
Tips

- Give a pump that a rule controls a starting status in its properties; rules change it from the first time step on.
- Use a gap between the start and stop levels of a tank (at least half a metre) so the pump does not start and stop at every step.
- Watch the result with **Show on Map** and the time bar, or plot the tank level and the pump flow under **Analysis > Advanced Graphs**.

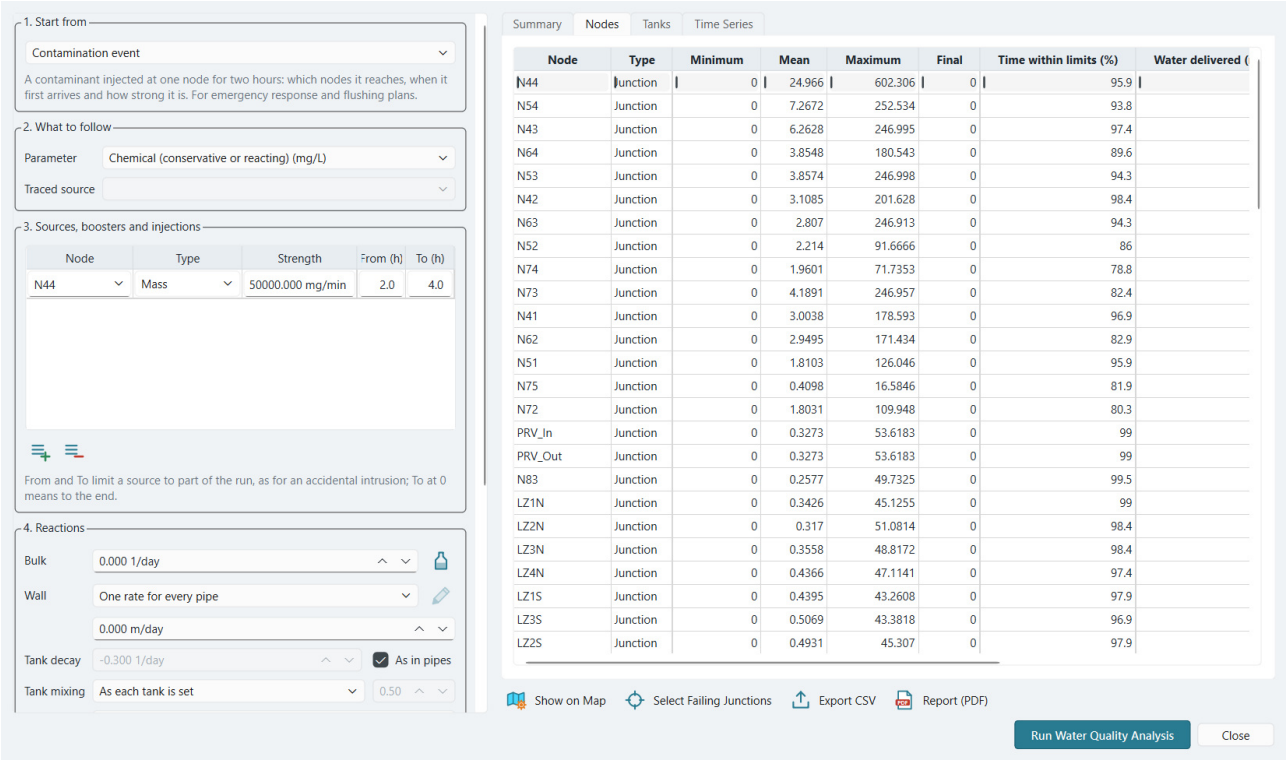
13

Water quality

Chlorine residuals, water age, source tracing and contamination events, from set-up to a report.



A chlorine compliance study: the set-up and the summary



A contamination event: junctions reached and arrival times

Analysis > Water Quality Analysis follows a substance or the age of the water through an extended-period run of the network. It is built for the questions a water utility answers every day: does every customer receive enough free chlorine, where does the water stay too long, which works feeds which zone, and what happens if something enters the network.

The window has the study set-up on the left, in the order you fill it, and the results on the right. Everything you set is kept with the project, so the next time the window opens the same study is ready to run again.

1. Start from a ready study

The first box offers ready studies. Each one fills every part of the set-up with this network's own reservoirs, tanks and nodes, so it runs at once and you only change the numbers.

Study	What it answers	Main settings
Chlorine residual compliance	Does every junction keep enough free chlorine?	Chlorine 1 mg/L leaving each reservoir, bulk decay -0.55 per day, wall decay by pipe material, limits 0.2 to 5 mg/L, 72 h with 24 h of warm-up
Chlorine with a booster at the service tanks	Does re-dosing at the tanks keep the far end above 0.2 mg/L?	As above, plus a 0.6 mg/L set-point at every tank
Water age and tank turnover	Where is the water older than 72 hours?	Water age over 168 h, statistics from hour 96

Where each source's water goes	Which zone does each works serve, and where do sources blend?	Source tracing from the first reservoir
Contamination event	Which nodes does a contaminant reach, when, and how strongly?	A mass injection at a central junction from hour 2 to hour 4, 48 h at 15 minute steps, reached above 0.01 mg/L
Fluoride or another conservative chemical	Does any junction exceed a maximum?	A chemical that does not decay, limit 1.5 mg/L

Custom study keeps whatever you set below. Changing the parameter by hand switches back to a custom study and fills in the usual limits of the new parameter.

2. What to follow

Parameter	Unit	Meaning
Chlorine (free residual)	mg/L	A reacting substance: it decays in the water and at the pipe walls
Water age	h	The time since the water left a source; tanks and dead ends age it
Source tracing	%	The share of the water at each node that came from the traced source
Chemical	mg/L	Any substance, conservative (no decay) or reacting

For source tracing, choose the **traced source**: a reservoir, a tank or any node. The window lists reservoirs and tanks first.

3. Sources, boosters and injections

Each row is one source at one node:

Type	Strength means	Use it for
Inflow	the concentration of water entering the network at a reservoir or tank (mg/L)	treatment works, a borehole with its own dosing
Set-point	the concentration the node keeps in the water leaving it (mg/L)	a booster chlorinator at a tank or a pumping station
Flow-paced	the concentration the node adds to the water passing through (mg/L)	a dosing pump paced by a flow meter
Mass	the mass added each minute (mg/min)	a spill, an intrusion, a tracer test

From (h) and **To (h)** limit a source to part of the run: an accidental intrusion from hour 2 to hour 4, a booster switched

on at hour 12, a dosing failure from hour 30. To at 0 means to the end of the run. The source is switched on and off on the time step of the demand profiles (one hour unless you changed it under Analysis Options, Times).

Add adds an empty row, or one booster for every node selected on the map; **Remove** deletes the selected row. Sources only matter for chlorine and chemicals: water age and tracing need none.

4. Reactions

- **Bulk**: the first-order decay rate of the water itself, per day (negative for decay). Free chlorine usually lies between -0.1 and -1.5 per day; warm water and water rich in organic matter decay faster.
- **From a bottle test**: fill a dark bottle with water leaving the works, keep it at a steady temperature and read the chlorine twice. Enter the two readings, the hours between them, the bottle temperature and the usual temperature of the water in the network. FlowCHAOS works out the rate, corrects it to the network temperature (the rate changes by a factor of about 1.07 for every degree; you can change the factor) and shows how long chlorine takes to halve.
- **Wall**: the rate at which pipe walls consume chlorine, in m/day. **One rate for every pipe** uses the value below. **By pipe material** gives each pipe the rate of its material; **Rates** shows every material class, how many pipes of the network belong to it, and its rate:

Material	Typical wall rate (m/day)
HDPE / PE, uPVC / PVC	-0.01
GRP	-0.02
Copper	-0.05
Ductile iron (lined), asbestos cement, concrete	-0.1
Steel	-0.3
Galvanised iron	-0.4
Cast iron (unlined)	-0.5

  These are starting values to calibrate with field readings. A pipe whose material is not recognised uses the single wall rate; a pipe with its own wall rate in its properties keeps it.
- **Tank decay**: the bulk rate inside tanks, the same as in pipes unless you untick **As in pipes**.
- **Tank mixing**: how the water mixes inside the tanks. **Complete mixing** suits a tank with one pipe for inlet and outlet; **Two compartments** splits the volume into a zone that mixes with the inflow (its share is the number beside it) and a zone that does not; **Plug flow** (first in, first out) suits a tall tank with separate inlet and outlet; **Stacked** (last in, first out) suits a tank where new water floats on old. **As each tank is set** keeps the model in each tank's properties.
- **Initial value**: the value everywhere at the start; tanks start full of it.

5. Limits at junctions

- **Minimum** and **Maximum**: the range every junction must keep, for example free chlorine from 0.2 to 5 mg/L (WHO guidelines), water age at most 72 hours, fluoride at most 1.5 mg/L.
- **Reached above**: a junction counts as reached the first time its value rises above this level. The results then give the hour each junction is first reached: the arrival time of a contaminant, or of water from a new source.

6. Time

- **Duration**: long enough for tanks and remote pipes to settle; three days for chlorine, a week for water age.
- **Warm-up**: the first hours are left out of the statistics while values settle from their starting state.
- **Results every** and **Quality step**: how often results are kept, and the transport step (shorter is more accurate and slower).

Run Water Quality Analysis solves the hydraulics and the transport together with the advanced engine. The window stays usable while it runs.

The results

Summary opens with a coloured verdict: green when every junction stays within the limits, amber when a few leave them, red when many do or when a contaminant reaches junctions. Below it:

- the number of junctions, the mean, the lowest and the highest value;
- how many junctions stay within the limits at all times;
- **water delivered within the limits**: the share of the water actually delivered to customers (each junction's delivered demand over the time) that met the limits. A dead end that fails but serves few people weighs little; a busy zone that fails weighs a lot. This is the figure a board or a regulator asks for;
- the junctions that leave the limits, with the share of time they stay within them;
- for a contamination event, the junctions reached, the first to be reached and the arrival hour at each, and the water delivered at reached junctions;
- the tanks, and the study set-up, so a printed summary says how it was obtained.

Nodes lists every node with its minimum, mean, maximum and final value, the share of time within the limits, the water delivered after the warm-up and the hour it was first reached. Sort by any column; double-click a row to find the node on the map.

Tanks lists the tanks with their minimum, mean, maximum and final value.

Time Series plots the selected nodes (up to six) over the run, with the limits as dashed lines and the end of the warm-up as a dotted line.

Under the tabs:

- **Show on Map** puts the run on the map: Map Styles colours nodes and pipes by concentration, and the time bar and Animate Map play it.
- **Select Failing Junctions** selects on the map every junction that leaves the limits.
- **Export CSV** writes the node table and the full time series.
- **Report (PDF)** writes a printable report: the title, the network, the date, the verdict, the figures, the set-up and the junction table.

A worked example: chlorine in a town network

1. **File > Open Sample Projects > Chlorine residual in a town network**.
2. **Analysis > Water Quality Analysis**. Choose **Chlorine residual compliance**.
3. Click **From a bottle test**: first reading 1.0 mg/L, second reading 0.6 mg/L after 48 hours, bottle at 25 degrees, network at 22 degrees. The bulk rate becomes about -0.21 per day. Click OK.
4. Run. The summary shows the lowest residual and where it occurs.
5. Choose **Chlorine with a booster at the service tanks** and run again to see what re-dosing at the tanks changes.
6. **Report (PDF)** for the file.

A worked example: a contamination event

1. Select the junction where the contaminant could enter, then choose **Contamination event**.
2. In the sources table, set the node, the mass (mg/min) and the hours it lasts.

3. Run. The verdict gives the number of junctions reached and the first arrival; the Nodes tab, sorted by **First reached**, gives the order in which to warn customers or close valves.
4. **Show on Map** and play the time bar to watch the plume travel.

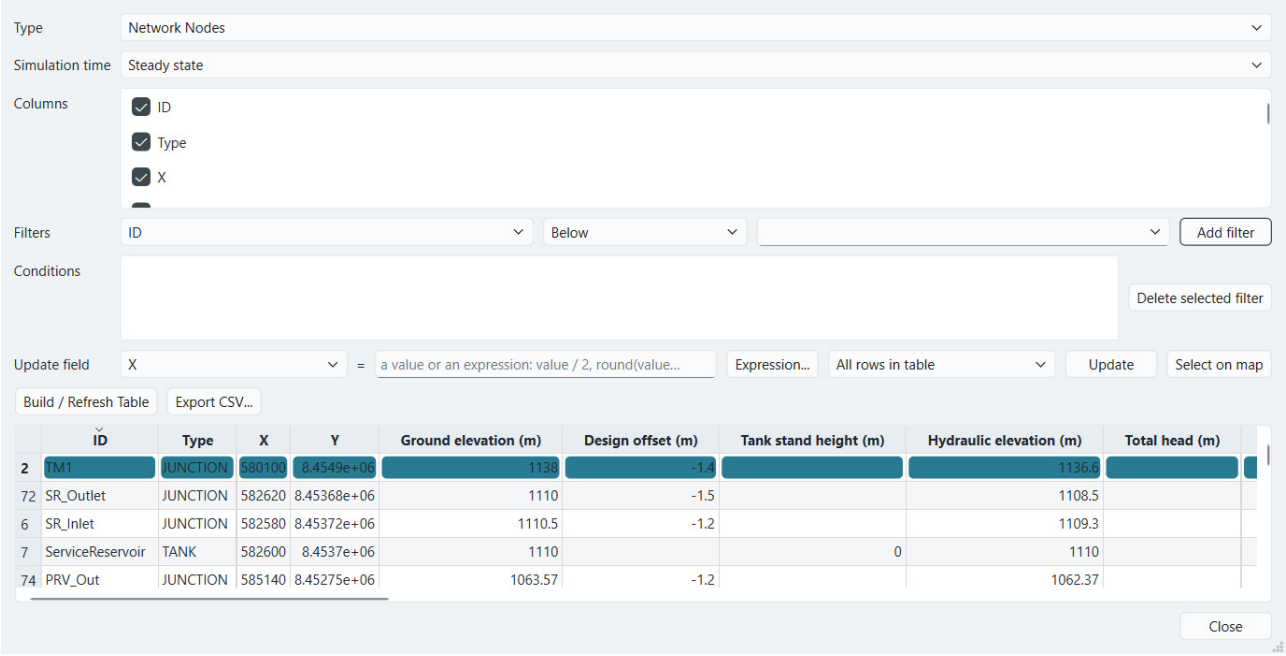
Tips

- Run the hydraulics first (Run Extended Period) and fix any warnings: water quality follows the flows, so unbalanced hydraulics give meaningless concentrations.
- Use a warm-up of at least a day, longer when tanks are large compared with the demand.
- Calibrate the bulk rate with bottle tests, then the wall rates with chlorine readings at a few hydrants; the **Field Observation Set** (Analysis menu) holds the readings.
- The same settings drive every extended-period run when the parameter is set under **Analysis Options > Water Quality**; the Water Quality Analysis window adds the study tools on top.

14

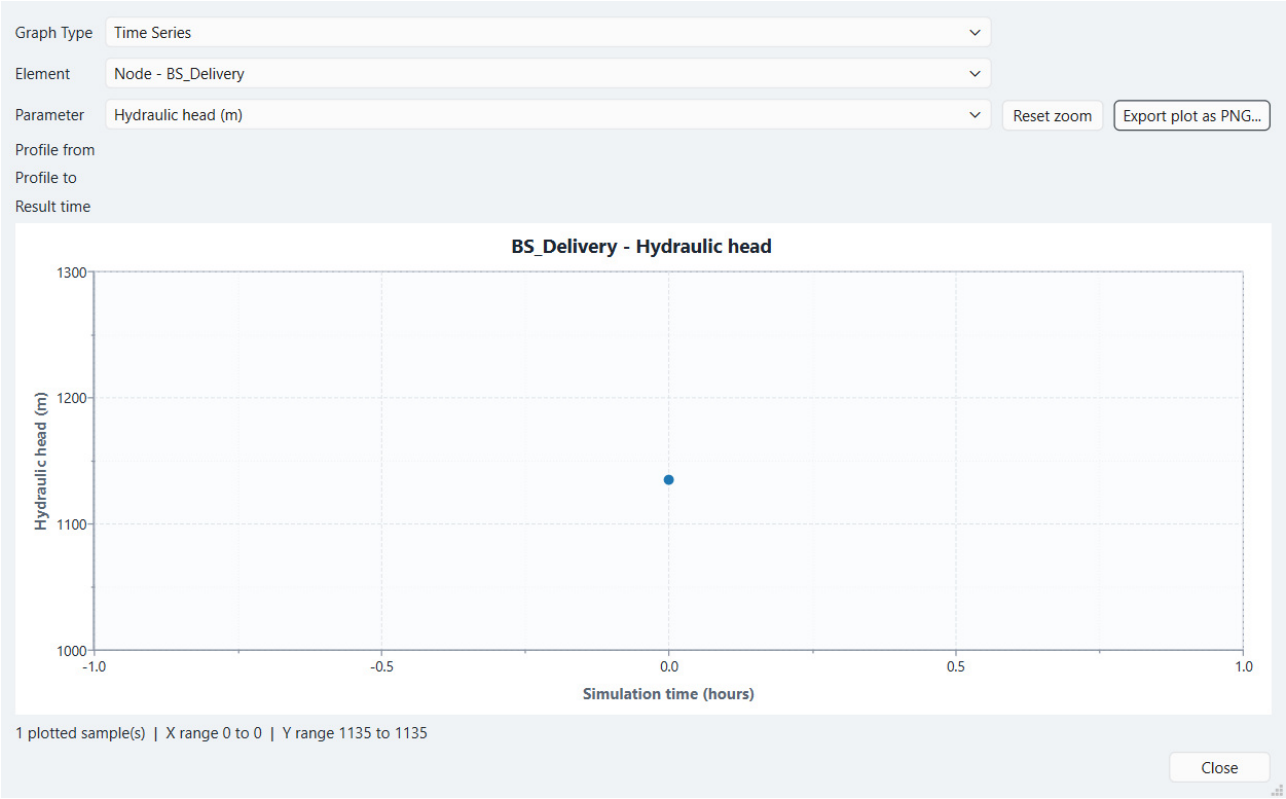
Reading the results

Reading what an analysis found: the map, the tables, the graphs, the warnings and the reports.



	ID	Type	X	Y	Ground elevation (m)	Design offset (m)	Tank stand height (m)	Hydraulic elevation (m)	Total head (m)
2	TM1	JUNCTION	580100	8.4549e+06	1138	-1.4		1136.6	
72	SR_Outlet	JUNCTION	582620	8.45368e+06	1110	-1.5		1108.5	
6	SR_Inlet	JUNCTION	582580	8.45372e+06	1110.5	-1.2		1109.3	
7	ServiceReservoir	TANK	582600	8.4537e+06	1110		0	1110	
74	PRV_Out	JUNCTION	585140	8.45275e+06	1063.57	-1.2		1062.37	

The results table of the junctions, with its tools



Results of chosen elements in a graph

Where the results appear

After a run the results are kept with the project and appear in several places at once:

- **the map**: the Map Styles panel colours the nodes and links by any result, with a legend on the map (Map views, legends and animation);
- **Element Properties**: click a node or a link to read its values under **Results**;
- **the Simulation Results panel**: the time bar and the result tables;
- **Analysis > Advanced Graphs**: time series, profiles, contour plots, frequency plots and the system's flow balance;
- **tool windows**: pipe sizing, fire flow, criticality, water quality and the energy report each show their own results.

The time bar

An extended period gives one set of results at every report step. The time bar at the top of the Simulation Results panel chooses the time shown; the map, tables, legends, labels and Element Properties all follow it. Play the run with the play button, step with the arrows, or choose a time from the list. **Animate Map** moves water along the pipes at the time shown, or through the run.

Below the times, the list holds **statistics over the run**: the **Minimum**, **Maximum**, **Average** and **Range** of every result at every element. Choose one and the map, the tables and the legends show it, for example the lowest pressure each junction sees in the day.

Reports

- **Analysis > Status Report** lists what happened during the run, in time order: pumps that started and stopped,

valves that changed state, tanks that filled or emptied, controls and rules that acted, and warnings. For a water quality run it ends with the mass balance: what entered, left, reacted and is still stored, and the difference as a percentage.

- **Analysis > Calibration Report** compares the results with field measurements entered in **Analysis > Field Observation Set** (pressures, flows, levels or quality, at times you give): the mean and root mean square error at each location, the correlation of computed and measured values, and a chart of one against the other.
- **Analysis > Full Report** writes a text file with the options, every node and every link at every report time, for archives and checking by hand.

Result tables

The **Nodes** table lists each node's head, pressure, delivered withdrawal and water quality; the **Links** table lists each link's start and end nodes, diameter, length, flow, velocity, velocity head, heads at both ends, head loss, friction gradient and status.

- **Columns** chooses the columns shown.
- Type in the filter box to show the rows whose name contains it.
- Click a column heading to sort by it; click again to reverse.
- Double-click a row to find the element on the map.
- Select cells, rows or columns and right-click to **Copy** or **Copy with Headers**; they paste as a table into spreadsheets and documents (Ctrl+C works too).

Analysis > Export Results writes a spreadsheet with a sheet for the nodes and one for the links; for an extended period, the time chosen in the time bar.

Graphs

Analysis > Advanced Graphs draws:

Graph	What it shows	Use it to
Time Series	one value of one node or link through the run (pressure at a junction, level of a tank, flow of a pump)	check that tanks fill and empty, pumps cycle sensibly and pressures stay up at peak
Profile Plot	a value along a path of nodes, from a start node to an end node	see the hydraulic grade along a main, and where pressure is lost
Contour Plot	a value over the whole network, as coloured areas	find the low-pressure zones at a glance
Frequency Plot	the share of nodes or links below each value	state compliance: 95 % of junctions above 15 m
System Flow	the total production and consumption through the run	check that supply meets demand and see the daily cycle

Hover a point to read its value; the export button saves the graph as a PNG image at twice screen resolution.

Edit > Longitudinal Profile draws the ground, the pipe invert, the cover and the hydraulic grade line along selected connected pipes, ready for a drawing sheet.

Solver warnings

The solver reports what it noticed during the run: a time step that did not balance, negative pressures, disconnected nodes, pumps working outside their curve or unable to deliver the head, valves that cannot hold their setting, tanks that are full or empty. The completion message says how many there are; **Analysis > Solver Warnings** lists them with their

times, and a double-click selects the element named. See Errors and warnings for what each one means and what to do.

Judging the results

A useful order for checking a design:

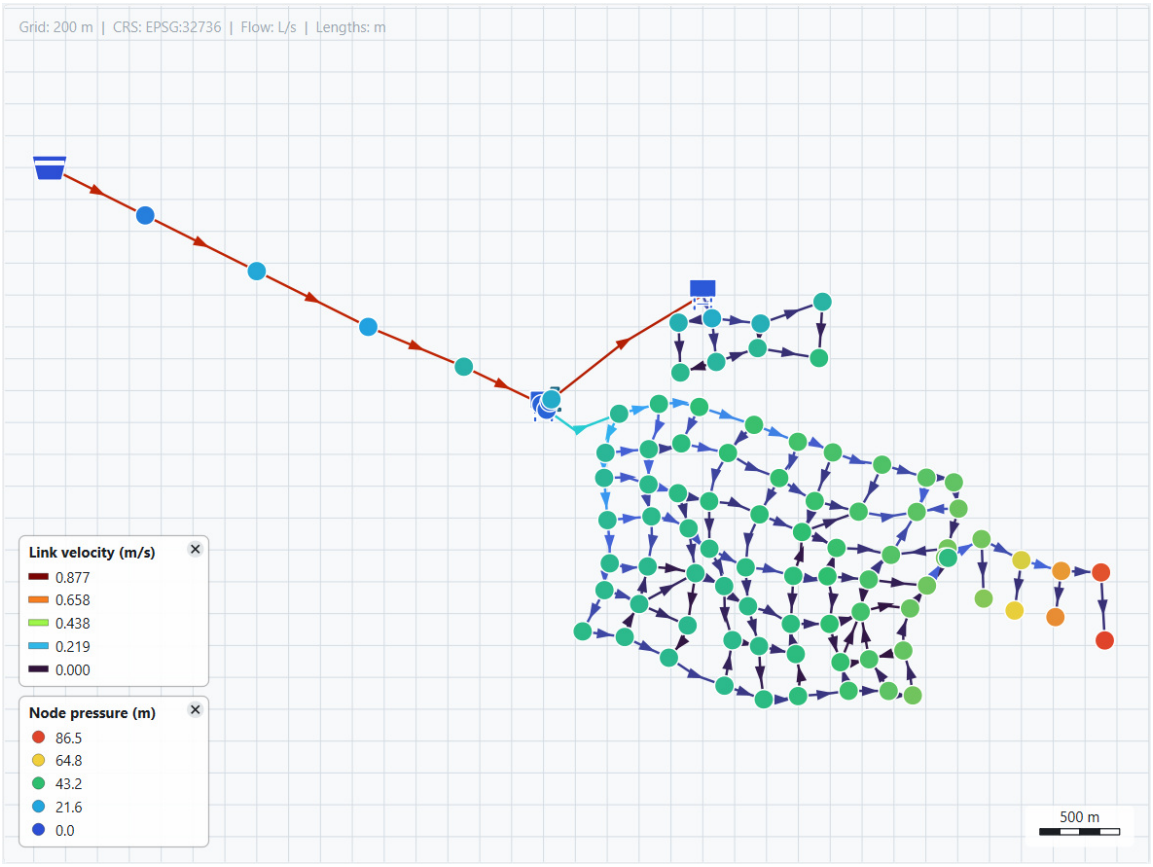
1. **Warnings first**: a network that does not balance gives meaningless numbers.
2. **Pressures**: every junction between the minimum (often 10 to 15 m for a public supply, 5 to 10 m at communal taps) and the maximum (often 60 to 80 m, to limit leakage and bursts), at the peak and at night.
3. **Velocities**: typically 0.5 to 1.5 m/s in distribution and up to 2 to 2.5 m/s in short mains; very slow pipes keep water too long.
4. **Head loss**: the friction gradient (m/km) shows undersized pipes; a few metres per kilometre is comfortable in a main.
5. **Tanks**: levels that cycle within their range every day, neither emptying nor staying full.
6. **Pumps**: running near their best efficiency, starting a few times a day rather than every hour.
7. **Water quality**: chlorine above the minimum everywhere, water age below the limit.

The limits come from the design standard of the project and the standards of your country or utility; FlowCHAOS shows the values, the engineer judges them.

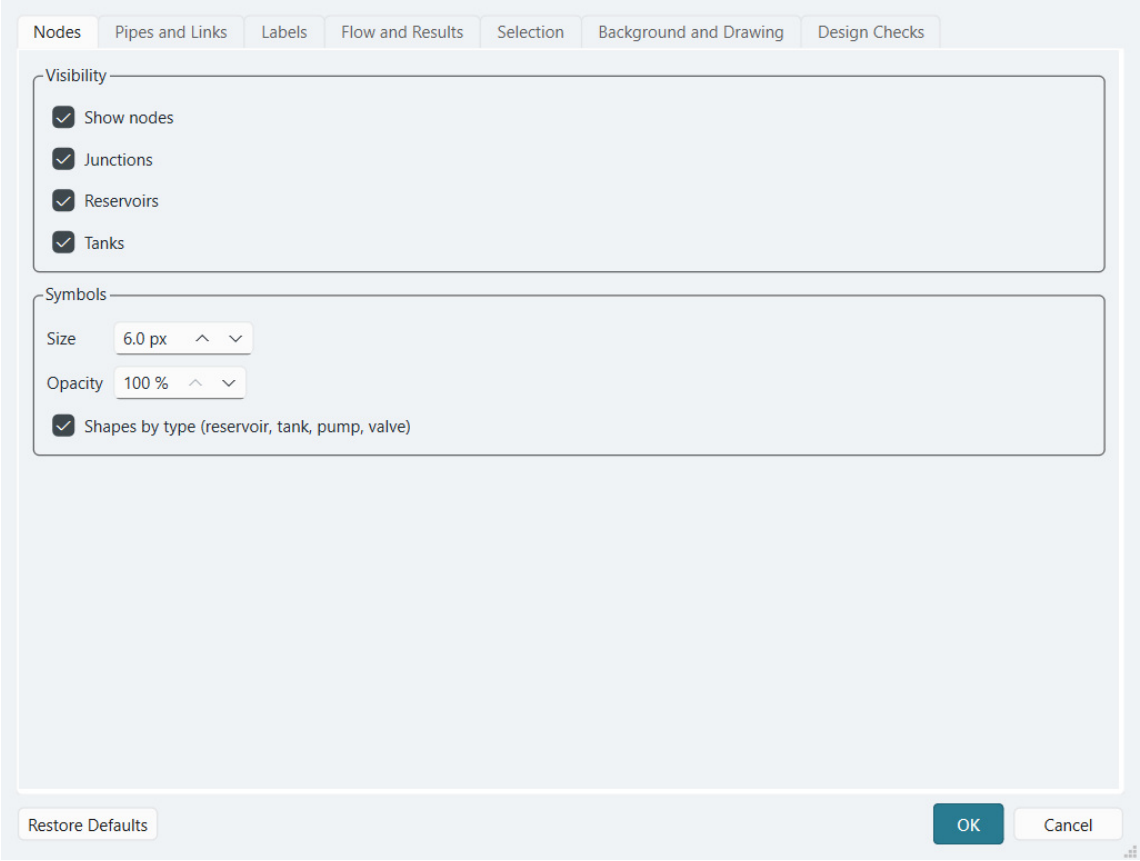
15

Map views, legends and animation

Colour the network by any value, read it with legends, and bring the results to life.



Map styles: pressure and velocity with their colour bands



Map Options: what the map shows and how

Map views

The **Map Styles** panel (View > Map Styles, or its toolbar button) has a view for **Nodes** and one for **Links**, chosen independently. Views of the elements' own data, such as diameter or ground elevation, work at any time, before any run; result views become available after a run. Each list is grouped into **Data** and **Results**. A project opens with **Layer colour**, so each layer keeps its own colour until you choose a view.

Nodes	Links
Data: ground level, hydraulic elevation, design withdrawal, starting concentration, pipe cover	Data: diameter, model length, roughness, criticality, bulk and wall decay rates
Results: pressure, hydraulic grade, delivered withdrawal, concentration, pressure above the minimum, withdrawal not delivered	Results: velocity, flow, friction gradient (m/km), head loss, friction factor, decay reaction rate, concentration, travel time of the water, pressure class used (%), Reynolds number, velocity head

Apply to at the top of the panel decides which layers take the view: the selected layer in the Layers panel, every network layer, or one layer you pick. The other layers keep their own colours, so two networks can be shown in different ways side by side.

Colour the pipes by diameter to check a design at a glance, or the nodes by ground elevation to read the terrain under the network.

Colour ramps

Each view has its own colour band, with the title above a full-width ramp; click it to choose a ramp, reverse it or split it into classes. The ramp list opens below the band, or above it when there is no room, and scrolls; every ramp chooser in FlowCHAOS works the same way. Nodes and links keep separate ramps, so changing one never changes the other.

When the view's legend has classes, the band shows those classes and their break values. Double-click a class to change its colour, or choose **Edit Classes and Colours** in the ramp list to set the number of classes, the break values and every colour in the Legend Editor.

Map Options

View > Map Options has a page for each part of the map: **Nodes** (which types are shown, size, opacity, shapes), **Pipes and Links** (pumps and valves shown, width, opacity, outline), **Labels**, **Flow and Results** (arrows and their size, sizing symbols by value), **Selection** (colour, halo and pulse) and **Background and Drawing**. Restore Defaults resets them.

More settings on these pages:

- **On the map**: a **scale bar** and a **north arrow**, each in the corner you choose. **View > Scale Bar** and **View > North Arrow** switch them on and off quickly.
- **Closed links dashed**: closed pipes, stopped pumps and shut valves are drawn dashed, at the time shown.
- **Hover information**: resting the pointer on an element shows its name, type and main results.
- **Design Checks**: after a run the map marks what lies outside your limits, each check switched on by itself:
    - pressure below the minimum, above the maximum, or negative;
    - velocity below the minimum (water standing) or above the maximum;
    - head loss per kilometre above the limit (the pipe is too small);
    - pressure above the pipe's class (PN10 allows about 102 m);
    - cover below the minimum where the pipe level was designed;
    - pipes carrying almost nothing (water ageing);
    - dead ends and nodes joined to nothing.

  A summary box in the corner you choose counts what each check found.

Legends

The legend button beside each band opens the **Legend Editor**:

- **Continuous** colours, **equal intervals**, **equal counts** (the same number of elements in each class) or **your own break values**;
- a colour per class (double-click a swatch) and a title.

Legends are drawn on the map. Drag one to move it; click its small round x to hide it (the Legend Editor shows it again). Positions and classes are saved with the project, one legend per view.

Colour relief and surface maps have their own **colour band legend**: right-click the layer and choose **Show Legend on Map**. Their icons in the Layers panel follow their colours, with a swatch for each class beneath them.

Animate Map

A legend in the top right corner of the map says what the moving marks mean: for the flow, the size of a mark at three flows and the fastest velocity; for the other animations, their colours and the values they stand for. It is drawn into saved videos and GIFs too; **View > Animation > Show Animation Legend** turns it off. Flow is shown as dots when FlowCHAOS starts; arrows and drops are in the same menu.

Animate Map (run toolbar and results time bar) brings the map to life. The arrow beside it chooses what moves:

Animation	What you see
Flow	particles travelling along every pipe in its flow direction, faster where the water is faster; as dots, small arrows or water drops
Head loss	warm sparks along the pipes, denser and brighter where more energy is lost
Water quality	particles moving with the flow, coloured by the concentration
Delivered withdrawal	ripples spreading from the nodes, larger where more water is taken
Pressure	halos breathing around the nodes, in the colours of their pressure
Design withdrawal	ripples from the design withdrawal (no run needed)
Roughness	a texture drifting along the pipes, denser on rougher pipes (no run needed)

During playback of an extended-period run the animation follows the time shown.

Save Animation as Video or GIF (the arrow beside Animate Map, or the Analysis menu) saves the time steps of the results, the moving flow, or both together as an MP4 video or an animated GIF, at the speed, smoothness and size you choose, with the time written in a corner.

Animate Map is available when there is something to animate: results of a run, or Design withdrawal or Roughness chosen in **View > Animation** (which works before any run).

Clearing results

Stop / Clear Results stops a running analysis, or clears the results so the map shows the network as drawn; Undo brings them back. A run itself is not an undo step (it is a calculation, not an edit of the network), so the results of earlier runs are not kept in memory; the undo history keeps the last 150 edits, so a long session never fills the memory.

The map grid

Grid on the Drafting toolbar shows or hides the grid lines. Its arrow has **Coordinate Labels** and **Rulers** (bands along the top and left edges with a tick at every grid line), each on or off on its own, so the map can show lines only, labels or rulers only, or any mix (the grid lines alone at first); and **Map Grid Settings**. The grid is a preference of FlowCHAOS, the same in every project: changing it never leaves a project with unsaved changes.

- **Spacing**: automatic (round steps of 1, 2 and 5 that follow the zoom, a guide to the eye; points snap every 10 m) or a fixed spacing, which points snap to.
- **Style**: lines, dots at the crossings or small crosses, with a stronger major line every few lines.
- **Colour** and **opacity**.
- **Coordinates**: the easting and northing (or longitude and latitude) of the major lines along the top and left edges of the map, read as the map moves.

Refresh (Ctrl+R, on the Navigate toolbar) draws every layer again: imagery and terrain read anew, vector layers and the network rebuilt, the Layers panel and the open attribute tables brought up to date.

The Browser

The **Browser** tab beside **Layers** shows the computer's folders and the files FlowCHAOS can open (networks, projects, shapefiles, drawings, GeoPackages, KML, rasters, survey points); a shapefile's other parts are not listed. Drag a file onto the map or the Layers panel, double-click it, or right-click it and **Add to Project**. The list at the top goes to the

drives, your folders, the project's folder and your favourites; the star keeps the folder you are in as a favourite.

Working with large data

FlowCHAOS keeps large projects responsive: an edit of the network does not copy the vector layers it does not touch, so moving a node is as quick with a large drawing loaded as without it; line layers are drawn as one path for each colour; shaded DEMs are worked out in the background and in limited memory; a hidden raster keeps only a small overview; and a tool window that is closed (a 3D view, a profile, the planner) gives its memory back, and opens again from the project when it is needed.

16

Terrain and surface maps

Turn survey points, network levels or a DEM into contours, colour relief, 3D surfaces and more.



A DEM drawn as shaded terrain under the network

Making a surface map

The **Contours** button (or **Add Layer > Surface Map**) opens the map maker. Choose the source from the map (a point layer, an earlier surface map, a DEM or the network's node levels) or from a file, then the map type:

Map type	What it shows
Contour map	lines, optionally filled bands, with labels that read uphill
Colour relief	continuous colours with hill shading

Post map, classed post map	the points with their values or classes
3D surface, 3D wireframe	the terrain in its own 3D window
Point cloud	the points coloured by level
Peaks and depressions	high and low points of the surface

The surface is made in the background with a progress window. The automatic cell size is fine enough for smooth surfaces and edges (at least 250 cells across); set your own in the properties.

Contour lines drawn in another program can be used directly: choose a **DXF** file as the source and its contour polylines (with their elevation), 3D lines and points become the points of the surface.

The work is done in the background: the status bar shows the task, its progress and a Cancel button, and the map stays usable meanwhile.

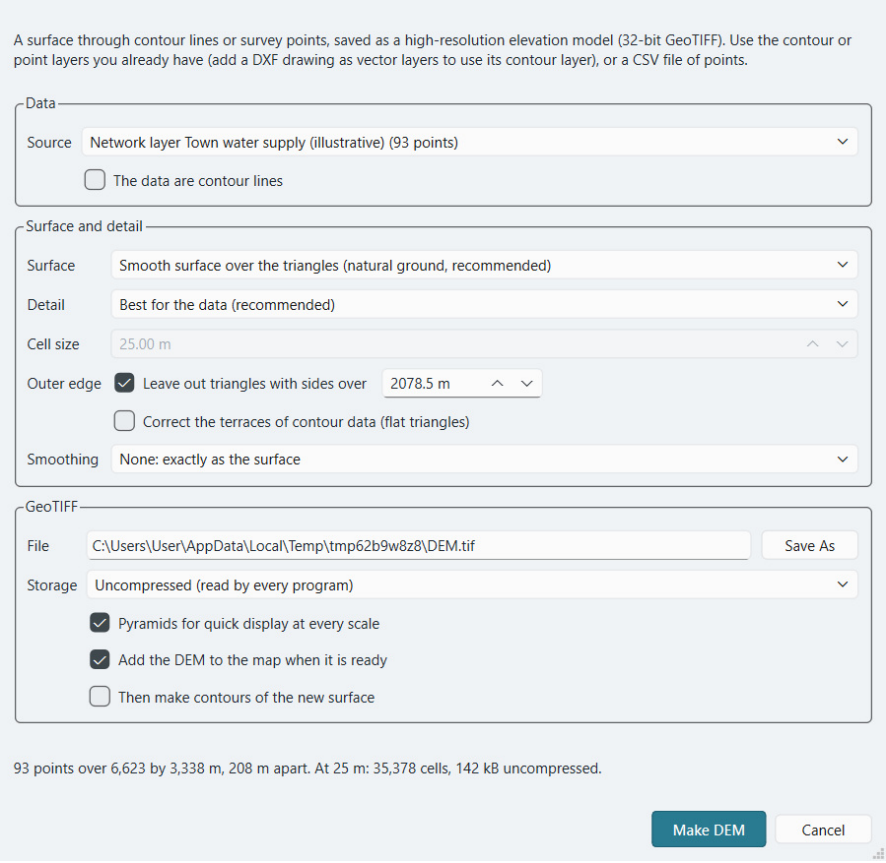
DEMs on the map

A DEM is drawn as terrain: a colour for its height (greens low, yellows and browns on the slopes, grey and white on the heights) lit by a hillshade from several directions, so every slope is modelled and none is lost in shadow. Zoomed in past the cells of the data the surface stays smooth; zoomed out, each pixel shows the average of the cells under it, so there is no speckle. On gentle ground the shading is raised (the heights are not changed), so the channels and banks of a floodplain still show. Choose hillshade, colour only or grey in the layer's properties.

A DEM from points, contours or a surface

Layer > Make DEM (GeoTIFF) builds a terrain model as a GeoTIFF from data in the project: a contour or surface layer, a line layer of contours or a point layer with an elevation field, the levels of a network layer or of all junctions; or from a CSV file of points, choosing which columns hold X (easting), Y (northing) and Z. To use the contours of a DXF drawing, add the drawing as vector layers first (only its contour layer, if you like) and choose that layer.

The data are read and analysed in the background, so the window keeps responding, and the settings are suggested from them: the cell size, the longest triangle side that still belongs to the survey, and the terrace correction for contours. The surface is smooth by default: it passes through every point and curves between them, with no sharp corners where the triangles meet; a flat triangulated surface (TIN) can be chosen instead. A 1 m DEM of a large scheme takes about a minute and runs in the background. The new DEM can be added to the map and, if asked, contoured at once.



Make DEM: the data analysed and the settings suggested

Gridding and smooth contours

Gridding method	Character
Kriging (linear variogram)	the default: honours every point, smooth between them
Minimum curvature (thin-plate spline)	the smoothest surface; may overshoot between sparse points
Triangulation (TIN)	linear between points: exact but faceted, so contours bend at triangle edges
Inverse distance to a power	weighted by distance; flat spots around the points

Line smoothing (None, Low, Medium, High, Very high) draws smooth contour lines: the surface is resampled finer with cubic interpolation before the lines are traced, grid-cell jitter is removed, and the corners are rounded. The lines stay on their level; Medium suits most maps, Very high gives the softest lines.

Properties

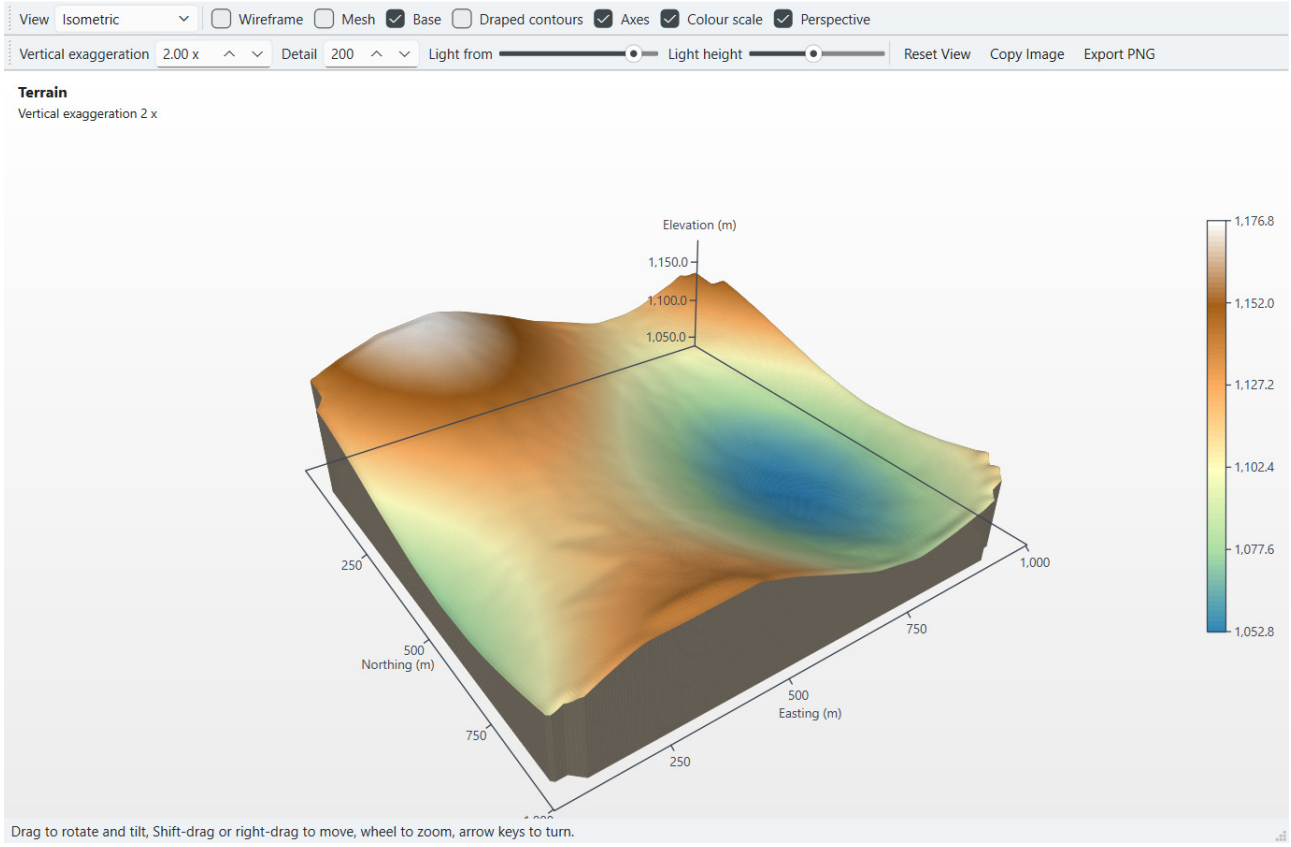
Each map type has the properties that make sense for it: gridding method and cell size, levels, fill, lines and labels for contours; colour, shading direction, smoothness, opacity and an optional contour overlay for relief; symbols for post

maps; surface, light and frame for 3D.

Right-click a surface map for **New Map from This Source** to make another type from the same data.

The 3D window

Double-click a 3D surface in the layers list, or use **Contours > Open 3D View**, to open it again at any time. Drag to turn and tilt (the surface turns the way the mouse moves), Shift-drag to pan, wheel to zoom. Preset views, vertical exaggeration, detail, light direction, walls to a base, axes with values, a colour scale, draped contours and high-resolution image export.



A terrain surface in the 3D view

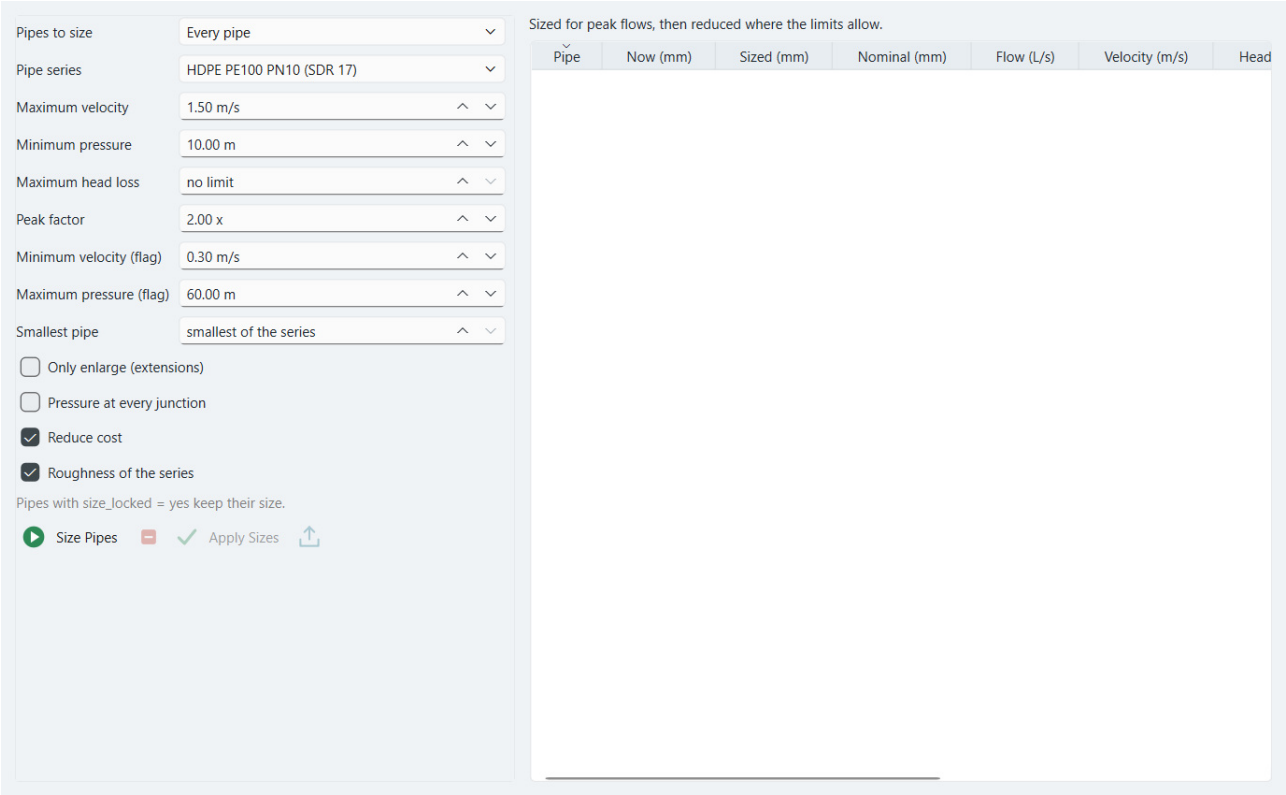
Exports

Contours export as DXF, Shapefile or GeoPackage; surfaces as an ASCII grid (.grd); maps go on sheets in the print layout.

17

Design tools

Size pipes automatically, check fire flow, draw long sections and set design levels.



Pipe sizing: the diameters that meet the velocity and pressure limits

Automatic pipe sizing

Analysis > Automatic Pipe Sizing finds the cheapest sizes of a pipe series that keep the network within your limits. It works in stages, each with the advanced solver:

1. **Feasibility**: with every pipe at its largest size, junctions still short of pressure are limited by the source level. They are reported (a pump or a higher tank is needed) and not chased.
2. **Sizes from flows**: the flows at peak demand give each pipe the smallest size that carries its flow below the velocity limit.
3. **Repair**: pipes that are too fast or lose too much head go up a size; where junctions are short of pressure, the pipes on the paths that supply them go up where a size gains the most head for its extra cost.
4. **Cost reduction**: each pipe is tried one size smaller, largest savings first, and kept smaller only where every limit still holds.

Settings

Setting	Meaning
Pipes to size	every pipe, or the selected ones
Pipe series	45 series, the most used first: HDPE PE100 and PE80 in every pressure class, uPVC by PN and by Class 6, 9, 12 and 16, PVC-O, irrigation laterals and mains, PPR, PEX, copper and galvanised steel for buildings, ductile iron, steel, GRP, asbestos cement and concrete for large mains, and North American sizes; or your own
Maximum velocity	no pipe may run faster
Minimum pressure	every junction that takes water must keep it (or every junction, if chosen)
Maximum head loss	per kilometre, optional
Peak factor	every withdrawal times this factor, to design for the peak hour
Minimum velocity	slower pipes are flagged (water may stagnate)
Maximum pressure	higher junctions are flagged (a pressure reducing valve may help)
Smallest pipe	no pipe below this diameter
Only enlarge	no pipe becomes smaller than it is (for extending a network)
Reduce cost	the cost reduction stage

Pipes with the field size_locked = yes keep their size. A main that fills a tank is sized for the demand the tank supplies (in a steady run a tank fills as fast as its inlet allows, which is not a design flow).

Results

The table shows each pipe's current and new size, nominal size, flow, velocity, head loss and a note. The summary gives the cost of the design (from the series' prices, or a relative cost), lengths by size, the lowest pressure, and the pipes and junctions that need attention. **Apply Sizes** writes diameter, roughness, material, pressure class and nominal size in one step that can be undone.

Add prices to a series to see real costs: see reference data.

Fire flow analysis

Analysis > Fire Flow Analysis gives, for every hydrant, the largest flow it can deliver while its residual pressure and the pressure at every other junction stay above the minimums. Test every junction, the selected ones, or those whose hydrant field is yes. The table shows the static pressure, available flow, residual pressure and the node that limits each hydrant; **Select Failing Hydrants** shows on the map where the network needs strengthening.

How it works: for each hydrant the fire flow is added as an extra withdrawal and the network is solved again, raising the flow until a pressure limit is reached (a search that needs only a few runs). The model stays open between runs, so even large networks finish quickly, and **Stop** ends the analysis at once, keeping the hydrants already tested.

Criticality analysis

Analysis > Criticality Analysis asks, for each pipe, pump or valve: what if it were out, from a burst or a repair? Each tested link is closed in turn.

1. Junctions that lose every path to a reservoir or tank are **cut off**: all their withdrawal is lost.
2. The rest of the network is solved with **pressure-dependent withdrawal**: a junction below the service pressure delivers only part of its withdrawal, and nothing at or below the pressure where delivery stops. What it misses beyond its normal shortfall counts as lost too.

Setting	Meaning
Test	every pipe, the selected links, or every pipe, pump and valve
Smallest pipe tested	skip service pipes below this diameter on large networks
Service pressure	full withdrawal at or above this pressure
Delivery stops at	no withdrawal at or below this pressure
At design withdrawal	demand patterns held at 1, times the peak factor; off, the patterns at the start time

Each link is rated by the share of the total withdrawal not delivered while it is out: **High** from 20 %, **Medium** from 5 %, **Low** from 0.5 %, otherwise **None**. The table also shows the junctions cut off, the shortfall and the lowest pressure that remains. Double-click a row to find the link; **Select High and Medium** selects them all; **Show on Map** colours the links by criticality (the *Criticality* link view, kept with the project); **Export CSV** lists the affected junctions too.

Reservoirs and tanks keep supplying for the moment analysed, so a pump that fills a tank may show no effect: the tank covers the outage until it runs down. Use an extended period run to see how long.

Long sections

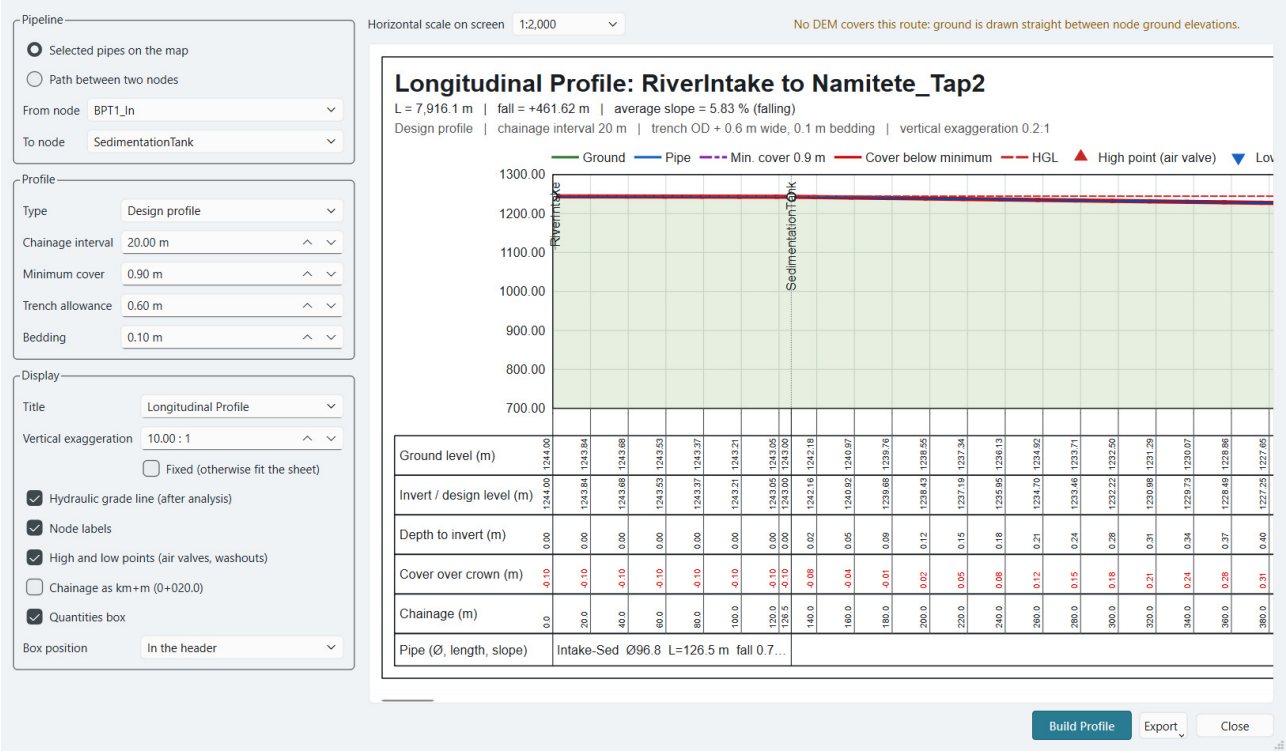
Select pipes and choose **L-Profile** (Ctrl+Shift+P). The profile shows ground, pipe invert, minimum cover and the hydraulic grade line, with tables of levels, depth, cover and chainage and the excavation quantities. It follows the selection on the map.

The sheet is drawn at the **Horizontal scale on screen** chosen above it (1:2,000 by default; 1:500 to 1:50,000, or Fit to window) and is as long as the pipeline at that scale: it scrolls sideways, and Ctrl with the mouse wheel zooms along it. The settings are on the left; **Build Profile**, **Export** (PDF, PNG or the chainage table as CSV) and **Close** stay at the foot of the window.

The values never overlap: where chainages crowd together they are spread apart with a leader line back to their true place, as in the planning network diagram, and node names do the same. If even that leaves no room the sheet says how many node values were left out (the CSV table has them all).

- **Vertical exaggeration**: fit the sheet, or fix it (for example 10:1) when the sheet has room.
- **Chainage as km+m**: 1+234.5 instead of 1234.5.
- **High and low points**: peaks of the pipe (for air valves) and dips (for washouts) are marked; fittings already on the map are filled, points found from the levels are open.
- **Quantities box**: in the header, in a corner of the chart, or dragged to any place on the sheet.

The trench is the pipe's outside diameter plus the **trench allowance** wide, and reaches the **bedding** depth below the invert; excavation is summed along the profile from these. The grade line is drawn between every node that has a result, so a node left out of the analysis does not remove it.



A longitudinal profile of a gravity main with its bands

Design levels

Design Levels sets invert levels from the ground and a cover depth, or tank stand heights, for the selected nodes.

Area Demand Allocator

Design > Area Demand Allocator (also in the Irrigation menu and the Processing Toolbox) sets the design withdrawals of junctions from the polygons they supply: the blocks of an irrigation scheme, the zones of a town. The settings are on the left and the matches on the right; **Match**, **Apply** and **Close** stay at the foot of the window.

The areas are measured from the polygons themselves (holes left out, on the ellipsoid like the areas of a GIS), in hectares, square metres, acres or square kilometres, so the layer needs no area field. The demand of each area comes from one of two bases:

- **A rate for each unit of area** (unit area loading): its area times the rate, for example the peak irrigation requirement in l/s per hectare, and a **factor**.
- **A total flow shared by area**: the scheme's design flow shared among the areas in proportion to their areas, times the factor.

Tick **Share by a number stored for each area** to use a field instead of the measured area, for example a population or a number of plots (the attribute table's field calculator can work one out first). The withdrawals are rounded to the **Decimal places** chosen (2 by default), in the table and when set.

The demand of each area goes to its junctions by one of the methods of the design literature:

- **The junction in each area** (nearest node): the junction inside it nearest its centre; with none inside, the junction nearest the area within the search distance.
- **Shared equally among the junctions in each area** (equal distribution).
- **Each junction takes the area it lies in.**

- **Thiessen service areas**: each junction supplies the part of the areas nearer to it than to any other junction (Voronoi polygons), so an area's demand is split by the share of it each junction serves.
- **Shared by half the length of the pipes at each junction** (the unit length method): a junction with more pipe around it takes a larger share of the area's demand.

Match lists every area (numbered as in its layer) with its area, its withdrawal, the junction's current withdrawal, its junction (a shared area shows each junction's share) and how it was found, and draws an arrow on the map from each area to its junction, coloured by how it was found; the colours of inside, nearest outside, set by hand and no junction can be chosen. Change a match in the table, select a junction on the map and **Use Selected Junction**, or drag on the map: an arrow's head dropped on another junction gives the area to that junction, and an end dropped inside another area has that junction supply that area instead; the table follows. **Clear the Map** takes the arrows and highlights off the map. The summary counts what was found and names junctions that supply more than one area, to check.

Apply replaces the design withdrawals (or adds to them), and can set the junctions that supply no area to zero; it is one step of Undo. A banner above the table says how it went: green when every junction has its new withdrawal (shown in green under Current), amber when some areas had no junction, red with the junctions that could not be set.

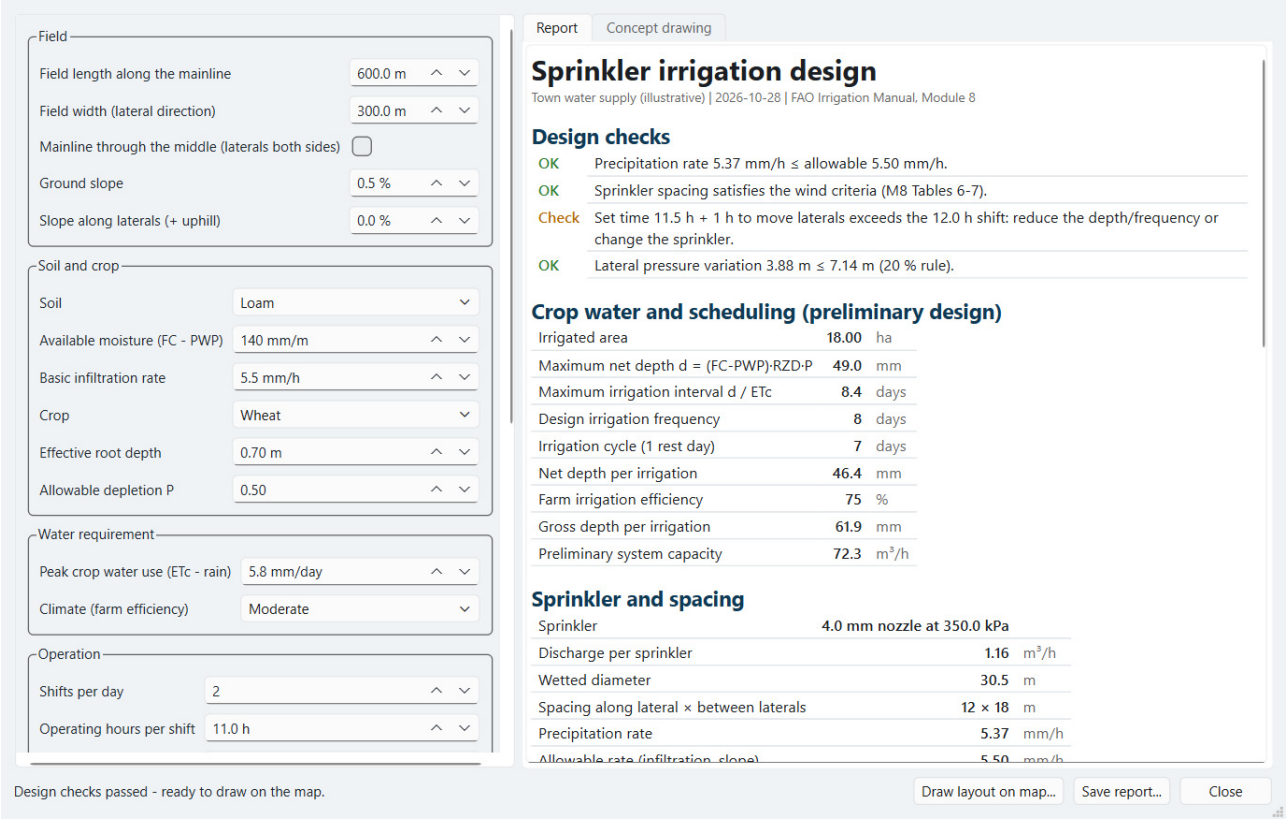
Irrigation

The **Irrigation** menu holds sprinkler, drip and centre pivot design tools that start from crop water needs and soil and draw the layout on the map.

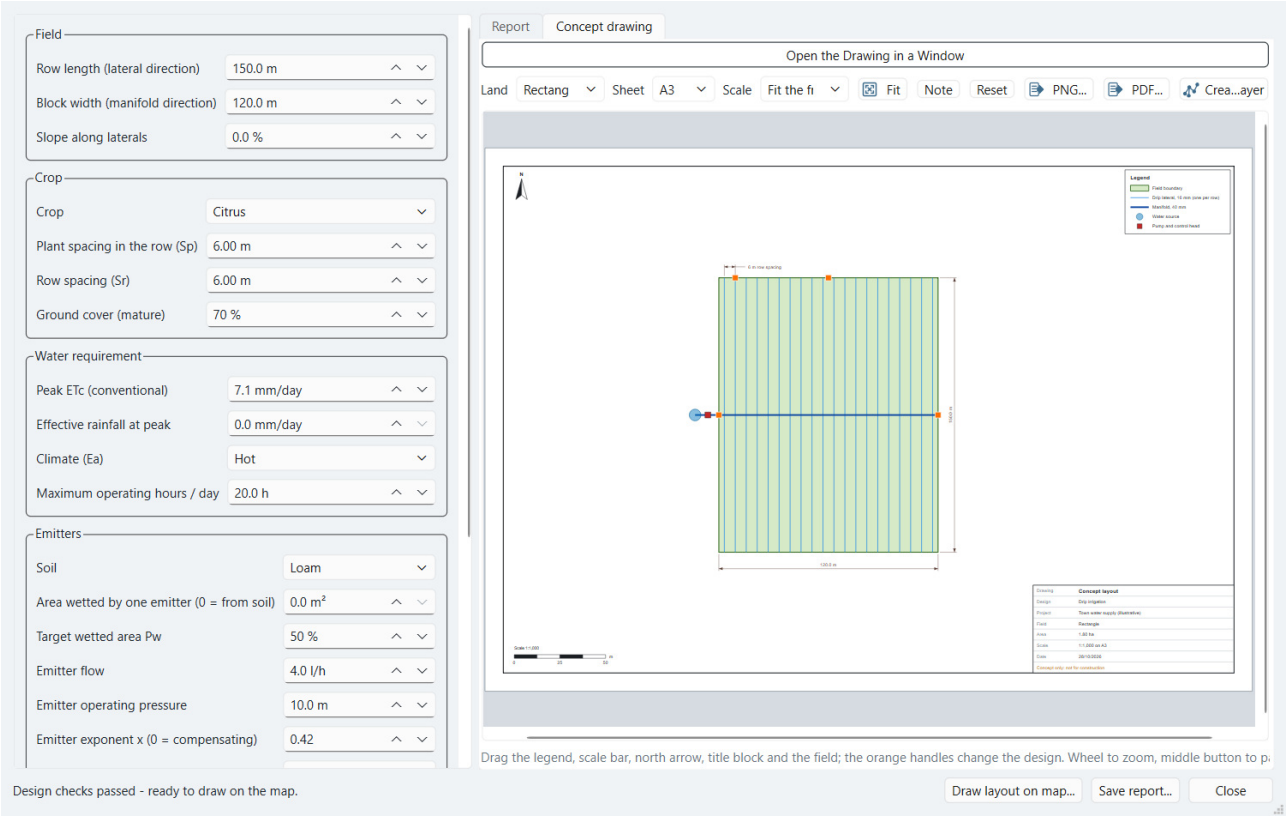
Each design tool has a **Concept drawing** page beside its report: the field to scale with the mainline or manifold, the laterals, the sprinklers and their wetted circles or the drip rows, the blocks watered one after another, the source and pump, dimensions, a legend, a north arrow, a scale bar and a title block. Choose the **land shape** (rectangle, L-shaped, tapering, triangle, irregular; for a pivot a circle in a square field, a full or a half circle): every shape keeps the area of the designed field. Export it as PNG or PDF (A3); a saved report includes it.

Quick Design Estimates gives a first look from a few inputs: for drip the daily need reduced for ground cover, the emitters, the application rate, the watering time and the blocks that fit the hours available; for sprinklers the depth the soil and roots hold, the interval, the precipitation rate (checked against the soil's intake), the set time, the sprinklers running together and the pump flow. **Irrigation > Climate Data** reads monthly climate records (.clim) for crop water needs, and **Apply Climate to Withdrawals** turns them into the withdrawals of the irrigated junctions.

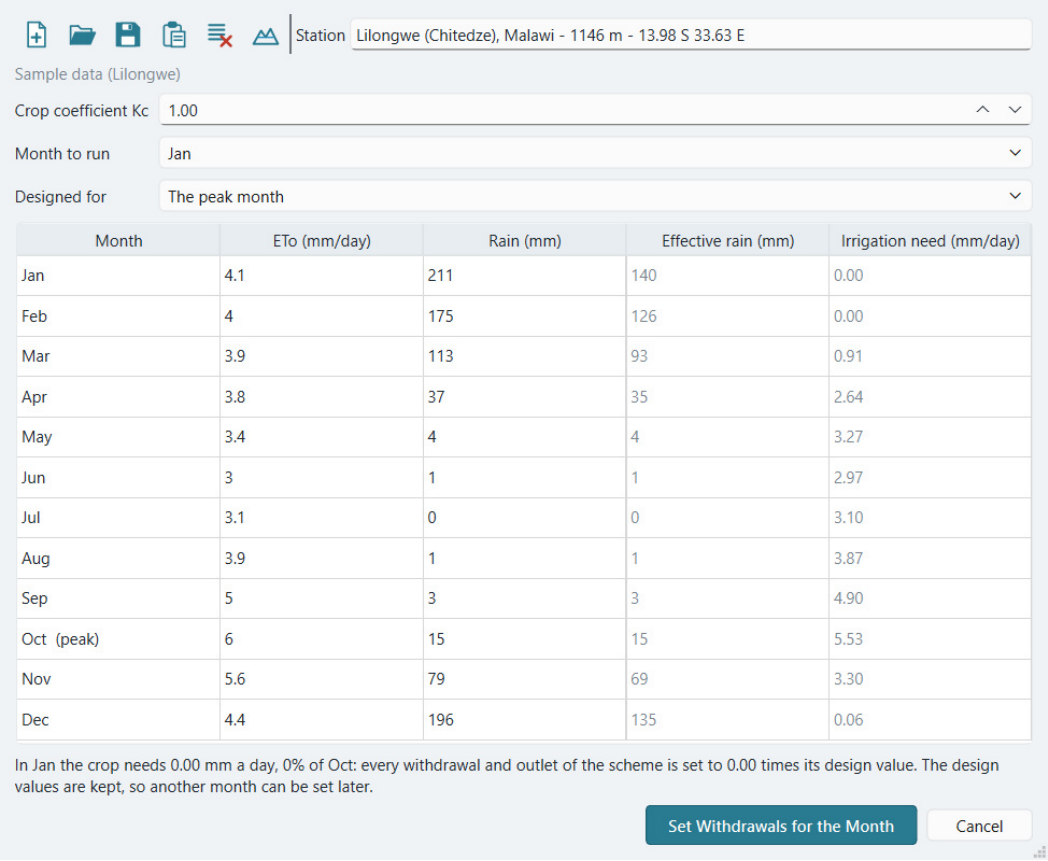
The climate window is a sheet of the twelve months: each month's reference evapotranspiration and rainfall, with the effective rainfall and the net need worked out, and the peak month marked. Type the values, or paste them from a spreadsheet or a text (tab, comma or semicolon separated, with or without a month column); **New Climate Sheet** starts a blank one, **Save Climate File** keeps it as a .clim file to use again, and **Sample Data** loads the Lilongwe means after asking before it replaces entries. Choose a month and FlowCHAOS scales the outlets for it; the design flow is kept with the project. **Sample Projects** has a **sprinkler scheme in July** with the Lilongwe (Chitedze) climate to try it with.



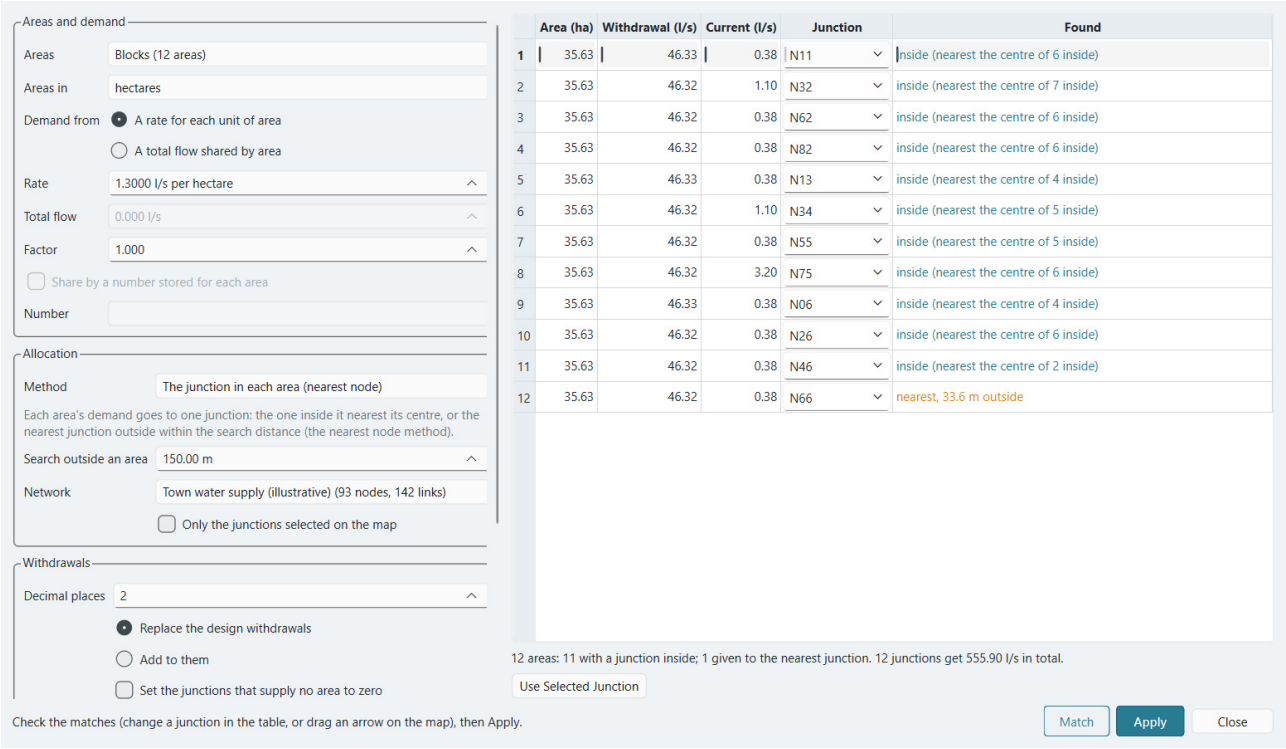
An irrigation design tool: a sprinkler layout and its figures



The concept drawing of a drip block, to scale



The climate sheet: monthly ETo and rain, and the need of each month

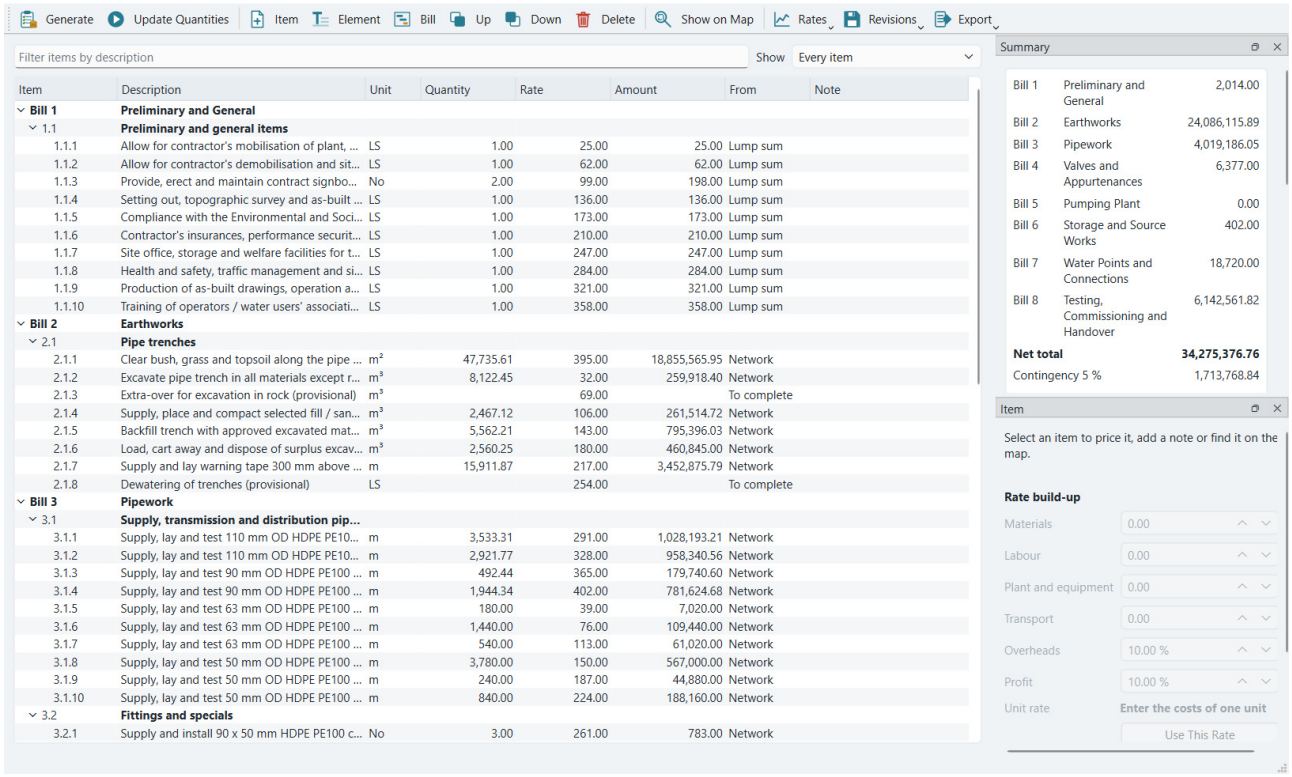


Area Demand Allocator: each block matched to the junction that supplies it

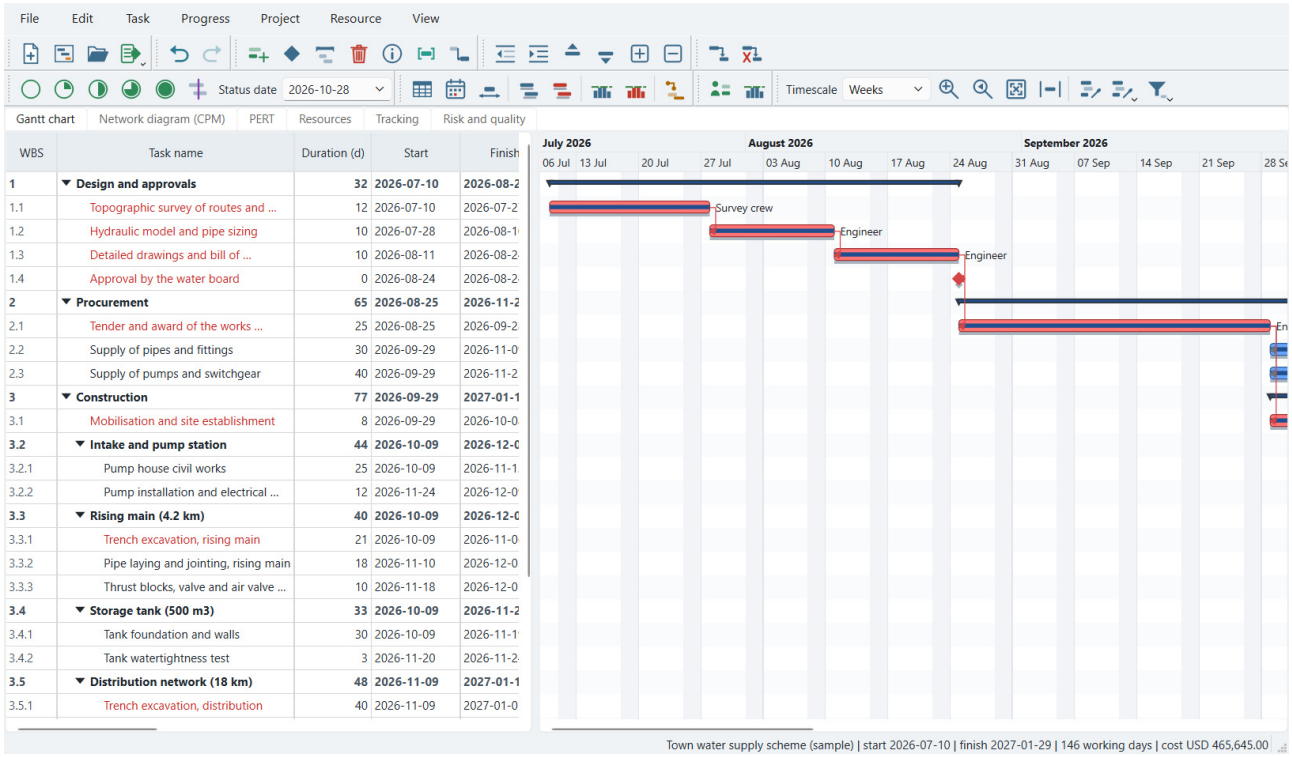
18

Quantities and planning

Bills of quantities measured from the design and kept with it, and the construction programme that follows them.



The Bill of Quantities workbench with its items and rates



Planning: the works schedule as a Gantt chart

The BOQ Workbench

File > Export > Bill of Quantities (or the BOQ button) opens the BOQ Workbench in its own window. The bill is part of the project: it is saved with it, and every change can be undone from the main window.

Generating a bill

Generate asks for the BOQ type (water supply, irrigation, building plumbing), the contents (every section of the type with blanks to complete, or only the work found on the map), the company and client details, the regional preset (currency, tax, levies, contingency), the default pipe materials and the trench sizes. The bill is then measured from the network:

- pipes by material, outside diameter and pressure class (from the modelled or static pressures);
- fittings from the geometry: bends by angle, tees, crosses, reducers, end caps, flange adaptors;
- valves by type and size, chambers, air valves, washouts and hydrants;
- earthworks from the trench width and depth: clearing, excavation, bedding, backfill and surplus;
- pumps with their duty point, tanks with their volume and stands, sources and connections.

Editing and tracing

The bill is a tree of bills, elements and items. Double-click a description, unit, quantity, rate or note to change it; add items, elements and bills, move rows up and down, or delete them. The **From** column tells where each quantity comes from:

From	Meaning
Network	measured from the map
Lump sum	a fixed quantity
To complete	a provisional item: type its quantity
Typed	added or changed by hand

Select a measured item and press **Show on Map**: the pipes and nodes it was measured from are selected on the map. When the network changes, **Update Quantities** measures it again: measured items follow, items typed by hand and every rate are kept, new work is added and work that left the network is marked. *Show: Changed by the last update* lists what moved.

Pricing

- Type a rate in the **Rate** column.
- Build it up in the **Item** panel: materials, labour, plant and equipment and transport for one unit, plus overheads and profit; **Use This Rate** prices the selected items.
- **Rates > Import Rates from a Priced Spreadsheet** reads any workbook with *Description* and *Rate* columns (a contractor's priced bill, a company rate list) into the project's rate library and prices the matching items. **Add This Bill's Rates to the Library** keeps your rates for the next bill.
- Plugins can add their own rate sources (see *Writing plugins*).

Items with a quantity but no rate are shaded; *Show: Quantity but no rate* lists them.

Summary

The **Summary** panel gives each bill's total, the net total, contingency, levies, tax and the grand total, the cost per kilometre of pipe and per connection, how complete the pricing is, and a chart of each bill's share of the cost.

Revisions and export

Revisions > Save a Revision keeps a copy (Rev A, Rev B, ...). **Compare with a Revision** lists every item whose quantity or rate changed and the change in the grand total; **Restore a Revision** brings one back. **Export** writes the bill as a spreadsheet (.xlsx) with the company letterhead, a document (.docx) or CSV.

Fittings as they are made

Fittings follow the pipe material (the materials section of resources/data/boq/catalog.json, which you can edit):

Material	Joint	Bends made	Turns taken without a fitting
Polyethylene up to 90 mm	compression	90 degree elbows	up to 60 degrees (laid to a curve from the coil)
Polyethylene 110 mm and over	butt fusion or electrofusion	11.25, 22.5, 30, 45, 60, 90	up to 11.25 degrees

uPVC, PVC-O	rubber ring socket	11.25, 22.5, 45, 90	up to 2 degrees (joint deflection)
Ductile iron	push-in socket	11.25, 22.5, 45, 90	up to 4 degrees
GRP	double bell coupling	11.25, 22.5, 30, 45, 60, 90	up to 2 degrees
Steel	welded	cut to the angle	up to 3 degrees
GI, PPR, copper	threaded, socket fusion, capillary	45, 90	up to 5 degrees

A turn that is not one of the angles made is built from two or three bends (67 degrees in PVC is a 45 and a 22.5). Bends of 11.25 and 22.5 degrees are standard catalogue items for PVC, ductile iron and large polyethylene. The bill also counts:

- joints between pipe lengths or coils (butt fusion joints every 12 m, compression couplings every coil, threaded sockets, double bell couplings; none for pipes with their own sockets);
- flange adaptors at valves and pumps as the material needs (stub flange and backing ring for polyethylene, flange adaptor for PVC, flanged spigot for ductile iron);
- thrust blocks at bends of 22.5 degrees and more, tees and dead ends of 75 mm and larger, but not on welded polyethylene, which carries the end load itself;
- route marker posts every 100 m, at turns and at valves.

Each pipe takes the lowest pressure class its material is made in that covers the pressure with the margin (polyethylene PN6 to PN25, ductile iron to PN64); a line above the highest class is reported in the notes. The notes also list what to order: the number of lengths or coils of each pipe, with an allowance for waste and cutting.

Construction planning

Analysis > Project Planning opens the planner:

- a task table and a Gantt chart side by side, with drag to move, resize and link tasks;
- constraints, deadlines, calendars and working time;
- critical path, PERT probabilities, resources and levelling;
- baselines, progress, status date and earned value with the cost curve;
- schedule risk and a schedule quality check (the **Risk and quality** tab).

Schedule risk. The plan is scheduled many times (1,000 by default) with each duration drawn at random: from the task's optimistic, most likely and pessimistic days when they are given, otherwise within a range around its duration (90 to 125 % unless you change it). The tab shows the finish date met in half the runs (P50), in 80 (P80) and in 90 (P90), the chance of finishing as planned, the spread of the finish as a chart and how often each task was on the critical path.

Schedule quality check. The 14-point assessment used to judge whether a plan can be relied on: tasks without a predecessor or successor, leads, lags, links other than finish to start, hard constraints, high and negative float, long tasks, invalid progress, tasks without resources, missed tasks, a critical path that runs to the finish, the critical path length index and the baseline execution index. Each row gives what was found, the limit, Pass or Fail and the tasks concerned; checks that need a baseline, a status date or a deadline say so until they have them.

Manual and automatic scheduling. An auto scheduled task takes its dates from its links, constraints and the calendar. **Task > Manually Schedule** pins the selected tasks to their start dates: they stay there when their predecessors move (the plan warns when a predecessor finishes later), and dragging the bar moves the pinned date. **Task > Auto Schedule** gives them back to the scheduler. **Project > New Tasks Manually Scheduled** decides how new tasks start. Manual

tasks are drawn with brackets at both ends of the bar and can be listed with **View > Filter > Manually Scheduled Tasks**.

Formatting the Gantt chart. View > Gantt Chart Style sets the theme (the colours of task, critical, summary and manual bars, milestones, progress, baseline and links, each of which can be changed), the bar shape (rounded, square, shaded or flat) and height, how progress is drawn, the text beside the bars (task names, resource names, durations, dates, percent complete) and the parts shown (links, baseline, critical path, float, non-working days, gridlines, the today and status date lines). **View > Gantt Presets** applies a whole format at once (Standard, Tracking, Critical path, Presentation, Detailed, Print in black and white); the themes and presets are in resources/data/planning/gantt_styles.json. The style is kept with the plan, and printed and exported charts use it.

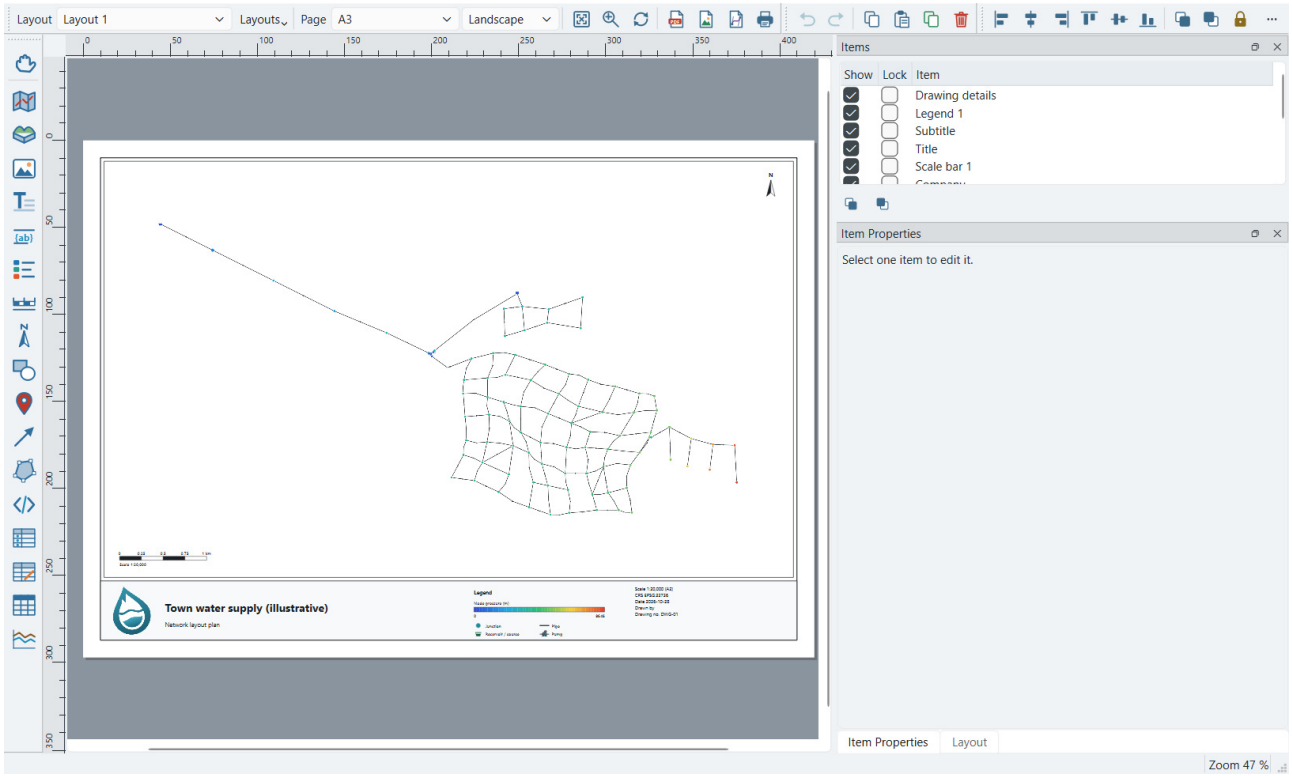
View also filters the tasks (critical, milestones, incomplete, late, manually scheduled) and shows the outline to a level. **Project > Move Project** moves the whole plan to a new start date with every constraint, deadline and manual start. The **Resource** menu assigns resources to the selected tasks, adds and removes resources, levels them and opens the resource sheet and usage.

The Gantt chart and the network diagram (CPM) each export as PNG, JPEG, TIFF, BMP, SVG or PDF (**File > Export**). **File > New Plan from Template** starts from a complete sample, a town water supply scheme with phases, crews and machines, costs, overlapping links, a baseline and progress, or from a template of a smallholder irrigation scheme, a borehole with solar pumping, a pipeline rehabilitation, or a feasibility study and detailed design. A schedule can also be generated from the project's quantities. Plans are saved with the project or as .chaosproj files, and exchanged as project schedule XML.

19

Map sheets and reports

Compose map sheets, profiles and reports for printing and sharing.



The print layout designer with a map, legend and title block

The print layout

File > Export > Print Layout opens the layout designer in its own window. Pick an item on the left toolbar, then drag a rectangle on the page (or click for the default size).

Item	What it shows
Map	everything the main map shows (the network, vector layers with their labels, contours, terrain, imagery and basemaps) at a chosen scale; can follow the main map
3D Map	a lit 3D view of a surface map, with axes and colour scale
Picture	a logo or photo

Label	text with fields such as `{project}` and `{scale}`
Dynamic Text	text filled when printed: date, time, scale, author, file, pipe length, withdrawal
Legend	the network symbols, vector layers (and their classes), terrain and imagery, contour layers and the result colour ramp; choose which appear and rename them
Scale Bar	linked to a map: alternating boxes, single box or line with ticks
North Arrow	classic, compass rose or triangle
Shape	rectangle, ellipse or line
Marker	a map pin, circle, square, triangle, diamond, star or cross, with a label
Arrow	an arrow in any direction, with heads at the end, the start or both
Node Item	a polyline or polygon whose points you type in millimetres
HTML	formatted text: headings, lists, tables and fields
Attribute Table	nodes or links with the columns you choose, filtered and sorted
Fixed Table	a table you type: revisions, notes, a title block
Summary Table	pipe lengths by diameter, node results or link results
Elevation Profile	ground, pipe and hydraulic grade along the pipes selected on the map

Rulers, guides, smart snapping, alignment, rotation, locking, copy and paste, and undo help you compose. Every item has its own properties panel; the layout redraws when the network changes, and **Refresh** (Ctrl+R) draws every item again from the project as it is now.

A legend lists what the project holds. In its properties, untick an item to leave it out, double-click one to rename it, or use **Show All** and **Hide All**.

Export as PDF, image or SVG, or print. Layouts are saved with the project, and a project can hold several.

Moving the map inside its frame

Double-click a map on the page to work inside it: drag to pan the map without moving the frame, turn the wheel to zoom around the pointer. A banner says you are moving map content; Esc or another double-click returns to arranging items. The same mode is the first button of the tools bar, **Move Map Content** (C). Ctrl and the wheel zoom a map at any time. Locked maps keep their view.

In the map's properties, **Use map view** takes the area shown in the main window, **Whole network** fits the network, and **Show on main map** zooms the main window to the area of this map.

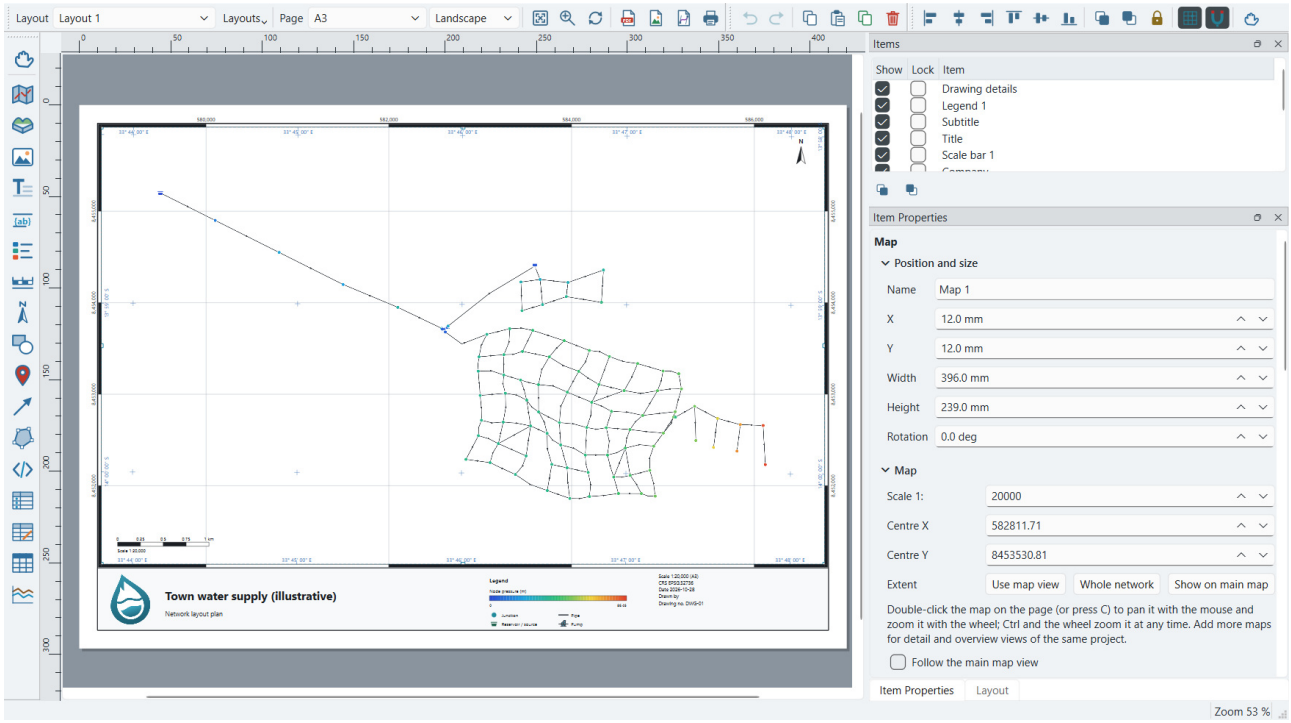
Map grids

A new map has no grid. A map on the sheet can carry several coordinate grids: select the map and use **Grids** in its properties. **Add Grid** adds a grid in the project's coordinates (metres); **Add Latitude and Longitude Grid** adds one in degrees, whose lines curve where the projection makes them curve. Tick a grid in the list to show it, choose it to

change it:

- **Draw as**: lines, crosses at the intersections, dots, or none (only the frame and labels).
- **Interval** and **Offset** for each direction, or automatic, a round step chosen from the map's scale.
- **Line colour**, **width** and **style**, and the size of the crosses.
- **Frame**: zebra bands, ticks inside or outside the map (or both), or a line round it.
- **Labels** on the sides chosen, outside or inside the map, along the left and right edges or level, as numbers, numbers with E and N, degrees, minutes and seconds, degrees and decimal minutes or decimal degrees, with their decimals, size and colour.

A metre grid labelled outside and a latitude and longitude grid labelled inside sit together without their labels meeting. Maps of layouts made before keep their edge ticks as their first grid.



A map with a metre grid on a zebra frame and a latitude and longitude grid

Overview maps

A sheet can hold several maps: a detailed one and a small one of the whole study area. In the small map's properties, **Show the extent of** names the detailed map, and its extent is drawn as a frame on the overview (a marker when it is too small to see). The frame follows as you pan and zoom the detailed map.

Fields

Labels, dynamic text and HTML items replace these fields when drawn: {project}, {title}, {date}, {date_long}, {time}, {scale}, {crs}, {author}, {company}, {layout}, {paper}, {file}, {nodes}, {links}, {pipe_km}, {withdrawal}, {analysis} and {version}.

Other outputs

- **Long sections** export as A3 PDF, PNG or a chainage table (CSV).
- **Contour map sheets** with axes, ticks and grid.

- **Results** export as spreadsheets; **graphs** as images.
- **File > Export > Canvas as Image** or **Canvas as PDF**: the map view as it is shown, every visible layer included, at print quality (PNG, JPEG, TIFF, BMP, WebP, SVG or PDF).

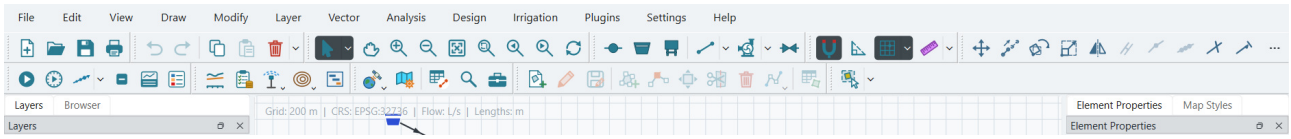
Part III. Reference

Keyboard shortcuts, File formats, The network file, Units of measurement, Coordinates and regional settings, Errors and warnings, How a network is solved, Glossary, Questions and answers, Troubleshooting

20

Keyboard shortcuts

Every keyboard shortcut in one place.



The toolbars: each command shows its shortcut in its tool tip

Project

Keys	Command
Ctrl+N	New project
Ctrl+O	Open
Ctrl+S	Save
Ctrl+Shift+S	Save As
Ctrl+Shift+V	Add a vector layer from a file
Ctrl+Alt+P	Project Properties
Ctrl+P	Print Layout
F1	Help

Editing

Keys	Command
Ctrl+Z / Ctrl+Y	Undo / Redo
Ctrl+X, Ctrl+C, Ctrl+V	Cut, Copy, Paste
Ctrl+A / Ctrl+Shift+A	Select All / Deselect All

F2	Element Properties
Delete	Delete the selection
Ctrl+Shift+O	Offset the selected pipe
Ctrl+Shift+D	Divide the selected pipe
Ctrl+Shift+T	Trim pipes
Ctrl+Shift+E	Extend pipes
Ctrl+Shift+L	Open or close the selected link

Tools

One key picks a tool on the map (not while typing in a box):

Key	Tool
S	Select
H	Pan
J, R, T	Junction, Reservoir, Tank
P	Pipe
U	Pump
V	Valve
M	Move
C	Copy with base point
Shift+R	Rotate
Shift+S	Scale
Shift+M	Mirror
Shift+T	Trim
Shift+E	Extend

Shift+D	Measure
Shift+V	Value Relay

While drawing or moving

Keys	Meaning
type a number	a length or a coordinate (see Drawing and editing)
Enter	accept the typed value
Esc	stop the current tool
R (rotate, scale)	reference angle or length
C (rotate, scale)	work on a copy
E (mirror)	erase the original

Map

Keys	Command
Ctrl++ (or Ctrl+=, Ctrl+Shift+=, keypad +) / Ctrl+-	Zoom in / out
Ctrl+0	Zoom to the network
Ctrl+J	Zoom to the selection
Ctrl+Left / Ctrl+Right	Previous / next view
Ctrl+F	Find / query
Ctrl+G	Go to coordinates

Analysis and design

Keys	Command

F5	Run steady state
Ctrl+Shift+P	Longitudinal profile
Ctrl+Shift+B	Bill of quantities
Ctrl+Shift+G	Project planning

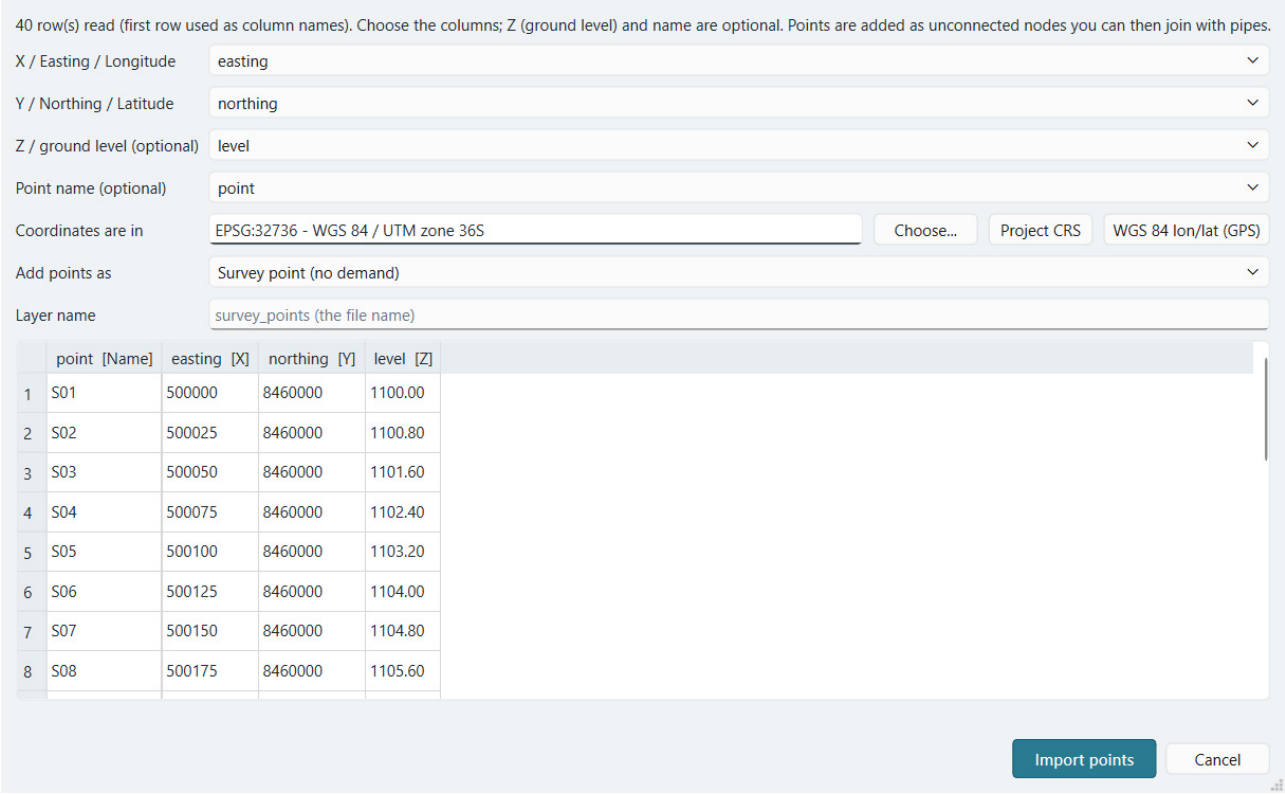
Tables

Keys	Command
Ctrl+C	Copy the selected cells (tab separated)
right-click	Copy, Copy with Headers, Select All

21

File formats

The files FlowCHAOS reads and writes.



Importing survey points from a text file

Projects

File	Use
`.fchaos`	a FlowCHAOS project: the whole work, as a GIS project file is: every network layer, vector layer, raster and basemap, surface map, results, labels, legends, layouts, plans and settings
`.fcnet`	a FlowCHAOS network: one network with every property, its profiles, curves, controls and rules, its coordinate system and its look; to share a network or add it to another project as a layer
`.chaosproj`	a construction plan on its own

Networks

Format	Read	Write	Notes
FlowCHAOS network (`.fcnet`)	yes	yes	one network layer, complete
Hydraulic model input (`.inp`)	import	export	the format of the widely used public-domain modelling program, supported for exchange; not a FlowCHAOS format
Binary network (`.net`)	yes		older binary model files
Network CSV	yes	yes	nodes and links in tables
Shapefile	yes	yes	nodes and links, with a `.prj`
GeoPackage	yes	yes	one file with nodes and links layers
GeoJSON	yes	yes	in WGS 84
KML	yes	yes	in WGS 84
GPX	yes		tracks and points
DXF	yes	yes	lines and polylines become pipes; export in layers
DXF as vector layers	yes		every drawing layer and kind of geometry, blocks drawn, texts labelled, levels kept

Survey and terrain

Format	Use
XYZ, CSV, TXT, XLSX	survey points with every column kept
GeoTIFF and other rasters	imagery and elevation models (DEM)
PDF	a page of a drawing or map, drawn as an image to georeference
Climate (`.clim`, TXT, CSV)	twelve months of ETo and rainfall, read and written
MBTiles	offline basemap tiles
ASCII grid (`.grd`)	surfaces made from points

Outputs

Format	Use
Spreadsheet (`.xlsx`)	results, bills of quantities, schedules
CSV	result tables, water quality results, field observations
PDF	water quality reports, the map canvas, map sheets
Document (`.docx`)	bills of quantities
PDF and images	map sheets, profiles, graphs, 3D views
Project schedule XML	plans for other scheduling programs
World file and GeoTIFF	georeferenced images

Every file is read in its own coordinate system and reprojected into the project's.

22

The network file

The FlowCHAOS network file (.fcnet): how a network is stored, part by part, so it can be shared, checked and read by other tools.

A network file holds **one network**: its elements, demand profiles, curves, controls, rules, analysis options, units and coordinate system, and optionally the look of its layer. FlowCHAOS writes one with **Export Layer As > FlowCHAOS Network** and opens it with **File > Open** or **Add Layer**, as a network layer. A project file (.fchaos) holds a whole project (many layers, rasters, plans, layouts); a network file is for sharing one network or adding it to another project.

How the file is written

- It is JSON text in UTF-8: readable in any text editor, and easy to read from a script or a spreadsheet tool.
- The parts always come in the same order, each with a plain name.
- Short lists (a point, the factors of a profile, an option) are kept on one line.
- Lengths are in metres, levels in metres above the datum, diameters in millimetres and flows in litres per second, unless `units` says otherwise; coordinates are in the `coordinate_system`.
- Every element has an `id` that never changes (references use it) and a `name` that you see on the map and that controls and rules refer to.

The parts

Part	What it holds
`format`, `version`	`flowchaos-network` and the layout version (2). A newer FlowCHAOS reads older versions.
`about`	the network's name, a description, the program and version that wrote it, and when (UTC)
`units`	the unit system and the units of flow, length, diameter and pressure
`coordinate_system`	the coordinate system of every x and y, as an EPSG code (for example EPSG:32736)
`network`	the elements, in six lists: `junctions`, `reservoirs`, `tanks`, `pipes`, `pumps`, `valves`
`demand_profiles`	each profile by name: the factors of its periods, in order
`curves`	each curve by name: its `kind` (Pump head, Pump efficiency, Storage volume, Component loss) and its `points`
`controls`	simple controls, one line each
`rules`	rule-based controls, one line each (RULE, IF, AND, OR, THEN, ELSE, PRIORITY)

`analysis`	the analysis options by group (`options`, `times`, `quality`, `energy`, `report` and others), one entry per setting
`layer_style`	optional: the symbols, colours and labels of the layer
`settings`	optional: FlowCHAOS settings that belong to this network

Nodes

Junctions, reservoirs and tanks share these fields:

Field	Meaning
`id`	the permanent identity of the node
`name`	the name shown on the map and used in controls and rules
`x`, `y`	the place, in the coordinate system of the file
`elevation_m`	the level used in the analysis (for a reservoir, its water level)
`demand_lps`	the design withdrawal (junctions)
`properties`	everything else the node has

Common properties: ground_elevation_m (the ground), design_offset_m (the pipe level below or above the ground), demand_profile, base_demand_lps, emitter_coefficient, description. A tank adds initial_level_m, minimum_level_m, maximum_level_m, diameter_m, minimum_volume_m3, stand_height_m and, when it has one, volume_curve. A reservoir may have total_head_m and a demand_profile that makes its level vary through the day.

Links

Pipes, pumps and valves share these fields:

Field	Meaning
`id`, `name`	as for nodes
`from`, `to`	the ids of the start and end nodes; flow is positive from `from` to `to`
`diameter_mm`	the inside diameter
`roughness`	the roughness for the friction formula of the analysis options

`status`	OPEN, CLOSED or CV (a pipe with a check valve)
`vertices`	optional: the bends between the two nodes, as x, y points
`properties`	everything else the link has

Common properties: material, minor_loss, model_length_m (a length given by the source of the network, used instead of the length on the map). A pump adds pump_type and pump_parameters (its head curve, power, speed and speed profile); a valve adds valve_type (PRV, PSV, PBV, FCV, TCV or GPV) and setting.

An example

A spring, a tank and one junction, with a demand profile, a control and two options:

```
{
  "format": "flowchaos-network",
  "version": 2,
  "about": {
    "name": "Village supply",
    "description": "",
    "application": "FlowCHAOS 1.0.0",
    "written": "2026-10-19T09:00:00Z"
  },
  "units": {
    "system": "metric",
    "flow": "L/s",
    "length": "m",
    "diameter": "mm",
    "pressure": "m"
  },
  "coordinate_system": "EPSG:32736",
  "network": {
    "junctions": [
      {
        "id": "j1",
        "name": "J1",
        "x": 500650.0,
        "y": 8460150.0,
        "elevation_m": 1150.0,
        "demand_lps": 1.2,
        "properties": {
          "ground_elevation_m": 1151.0,
          "design_offset_m": -1.0,
          "demand_profile": "RES"
        }
      }
    ],
    "reservoirs": [
      {
        "id": "r1",
        "name": "Spring",
        "x": 500100.0,
        "y": 8460400.0,
        "elevation_m": 1210.0,
        "demand_lps": 0.0,
        "properties": {
          "total_head_m": 1210.0
        }
      }
    ],
```

```
    "tanks": [
      {
        "id": "t1",
        "name": "Tank1",
        "x": 500400.0,
        "y": 8460300.0,
        "elevation_m": 1180.0,
        "demand_lps": 0.0,
        "properties": {
          "ground_elevation_m": 1180.0,
          "initial_level_m": 2.0,
          "minimum_level_m": 0.3,
          "maximum_level_m": 3.5,
          "diameter_m": 6.0
        }
      }
    ],
    "pipes": [
      {
        "id": "p1",
        "name": "P1",
        "from": "r1",
        "to": "t1",
        "diameter_mm": 110.0,
        "roughness": 140.0,
        "status": "OPEN",
        "vertices": [[500250.0, 8460380.0]],
        "properties": {
          "material": "HDPE"
        }
      },
      {
        "id": "p2",
        "name": "P2",
        "from": "t1",
        "to": "j1",
        "diameter_mm": 90.0,
        "roughness": 140.0,
        "status": "OPEN",
        "properties": {
          "material": "HDPE"
        }
      }
    ],
    "pumps": [],
    "valves": []
  },
  "demand_profiles": {
    "RES": [0.6, 0.5, 1.4, 1.8, 1.2, 0.9]
  },
  "curves": {},
  "controls": [
    "LINK P1 CLOSED IF NODE Tank1 ABOVE 3.3"
  ],
  "rules": [],
  "analysis": {
    "options": [["HEADLOSS", "H-W"]],
    "times": [["DURATION", "24:00"]]
  }
}
```

Reading a network file from a script

The file is ordinary JSON, so any language can read it. In Python:

```
import json
```

```
with open("village.fcnet", encoding="utf-8") as stream:
    network = json.load(stream)["network"]
for pipe in network["pipes"]:
    print(pipe["name"], pipe["diameter_mm"], pipe["properties"].get("material"))
```

Exchanging networks with other programs

For programs that do not read network files, FlowCHAOS imports and exports the hydraulic model input format (.inp, **Export Layer As > Hydraulic Model**). It is the format of the widely used public-domain network modelling program, not a FlowCHAOS format; FlowCHAOS supports it so models can be exchanged. That format has a fixed set of sections (junctions, reservoirs, tanks, pipes, pumps, valves, profiles, curves, controls, rules, options and times). FlowCHAOS keeps what the format cannot hold (ground levels, design offsets, materials, labels, layer styles) in the project, writes a harmless default for anything an imported model refers to but does not define, and renames names already used in the project on import (T1 becomes T1_2), with every reference to them. Use the network file whenever both sides are FlowCHAOS: it keeps everything.

23

Units of measurement

The units of every quantity, in metric and US customary projects.

The **flow units** chosen under **Analysis > Analysis Options > Hydraulics** decide the units of the other quantities. Metric flow units (L/s, L/min, ML/day, m3/hour, m3/day) make the whole project metric; US flow units (US gal/min, ft3/s, US Mgal/day, Imperial Mgal/day, acre-ft/day) make it US customary. FlowCHAOS shows lengths of pipes in metres or feet accordingly, and measurements in the units chosen in Project Properties.

Quantity	Metric	US customary
Concentration	mg/L or ug/L	mg/L or ug/L
Demand, flow	as the flow units	as the flow units
Diameter of pipes	mm	in
Diameter of tanks	m	ft
Efficiency	%	%
Elevation, head, level	m	ft
Emitter coefficient	flow units per m^n of pressure	flow units per psi^n
Energy	kWh	kWh
Friction factor	unitless	unitless
Hydraulic head	m	ft
Length	m	ft
Minor loss coefficient K	unitless	unitless
Power	kW	hp
Pressure	m of water	psi
Reaction coefficient, bulk	1/day (first order)	1/day (first order)
Reaction coefficient, wall	m/day (first order), mg/m2/day (zero order)	ft/day, mg/ft2/day
Roughness	Hazen-Williams C or Manning n unitless; Darcy-Weisbach e in mm	C or n unitless; e in millifeet

Source mass injection	mass per minute	mass per minute
Velocity	m/s	ft/s
Volume	m3	ft3
Water age	hours	hours

Flow unit conversions

Unit	In L/s
1 L/min	0.016667
1 m3/hour	0.27778
1 m3/day	0.011574
1 ML/day	11.574
1 US gal/min	0.063090
1 ft3/s	28.317
1 US Mgal/day	43.813
1 Imperial Mgal/day	52.617
1 acre-ft/day	14.276

Pressure

One metre of water is 9.81 kPa, 0.0981 bar or 1.42 psi; 10 m is about 1 bar. A pipe class PN10 (10 bar) is rated for about 100 m of water, including surge.

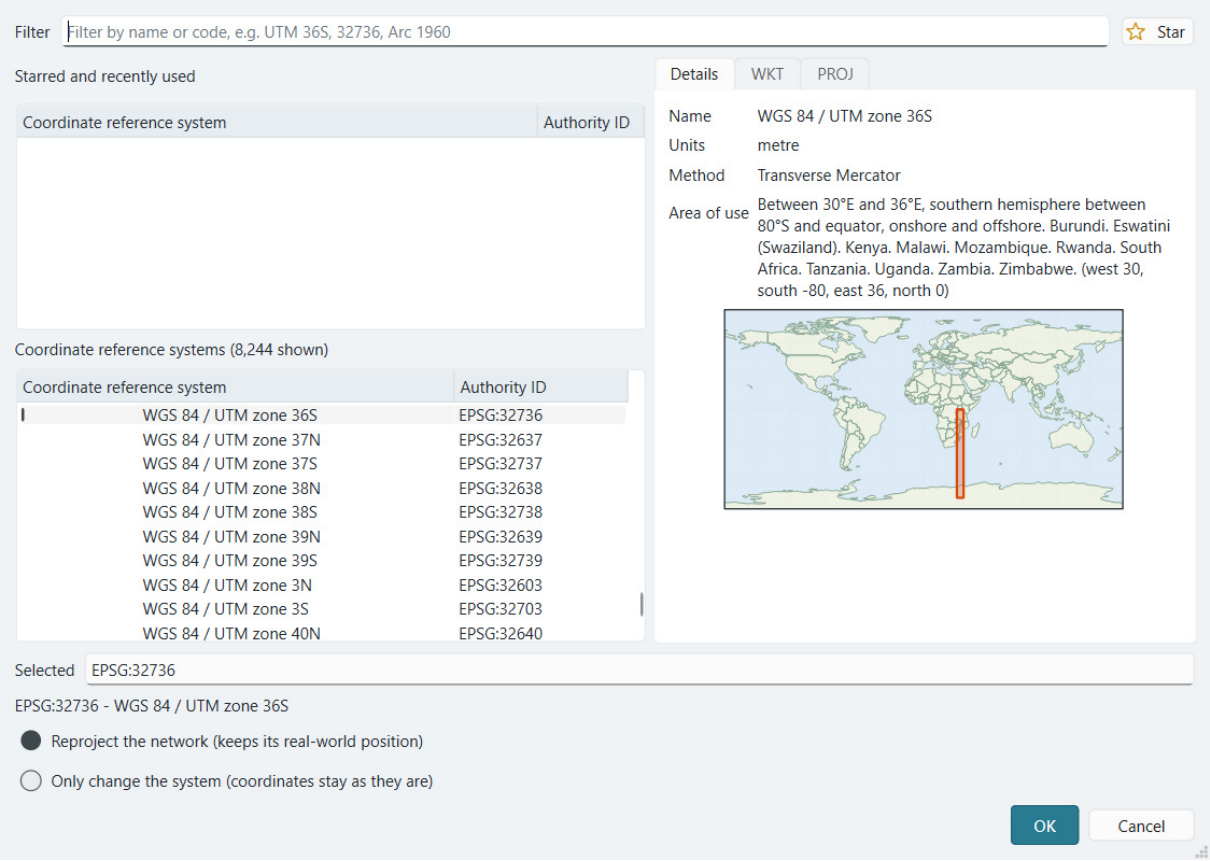
Measurements on the map

The Measure tool and the length and area fields show the units chosen in **Project Properties > Measurements**: metres, kilometres, feet, yards, miles or nautical miles for lengths; square metres, hectares, square kilometres, acres or square feet for areas; degrees, radians, gradians, mils or degrees-minutes-seconds for angles.

24

Coordinates and regional settings

Coordinate systems, units and how FlowCHAOS keeps every layer in the right place.



Choosing a coordinate reference system, with its area of use

The project's coordinate system

A project has one coordinate reference system (CRS), shown in the status bar. New projects use the default coordinate system, **EPSG:32736** (UTM zone 36 South) until you choose another. On the first start FlowCHAOS suggests one for the country the computer is set to (its national grid or UTM zone); change it at any time in **Settings > Default Coordinate System**. Click it to open **Project Properties > Coordinates**, see its details and area of use, and change it.

The coordinate system selector lists every geographic, projected and compound system of the EPSG registry and other authorities, grouped by kind and projection method; deprecated systems are left out. **Star** a system (the Star button, or right-click it) to keep it at the top of the list, above the recently used ones. Type part of a name, a code or a place in the filter. The selected system shows its units, method, its area of use on a world map, and its WKT and PROJ definitions. Any EPSG code, WKT or PROJ string can also be typed directly.

Choose a projected system in metres for design work (a UTM zone or your national grid): lengths, areas and slopes are then measured directly.

Layers in other systems

A file opened into an empty project keeps its coordinate system, which becomes the project's; it is reprojected only when you choose another system. Rasters and basemaps keep their own system and are reprojected on the fly; networks added to a project are transformed into its system. When importing survey points or a CSV, say which system the coordinates are in. A raster's system can be corrected in its properties without rewriting the file.

Coordinates on screen

Project Properties > Coordinates chooses how the cursor position is shown: map units (metres), degrees (latitude and longitude) or degrees, minutes and seconds, with the number of decimals.

Units

Quantity	Unit
Lengths and levels	metres (lengths can be shown in feet)
Diameters	millimetres
Flow	litres per second by default; set in Analysis Options
Pressure and head	metres of water
Velocity	metres per second

Project Properties > Measurements sets the units the measure tool shows: lengths in metres or kilometres (automatic), metres, kilometres, feet, yards, miles, nautical miles, centimetres, millimetres, inches, degrees or map units; areas in square metres or hectares (automatic), square metres, square kilometres, hectares, acres, square feet, yards, miles, nautical miles, centimetres, millimetres, inches, square degrees or map units; angles in degrees, radians, gradians, mils or degrees, minutes and seconds; and the number of decimals.

Language and regional formats

Settings > Language and Region chooses the language of the menus, commands and panel titles: English, Chichewa, Kiswahili, isiZulu, isiXhosa, chiShona, French, Portuguese, Spanish, Italian, German, Dutch, Polish, Russian, Turkish, Arabic, Hebrew, Hindi, Thai, Indonesian, Vietnamese, Chinese, Japanese and Korean (Arabic and Hebrew turn the whole window right to left). It also chooses how numbers and dates are written (for example 1,234.56 or 1 234,56, and 31/12/2026 or 12/31/2026) in measurements and bills of quantities.

Design standards

Project Properties > Analysis > Design standard sets the starting values of pipe sizing, fire flow, criticality and the withdrawal tools: water use per person, peak factor, losses, minimum and maximum pressure, velocity limits and fire flow. The presets describe typical schemes (rural standpipes, small towns, urban house connections, multi-storey areas, institutional areas); check them against the standard that applies to the project, and add your country's or utility's standard to design/standards.json in the reference data.

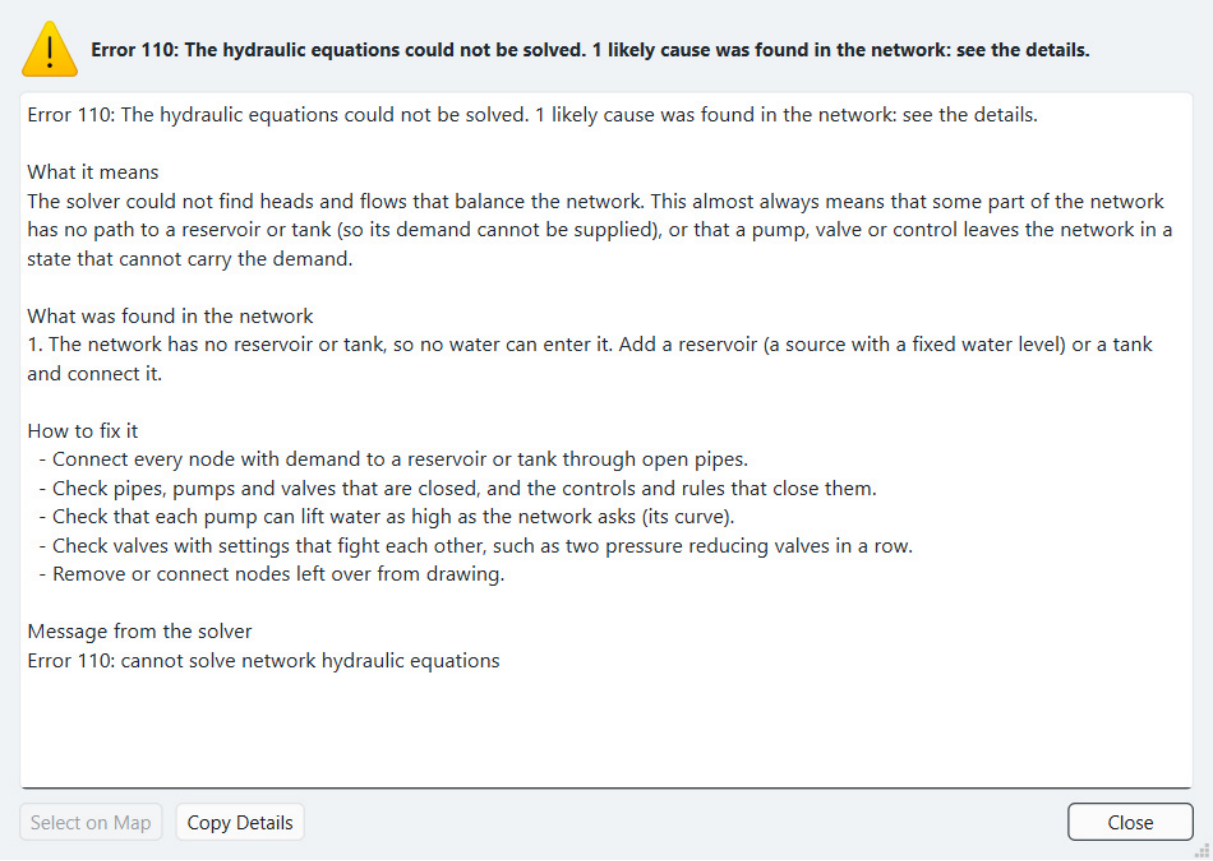
Map scale

The scale box in the status bar shows the map scale as 1:N. Pick a scale from the list or type one and press Enter to zoom to it. The list comes from **Project Properties > View**.

25

Errors and warnings

What the solver's errors and warnings mean, and what to do about each.



When a run cannot be completed: the reason and what to check

A run can end with an **error** (it could not solve, and there are no results) or finish with **warnings** (it solved, but something deserves a look). Warnings are listed under **Analysis > Solver Warnings**, where a double-click selects the element named.

An error opens **Analysis could not be completed**: the number and its meaning in plain words, the time into the run when it happened, what the solver reported, and what FlowCHAOS found when it checked the network for the usual causes: parts with no reservoir or tank, nodes cut off by closed pipes or valves (with the closed links named), pipes of zero length, diameter or roughness, diameters typed in metres, impossible tank levels, valves joined to a reservoir, a tank or each other, pumps without a curve or too weak for the lift, and controls that close links during the run. **Select on Map** selects every element named and zooms to them; **Copy Details** copies the text for a report or a question.

Warnings

Warning	What it means	What to do
The network is unbalanced at a time	the solver did not converge within the maximum trials	look for the cause below; raise **Maximum trials**; if it only happens briefly, choose **Continue** for unbalanced states
Negative pressures at a time	some junctions have a head below their elevation	the network cannot deliver the withdrawals there: larger pipes, more head at the source, a booster, or a pressure-driven analysis to see what is really delivered
Node disconnected at a time	a node has no open path to any source	a closed pipe or valve, a pump that closed, or a gap in the drawing; check **Select Connected Network** from a source
Pump closed because it cannot deliver the head	the system needs more head than the pump's shut-off head	the pump curve is too low for the lift, or a tank above it is full: check the curve, the levels, and the controls that run it
Pump open but exceeds its maximum flow	the pump runs beyond the end of its curve	the system head is lower than expected: a missing valve, a pipe too large, or a curve that needs its run-out point
Valve cannot deliver its setting	a pressure or flow control valve cannot reach its setting with the head available	the setting is out of reach: lower it, or give the valve more head upstream
Valve causes ill-conditioning	valves set against each other make the equations unstable	do not put two pressure valves in series without a pipe between them, nor a valve against a fixed-head node
System has negative pressures and disconnections together	several of the above at once	fix the disconnection first, then the pressures

Errors

Number	Meaning	What to do
101	not enough memory	close other programs; split a very large model
110	the hydraulic equations could not be solved	a disconnected part with no source, a pump that cannot lift, valves in conflict, or a pipe of zero length or diameter
200	one or more errors in the input data	the message lists them; fix each element named

201, 202, 203	a line of the model could not be read, a value is not a number, or a node referred to does not exist	usually from an edited or imported file: check the element named
204, 205	a link or a profile (pattern) referred to does not exist	create the missing profile or curve, or remove the reference
206	a curve referred to does not exist	add it in the Performance Curve Library
207	an attempt to control a check valve	a check valve opens and closes by itself; control another link
208	a value of a control or rule is out of range	check the control's node, value or setting
209, 211, 213	an illegal value for a node, a link, or an option	correct the property named (a negative diameter, a level below the minimum, an unknown option)
210	a valve joined to a reservoir or tank	put a short pipe between them
212, 214, 215	an unknown or repeated element, or a missing section	rename duplicates; in an imported file check its sections
216	an energy item refers to an unknown pump	correct the pump name in the energy settings
217, 219, 220	a source, a valve or a reservoir data conflict	a source on an element that cannot have one, two valves sharing a node illegally
221	a rule is badly written	use **Check** in the rules editor
222, 223	a pipe or valve with the same start and end, or a node without links	redraw the element
224	no tank or reservoir in the network	every network needs a source: add a reservoir or a tank
225	a tank's levels are not consistent (minimum above maximum, starting level outside them)	correct the tank's levels
226	a pump with no curve or power	give the pump a head-flow curve or a power
227	a pump curve that is not valid	head must fall as flow rises; start the curve at zero flow
230	a curve with x values not increasing	sort the points; remove repeated x values
233	a node not connected to any link	connect it or delete it (survey points are left out automatically)
302, 303, 304	a file could not be opened or written	check the folder exists and is not read-only, and the file is not open in another program
307, 308, 309	the results could not be saved	check the disk space and the folder

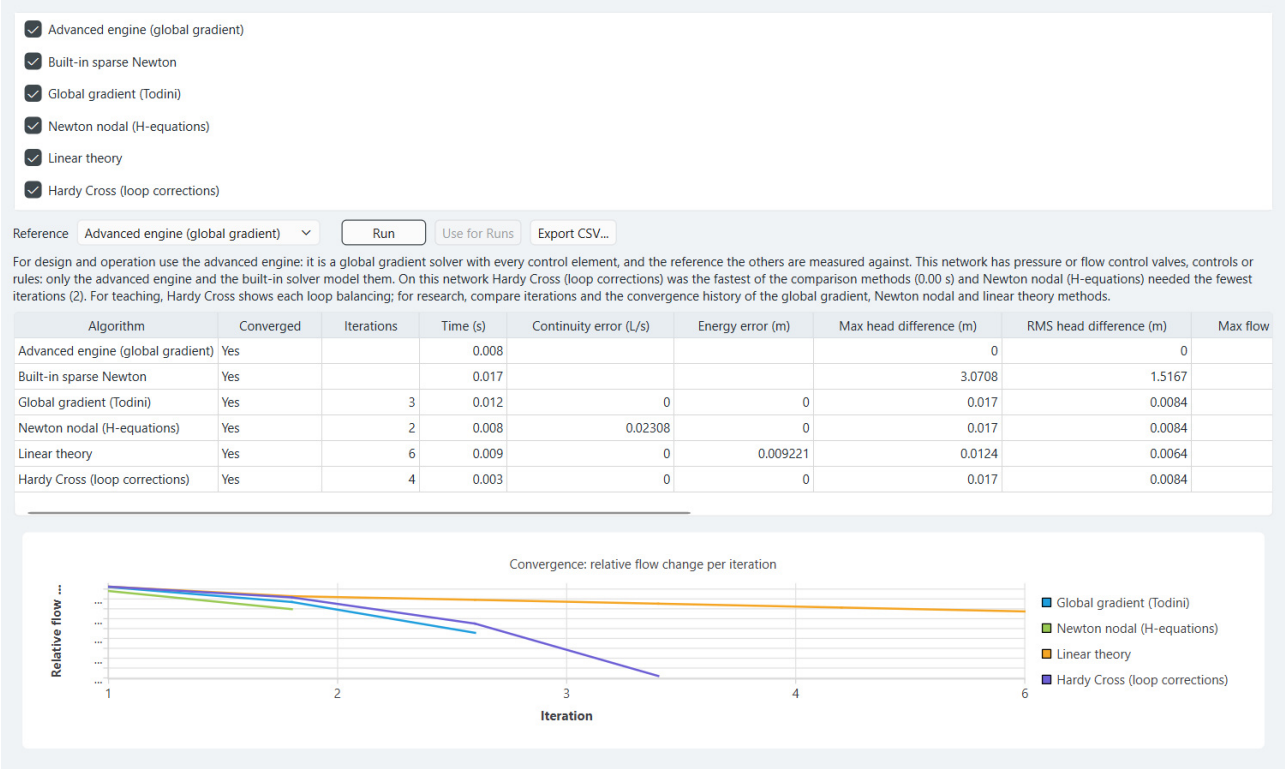
Before calling it a bug

1. Open **Analysis > Solver Warnings** and select the elements named.
2. Use the Processing Toolbox **Check** tools: disconnected parts, duplicate or very short pipes, nodes without links.
3. Run a steady state before an extended period; fix that first.
4. If a result still looks wrong, use **Help > Report an Issue**, with the project attached and the value you expected. Calculation questions are answered first.

26

How a network is solved

How FlowCHAOS solves a network: the equations, the methods, and how time and water quality advance.



Heads and flows of each algorithm compared on one network

The equations

For a network of n junctions and m links, a hydraulic state is found when:

- at every junction i, the flows Q in and out balance the withdrawal d: sum(Q in) - sum(Q out) = d(i);
- along every link from node i to node j, the heads H satisfy H(i) - H(j) = h(Q), where h is the head loss of a pipe or valve, or minus the head added by a pump.

The head loss of a pipe is h = r Q |Q|^(n-1) + m Q |Q|, with r from the friction formula and m from the minor losses. Because h depends on Q nonlinearly, the equations are solved by iteration.

The global gradient method

The advanced engine, the default, uses the global gradient method (Todini and Pilati, 1988). Each iteration:

1. Linearises every link around its current flow: h = h(Q0) + h'(Q0) (Q - Q0).
2. Eliminates the flows, leaving a symmetric, sparse system in the unknown heads, which is solved by sparse Cholesky factorisation.
3. Updates every flow from the new heads.

4. Checks the status of pumps, check valves and control valves at the set frequency.
5. Stops when the sum of the flow changes, divided by the total flow, is below the accuracy (0.001 by default), and any limits on head error and flow change are met.

Continuity holds exactly after every iteration, so even an unconverged state balances at the nodes. Convergence is usually reached in 5 to 20 iterations whatever the size of the network.

The other methods

Settings > Hydraulic Solver offers other algorithms, for checking a result or for teaching:

Algorithm	Unknowns	How it works
Built-in sparse Newton	flows and heads	Newton on the full system with least-squares safeguards; models pressure-driven withdrawal and control valves
Global gradient (Todini and Pilati, 1988)	flows and heads	the method above, written independently
Newton nodal (Martin and Peters, 1963)	heads	Newton on the junction heads alone, with flows from the heads; damped by a line search
Linear theory (Wood and Charles, 1972)	flows and heads	each element made linear and the system solved exactly; flows averaged between iterations
Hardy Cross (1936)	loop flows	one loop corrected at a time, with pseudo-loops between sources; slow, but every step can be followed by hand

Analysis > Compare Solvers runs them side by side and reports the iterations, time, and the largest differences in head and flow; extended-period runs always use the advanced engine.

Pressure-driven withdrawal

In a pressure-driven run each junction's withdrawal becomes an unknown that follows its pressure (see The network model). The advanced engine adds the withdrawals to its unknowns and solves them with the heads. The comparison algorithms model each withdrawal as an outlet link from the junction to a virtual reservoir at its elevation plus the minimum pressure, whose head loss reproduces the delivery curve, so every method answers the same question.

Extended periods

An extended period is a sequence of hydraulic states:

1. The state at the start time is solved with the tank levels and the controls at their starting values.
2. The time to the next event is found: the next hydraulic step, the next change of a demand profile, the time a tank would fill or empty, or the time a control or rule would act.
3. Tank levels are advanced over that time with the flows found (each level changes by the net flow divided by the tank's area at that level).
4. Controls and rules are evaluated; statuses change; the next state is solved.

Results are reported at every report step; between them, the run can take shorter steps so a tank never overshoots its limits and a control acts at the right moment.

Water quality

Water quality is solved with the hydraulics, by an event-driven Lagrangian method:

- each pipe holds a series of water segments of different quality; at each quality step, water enters at one end and leaves at the other, so segments move with the flow;
- segments whose qualities differ by less than the quality tolerance are merged;
- at each node, the water leaving is the flow-weighted mix of the water arriving, plus any source;
- reactions change each segment's concentration over the step: bulk decay at kb C^n, wall decay by a rate limited by the transfer of the substance to the wall (from the pipe diameter, velocity and diffusivity);
- tanks mix by their model (fully mixed, two compartments, plug flow or stacked).

Short quality steps (one to five minutes) keep the results accurate; the warm-up period of a study lets the starting values wash out.

References

- Todini, E. and Pilati, S. (1988). A gradient algorithm for the analysis of pipe networks. In Computer Applications in Water Supply, Vol. 1. Research Studies Press.
- Martin, D. W. and Peters, G. (1963). The application of Newton's method to network analysis by digital computer. Journal of the Institution of Water Engineers, 17.
- Wood, D. J. and Charles, C. O. A. (1972). Hydraulic network analysis using linear theory. Journal of the Hydraulics Division, ASCE, 98(HY7).
- Cross, H. (1936). Analysis of flow in networks of conduits or conductors. University of Illinois Engineering Experiment Station, Bulletin 286.
- Rossman, L. A. and Boulos, P. F. (1996). Numerical methods for modeling water quality in distribution systems: a comparison. Journal of Water Resources Planning and Management, 122(2).

27

Glossary

The words FlowCHAOS uses, and what they mean.

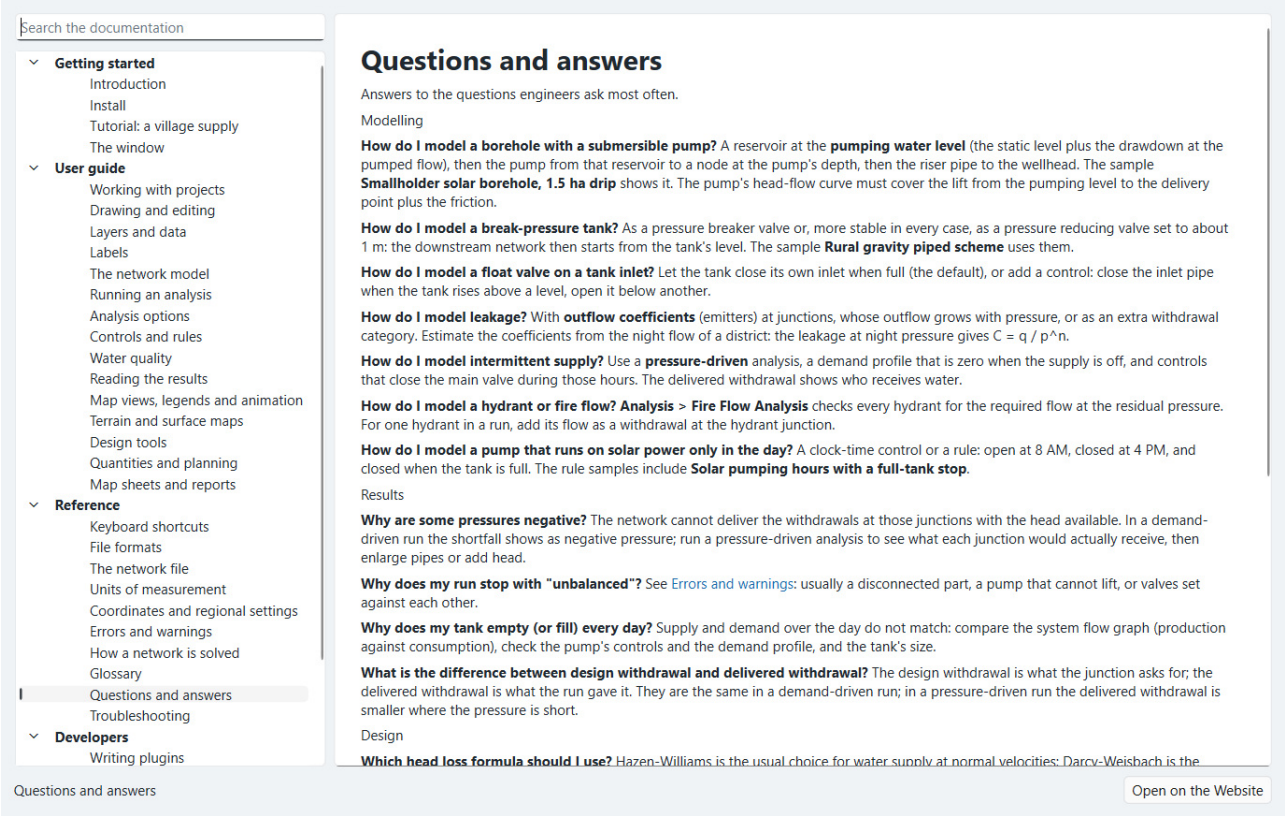
Term	Meaning
Junction	a node where pipes meet and where water can be withdrawn
Reservoir	a source with a fixed water level (a lake, a river intake, a bulk supply)
Tank	storage whose level rises and falls with inflow and outflow
Pipe, pump, valve	the links between nodes
Design withdrawal	the base demand of a junction, in L/s
Delivered withdrawal	the water a junction receives in a run (less than the design withdrawal when pressure is short under pressure-driven demand)
Demand profile	multipliers of the design withdrawal through the day
Ground elevation	the level of the ground at a node
Hydraulic elevation	the level used by the analysis (the pipe invert or the ground plus an offset)
Hydraulic head	the level water would rise to in a standpipe at a node
Pressure	head minus hydraulic elevation, in metres of water
Head loss	energy lost along a pipe by friction; per km it is the hydraulic gradient
Roughness coefficient	Hazen-Williams C, Darcy-Weisbach roughness height or Manning n
Starting concentration	the water quality at a node when an analysis starts
Bulk and wall decay rates	how fast a substance decays in the water and at the pipe wall
Peak factor	the ratio of peak to average withdrawal used for design
Residual pressure	the pressure left at a hydrant while it delivers a fire flow
Steady state	one moment in time
Extended period	a sequence of moments through time (for example 24 hours)

Long section	the profile of a pipeline with ground, invert, cover and the hydraulic grade line
HGL	hydraulic grade line: the hydraulic head along a pipeline
CRS	coordinate reference system
DEM	digital elevation model: a raster of ground levels
Map view	what nodes or links are coloured by
Layer key	the name under which a layer's labels and display settings are stored

28

Questions and answers

Answers to the questions engineers ask most often.



The Help window: every chapter of this manual, searchable

Modelling

How do I model a borehole with a submersible pump? A reservoir at the **pumping water level** (the static level plus the drawdown at the pumped flow), then the pump from that reservoir to a node at the pump's depth, then the riser pipe to the wellhead. The sample **Smallholder solar borehole, 1.5 ha drip** shows it. The pump's head-flow curve must cover the lift from the pumping level to the delivery point plus the friction.

How do I model a break-pressure tank? As a pressure breaker valve or, more stable in every case, as a pressure reducing valve set to about 1 m: the downstream network then starts from the tank's level. The sample **Rural gravity piped scheme** uses them.

How do I model a float valve on a tank inlet? Let the tank close its own inlet when full (the default), or add a control: close the inlet pipe when the tank rises above a level, open it below another.

How do I model leakage? With **outflow coefficients** (emitters) at junctions, whose outflow grows with pressure, or as an extra withdrawal category. Estimate the coefficients from the night flow of a district: the leakage at night pressure gives C = q / p^n.

How do I model intermittent supply? Use a **pressure-driven** analysis, a demand profile that is zero when the supply is off, and controls that close the main valve during those hours. The delivered withdrawal shows who receives water.

How do I model a hydrant or fire flow? Analysis > Fire Flow Analysis checks every hydrant for the required flow at the residual pressure. For one hydrant in a run, add its flow as a withdrawal at the hydrant junction.

How do I model a pump that runs on solar power only in the day? A clock-time control or a rule: open at 8 AM, closed at 4 PM, and closed when the tank is full. The rule samples include **Solar pumping hours with a full-tank stop**.

Results

Why are some pressures negative? The network cannot deliver the withdrawals at those junctions with the head available. In a demand-driven run the shortfall shows as negative pressure; run a pressure-driven analysis to see what each junction would actually receive, then enlarge pipes or add head.

Why does my run stop with "unbalanced"? See Errors and warnings: usually a disconnected part, a pump that cannot lift, or valves set against each other.

Why does my tank empty (or fill) every day? Supply and demand over the day do not match: compare the system flow graph (production against consumption), check the pump's controls and the demand profile, and the tank's size.

What is the difference between design withdrawal and delivered withdrawal? The design withdrawal is what the junction asks for; the delivered withdrawal is what the run gave it. They are the same in a demand-driven run; in a pressure-driven run the delivered withdrawal is smaller where the pressure is short.

Design

Which head loss formula should I use? Hazen-Williams is the usual choice for water supply at normal velocities; Darcy-Weisbach is the most exact for any pipe, velocity and temperature. Keep one formula for a project.

What velocities and pressures should I design for? Follow your standard; common values are 0.5 to 1.5 m/s in distribution, a minimum pressure of 10 to 15 m (5 to 10 m at communal taps), and a maximum static pressure of 60 to 80 m. The project's design standard holds the values the tools start from.

How do I choose pipe classes? The pipe class (PN) must exceed the highest pressure the pipe sees, static or in operation, plus an allowance for surge. Check the static pressure with the pumps off and the tanks full; in long rising mains, keep a margin for surge.

Data

Can I use my old models? Yes: FlowCHAOS imports hydraulic model input files (.inp) and binary model files (.net) of the widely used public-domain modelling program; each becomes a network layer, with its profiles, curves, controls and rules. Save your work as a FlowCHAOS project (.fchaos) or network file (.fcnet) to keep everything.

Can I work offline? Yes. Everything works offline except online basemaps, satellite imagery, place search and online terrain. Offline basemaps (MBTiles) and DEMs work without a connection.

Where are my files and settings? Projects are where you save them. Settings, logs and your own reference data are listed in Install, under Where FlowCHAOS keeps things.

The program

Is FlowCHAOS free? Yes. FlowCHAOS is freeware: it is licensed without a fee for commercial, academic and public-sector use under the FlowCHAOS Freeware Licence.

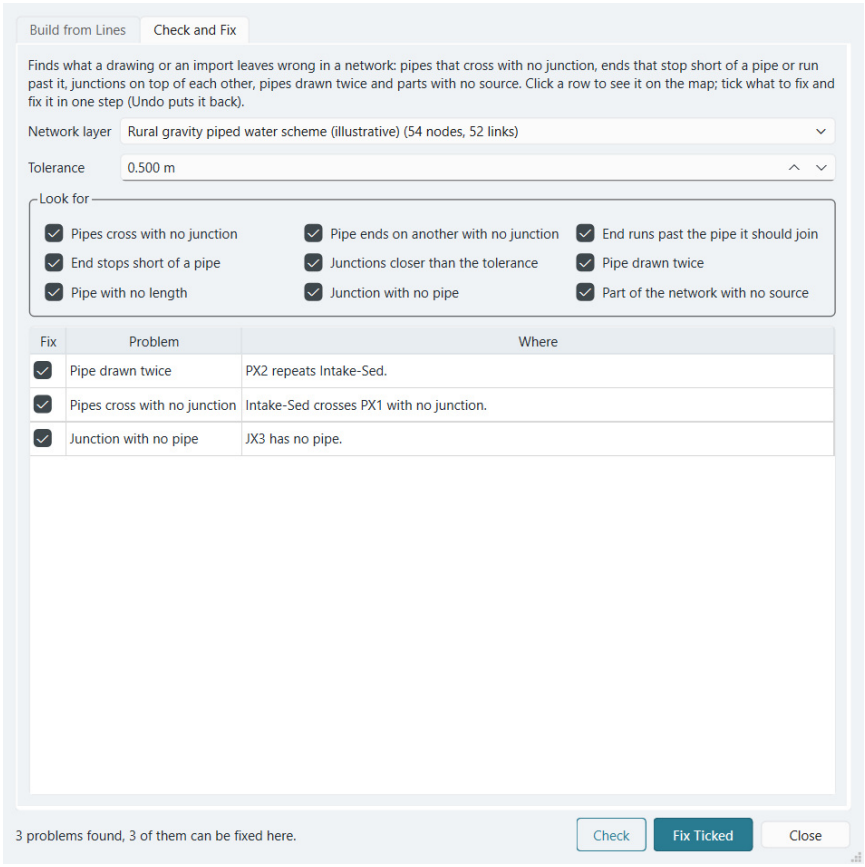
Which Windows do I need? Windows 10 or later, 64-bit, with 4 GB of memory (8 GB for large networks and terrain).

How do I report a problem? Help > Report an Issue, with the project and the value you expected. Calculation questions are answered first.

29

Troubleshooting

When something does not work as expected.



Check and Fix finds what stops a run: crossings, gaps, doubled pipes and parts with no source

The analysis does not solve

- Every part of the network must be connected to a reservoir or a tank. Nodes without pipes (survey points) are left out on purpose.
- Check reservoir levels and tank levels: a source lower than the junctions cannot supply them.
- Pumps need a head curve; valves need a setting.
- Very small or very large values (a 1 mm pipe, a 100 km pipe) often point to a units mistake.

Pressures are negative or very low

The source is too low or the pipes are too small. Run **Automatic Pipe Sizing**: it reports the junctions the source cannot serve whatever the sizes (a pump or a higher tank is needed there) and sizes the rest.

Layers are in the wrong place

The file was read in the wrong coordinate system. Remove it and import it again with its system, or correct a raster's system in its properties.

A command is greyed out

It needs something first: point at it to see what (a selection, results, a pipe, a DEM).

The map is slow on a very large network

Turn off labels and the flow animation while you work, hide layers you do not need, or show them only between two scales (Layer Properties > Display).

Reporting a problem

Open an issue with the steps, what you expected, what happened, the FlowCHAOS version (Help > About) and the last lines of the log: %LOCALAPPDATA%\FlowCHAOS\FlowCHAOS\logs\flowchaos.log. A small project that shows the problem helps the most. Suspected calculation errors are handled first.

Appendix. Extending FlowCHAOS

Writing plugins, Tutorial: build a plugin, Plugin API, Project model, Plugin examples

30

Writing plugins

Write your own tools for FlowCHAOS in Python: what a plugin is, how it is found, loaded and shared.

What a plugin can do

A plugin is Python code that FlowCHAOS loads at start-up (or when you enable it). It can:

- add commands to the **Plugins** menu;
- read the whole project: nodes, links, their values, every imported field, settings and results;
- change the network, as one step the user can undo;
- run analyses, use the design tools and read their results;
- open its own windows and dialogs with Qt (PySide6);
- add tools to the **Processing Toolbox**, with parameters, help, a log and a result table;
- add rate sources to the **BOQ Workbench** (a rate database, a supplier's price list);
- read and write files: reports, exports, imports from your own formats.

Plugins are grouped by what they add: **GIS** (data sources, formats, map tools), **CAD** (drawing and editing tools), **Optimisation** (design and operation optimisation), **Reporting** (reports and exports), **Cost estimation** (rates and estimates) and **Engineering tools** (any other calculation).

A plugin in two files

A plugin is a folder with a manifest and an entry point:

```
pipe_report/
    plugin.json
    __init__.py
```

plugin.json describes it:

```
{
  "id": "pipe-report",
  "name": "Pipe Report",
  "description": "Length of pipe by diameter, as a table you can copy.",
  "version": "1.0.0",
  "category": "Reporting",
  "min_flowchaos": "0.15.0",
  "startup": true
}
```

Key	Meaning
`id`	unique, lower case letters, digits, − and _, starting with a letter or digit
`name`	shown in the Plugins panel and menus
`description`	one sentence shown in the Plugins panel

`version`	your version number
`category`	GIS, CAD, Optimisation, Reporting, Cost estimation, Engineering tools or Other
`min_flowchaos`	the oldest FlowCHAOS version it works with (an older one refuses to load it)
`startup`	`true` to load it when FlowCHAOS starts (the user can switch it off)

__init__.py defines register(host), called once when the plugin loads:

```
def register(host):
    def run():
        project = host.document.project
        ...

    host.add_plugin_action("Pipe Report", run, "Length of pipe by diameter")
```

An optional unregister(host) is called when the plugin is switched off. Everything added through the host (commands, toolbox tools, rate sources) is removed automatically.

A Processing Toolbox tool

A tool declares its parameters; the toolbox builds its window, runs it as one undoable step when it changes the network, shows its message and table, and selects what it returns on the map:

```
from flowchaos.core.processing import Algorithm, Outcome, Param


def long_pipes(project, values, context):
    limit = values["limit"]
    rows, select = [], set()
    for link in project.links:
        if link.link_type == "PIPE" and project.link_length_m(link) > limit:
            rows.append([link.name, round(project.link_length_m(link), 1)])
            select.add(("link", link.id))
    return Outcome(f"{len(rows)} pipes longer than {limit:g} m.", ["Pipe", "Length (m)"], rows,


def register(host):
    host.add_toolbox_algorithm(Algorithm(
        "my_long_pipes", "Find long pipes", "My tools", "tb-plugin",
        "Selects the pipes longer than the limit.",
        (Param("limit", "Longer than", "number", 500.0, suffix=" m"),),
        long_pipes, needs="links"))
```

changes=True makes the run an undoable edit of the project; parameter kinds are number, integer, choice (with options), bool, text and file. A new group gets its own heading.

A rate source for bills of quantities

```
def register(host):
    def supplier_prices(boq):
        return {"Supply, lay, joint and test 110 mm OD uPVC PN10 pipe": 18.40}  # description -

    host.add_rate_source("Supplier price list", supplier_prices)
```

The BOQ Workbench lists it under **Rates**; matching items without a rate are priced from it.

Managing plugins

Plugins > Manage Plugins lists the installed plugins with their version, author and description. Its buttons install a

plugin from a ZIP file, open the **Plugin Builder** (a form that writes a new plugin with the parts you choose, ready to edit), reload the plugins after you change one, open the plugin folder and uninstall the selected plugin. A sample, the **Design Check Report**, is in docs/plugins and the tutorial *Build a plugin* walks through writing one.

Where plugins live

Location	For
`%USERPROFILE%\.flowchaos\plugins\<id>\`	your own plugins (copy the folder here)
`flowchaos/plugins_builtin/<id>/`	plugins shipped with FlowCHAOS

The **Plugins** panel lists every plugin, switches them on and off, installs a plugin from a .zip archive (one folder with one plugin.json) and removes user plugins.

Developing a plugin

1. Watch the log (%LOCALAPPDATA%\FlowCHAOS\FlowCHAOS\logs\flowchaos.log): a plugin's Python errors are written there.
2. Create the folder in %USERPROFILE%\.flowchaos\plugins\ and restart FlowCHAOS. After a change, switch the plugin off and on again in the Plugins panel to load the new code.
3. Errors while loading are shown in the Plugins panel and written to the log; FlowCHAOS keeps running.
4. Keep calculations in plain functions that take a project and return values: they are easy to test with pytest without opening a window.

Rules of thumb

- **Change the project through** host.document.apply, never directly: the change is undoable, saved and every view refreshes. See the plugin API.
- **Long work on a worker thread**: use flowchaos.core.workers.submit and update the window when it finishes, so the map stays responsive.
- **Plain, clear text** in your commands and messages.
- **Store your plugin's settings** in project.settings["plugin:<id>"] when they belong to the project, or in QSettings when they are personal.

Sharing

Zip the plugin folder (the zip contains pipe_report/plugin.json and pipe_report/__init__.py) and share it. Others install it from the Plugins panel. Plugins for everyone can be proposed for plugins_builtin through a pull request.

Reference data

Design tables are JSON files under flowchaos/resources/data/: pipe series for automatic sizing (design/pipe_series.json), regional settings and rates for quantities, irrigation tables. Each file starts with a _help entry explaining its fields. To change the data of an installed copy without rebuilding, put a file with the same relative path in %APPDATA%\FlowCHAOS\data; it is used instead of the shipped one.

A pipe series with prices, so sizing reports a real cost:

```
{"name": "HDPE PE100 PN10 (SDR 17)", "material": "HDPE PE100", "pn": "PN10", "sdr": 17,
 "outside_mm": [63, 75, 90, 110, 125, 160],
 "cost_per_m": [4.1, 5.6, 7.9, 11.8, 15.2, 24.6], "currency": "USD",
 "hazen_williams_c": 150, "darcy_mm": 0.007, "manning_n": 0.009}
```

Continue with the plugin API, the project model and the worked examples.

31

Tutorial: build a plugin

Tutorial: build a FlowCHAOS plugin step by step, from an empty folder to a ZIP your colleagues install.

This tutorial builds **Design Check Report**, a plugin that lists, after a run, every pressure, velocity and head loss outside your design limits, in a window of its own and as a tool in the Processing Toolbox. The finished plugin comes with FlowCHAOS as an example (FlowCHAOS-Design-Check-1.0.0.zip); building it yourself shows every part a plugin can have. You need FlowCHAOS and a text editor; Python knowledge helps but each step shows the full code.

Step 1: let the Plugin Builder make the folder

Plugins > Plugin Builder... writes a complete, working starting folder:

1. **Name**: Design Check Report.
2. **Description**: Pressures, velocities and head losses outside the design limits, after a run.
3. **Category**: Engineering tools.
4. Leave the three parts ticked (a command, a tool window, a toolbox tool), keep **Install it in FlowCHAOS and switch it on now** ticked and press **Build Plugin**.

The builder writes the folder (by default in Documents\FlowCHAOS plugins), installs it and switches it on: the Plugins menu now has its commands and the Processing Toolbox its tool. The folder opens in Explorer:

```
design_check_report/
    plugin.json          the manifest
    __init__.py          register(host): what the plugin adds to FlowCHAOS
    tools.py             the plugin's own work, without windows
    README.md            what it does and how to install it
    tests/test_tools.py  a first test
```

Step 2: the manifest

plugin.json tells FlowCHAOS what the plugin is:

```
{
  "id": "design-check-report",
  "name": "Design Check Report",
  "description": "Pressures, velocities and head losses outside the design limits, after a run.
  "version": "0.1.0",
  "category": "Engineering tools",
  "min_flowchaos": "1.0.0",
  "startup": true
}
```

The id must be unique; min_flowchaos is the oldest FlowCHAOS your plugin works with (an older one refuses to load it, with a clear message); startup loads it each time FlowCHAOS starts. The keys are explained in Plugins.

Step 3: keep the work apart from the windows

Put the calculation in a module of its own, checks.py, with plain functions on the project. They need no window, so they are easy to test and to reuse in a menu command, a window and a toolbox tool alike:

```
from dataclasses import dataclass

DEFAULTS = {"min_pressure_m": 10.0, "max_pressure_m": 60.0, "min_velocity_mps": 0.3,
            "max_velocity_mps": 2.0, "max_headloss_m_per_km": 10.0}


@dataclass
```

```
class Finding:
    kind: str          # "node" or "link"
    element_id: str
    name: str
    check: str
    value: float
    limit: float
    severity: str      # "High", "Medium" or "Low"


def check_design(project, limits=None):
    limits = {**DEFAULTS, **(limits or {})}
    results = project.results or {}
    nodes = {row["name"]: row for row in results.get("nodes", [])}
    found = []
    for node in project.nodes:
        row = nodes.get(node.name)
        if node.node_type != "JUNCTION" or row is None:
            continue
        pressure = float(row["pressure_m"])
        if pressure < limits["min_pressure_m"]:
            found.append(Finding("node", node.id, node.name, "Pressure below the minimum (m)",
                                 pressure, limits["min_pressure_m"], "Medium"))
    return found
```

What the project holds is in The project model: project.nodes and project.links with their values, project.results after a run (one row per node and link, with pressure_m, head_m, flow_lps, velocity_mps, headloss_m), and project.link_length_m(link). The full checks.py of the example also checks velocities, head loss per kilometre and dead ends.

Step 4: register what the plugin adds

__init__.py defines register(host). FlowCHAOS calls it when the plugin is switched on and gives it the **host**, the main window (Plugin API):

```
from . import checks


def register(host):
    host.add_tool_window("design-check-report", "Design Check Report", lambda: _window(host),
                         "Pressures, velocities and head losses outside the design limits")
```

add_tool_window puts a command in the Plugins menu that opens your window beside the map, one at a time. add_plugin_action(text, callback, tooltip) adds a plain command instead.

Step 5: a window of your own

A window is ordinary Qt (PySide6). Build it in a function that returns it:

```
def _window(host):
    from PySide6.QtWidgets import QLabel, QPushButton, QTableWidget, QTableWidgetItem, QVBoxLay

    window = QWidget()
    window.setWindowTitle("Design Check Report")
    layout = QVBoxLayout(window)
    summary = QLabel("Run a steady-state analysis, then Check.")
    table = QTableWidget(0, 4)
    table.setHorizontalHeaderLabels(["Severity", "Check", "Element", "Value"])
    check = QPushButton("Check")
    layout.addWidget(summary)
    layout.addWidget(table, 1)
    layout.addWidget(check)

    def run_check():
        found = checks.check_design(host.document.project)
        table.setRowCount(len(found))
        for r, f in enumerate(found):
            for c, value in enumerate((f.severity, f.check, f.name, round(f.value, 2))):
```

```
                table.setItem(r, c, QTableWidgetItem(str(value)))
        summary.setText(f"{len(found)} finding(s).")
        host.canvas.set_selection({(f.kind, f.element_id) for f in found})  # shown on the map

    check.clicked.connect(run_check)
    return window
```

host.canvas.set_selection({("node", id), ("link", id)}) selects elements on the map; the user sees them highlighted and can zoom to them with Ctrl+J.

Step 6: a tool in the Processing Toolbox

A toolbox tool declares its parameters; FlowCHAOS builds the form, the help, the log, the result table and the map selection:

```
def _toolbox_run(project, values, context):
    from flowchaos.core.processing import Outcome

    found = checks.check_design(project, values)
    return Outcome(f"{len(found)} finding(s).", ["Severity", "Check", "Element"],
                   [[f.severity, f.check, f.name] for f in found],
                   select={(f.kind, f.element_id) for f in found})


def register(host):
    from flowchaos.core.processing import Algorithm, Param

    host.add_toolbox_algorithm(Algorithm(
        "design_check_report", "Design check report", "Checks", "tb-check",
        "Every value of the last run outside the limits.",
        (Param("min_pressure_m", "Minimum pressure", "number", 10.0, suffix=" m"),),
        _toolbox_run, needs="results"))
```

needs keeps the tool greyed until it makes sense: network, links, selection, results or none. A tool that changes the network sets changes=True: the toolbox then runs it as one step of Undo, and new elements join the network layer being edited.

Step 7: changing the network

Every change goes through the document, as one step the user can undo:

```
def set_material(project):
    for link in project.links:
        if link.link_type == "PIPE" and link.diameter_mm <= 110:
            link.attributes["material"] = "HDPE"

host.network_edit("Set materials of small pipes", set_material)
```

host.network_edit(label, edit) makes the change in the network layer being edited (and asks which layer when several exist); host.document.apply(label, edit) changes the project as a whole (settings, results). Never keep project.nodes or a node between two edits: each edit makes a new project, so read host.document.project again each time.

Step 8: long work in the background

A calculation that takes more than a moment must not freeze the window. Run it on a worker and hand the result back with a Qt signal:

```
from PySide6.QtCore import QObject, Signal
from flowchaos.core import workers


class Done(QObject):
    finished = Signal(object)


done = Done()
done.finished.connect(show_results)        # runs on the window's thread
project = host.document.project            # read here, then work on it there
workers.submit(lambda: done.finished.emit(checks.check_design(project)), pool_name="compute")
```

Never touch windows from the worker: compute there, show the result in the connected function.

Step 9: try it while you write it

1. Edit the files of the installed copy (Plugins > Manage Plugins > **Open Plugin Folder**), or build again with the Plugin Builder.
2. Press **Reload** in Plugins > Manage Plugins: the plugin is switched off and on again, reading its files anew. No restart is needed.
3. An error while loading is shown in the Plugin Manager (point at the plugin's name) and written to the log (Help > Open Crash Log Folder, flowchaos.log).

Test the plain functions with pytest, without starting FlowCHAOS:

```
from flowchaos.core.sample_library import town_water_supply
from flowchaos.core.solver_backend import run_with_best_backend
from flowchaos.ui.main_window import prepare_results

import checks


def test_finds_high_pressures():
    project = town_water_supply()
    project.results, _message = prepare_results(run_with_best_backend(project))
    assert any(f.check.startswith("Pressure above") for f in checks.check_design(project))
```

Step 10: share it

Zip the folder (the folder itself, with plugin.json inside it) and send the ZIP. Your colleagues install it with **Plugins > Manage Plugins > Install from ZIP** and tick it. Raise version in plugin.json for each new release; installing a newer ZIP replaces the older one.

Good practice

- Name things as the engineer sees them: "Design check report", not "dcr_tool".
- Keep the calculation in plain functions and test them; keep windows thin.
- Every change through network_edit or document.apply, never by changing host.document.project directly, so Undo and the map stay right.
- Work longer than half a second goes to a worker (Step 8).
- unregister(host) stops anything the plugin started itself: timers, threads, open files. What was added through the host is removed by FlowCHAOS.
- A plugin runs with the user's permissions: write only to files the user chooses.

More examples are in Plugin examples; every call is in Plugin API.

32

Plugin API

The objects a plugin receives and the calls it can make.

The host

register(host) receives the host: the FlowCHAOS main window.

Member	What it is
`host.document`	the open project and its undo history (see below)
`host.main_window`	the main window, to parent your dialogs and windows
`host.canvas`	the map
`host.add_plugin_action(text, callback, tooltip="")`	adds a command to the Plugins menu and returns its `QAction`
`host.remove_plugin_action(action)`	removes a command
`host.statusBar().showMessage(text, milliseconds)`	a short message in the status bar
`host.show_tool_window(key, factory)`	opens a window of yours beside the map (one per key)
`host.add_tool_window(key, title, factory, tooltip="")`	a Plugins menu command that opens your window
`host.add_toolbox_algorithm(algorithm)`	adds a tool to the Processing Toolbox (`flowchaos.core.processing.Algorithm`)
`host.remove_toolbox_algorithm(id)`	removes it (done for you when the plugin is switched off)
`host.add_rate_source(name, provider)`	rates for the BOQ Workbench: `provider(boq)` returns `{description: rate}`
`host.rate_sources`	the rate sources currently available

The document

host.document holds the project and every change to it.

Member	What it does
`document.project`	the `NetworkProject` (see project model)
`document.apply(label, edit)`	runs `edit(project)` as one undoable step called `label`
`document.apply_field(label, name, value)`	sets one field of the project (for example `results`) undoably
`document.changed`	a signal emitted after every change
`document.projectReplaced`	a signal emitted when another project is opened
`document.file_path`	the project's file, or `None` before it is saved

Changing the project:

```
def double_demands(project):
    for node in project.nodes:
        if node.node_type == "JUNCTION":
            node.demand_lps *= 2

host.document.apply("Double the withdrawals", double_demands)
```

The step appears in Edit > Undo as "Double the withdrawals", the project is marked unsaved and the map, tables and panels refresh.

The map

Member	What it does
`canvas.selection`	a set of `("node", id)` and `("link", id)`
`canvas.set_selection(items)`	selects elements
`canvas.zoom_to_selection()`	zooms to them
`canvas.fit_network()`	shows the whole network
`canvas.set_map_views(node_view, link_view)`	colours the map (keys as in `flowchaos.core.map_views`)
`canvas.refresh()`	redraws after you changed display settings

Analysis

```
from flowchaos.core.solver_backend import run_with_best_backend

results = run_with_best_backend(host.document.project)
for node in results.nodes:          # name, head_m, pressure_m, demand_lps, quality
```

```
    print(node.name, node.pressure_m)
for link in results.links:          # name, flow_lps, velocity_mps, headloss_m, quality, status
    print(link.name, link.flow_lps)
```

Extended period: run_extended_with_best_backend(project, duration_seconds, timestep_seconds) returns frames, each with time_seconds and hydraulics (nodes and links as above).

Design tools

```
from flowchaos.core.pipe_sizing import pipe_series, size_pipes, apply_sizes
from flowchaos.core.fire_flow import fire_flow_analysis, hydrant_candidates

project = host.document.project
pipes = [l.id for l in project.links if l.link_type == "PIPE"]
sizing = size_pipes(project, pipes, pipe_series()[0], max_velocity=1.5, min_pressure=15, peak_f
print(sizing.message, sizing.cost)
host.document.apply("Size pipes", lambda p: apply_sizes(p, sizing))

hydrants = fire_flow_analysis(project, hydrant_candidates(project), target_lps=25)
```

Background work

Long calculations must not freeze the window:

```
from flowchaos.core import workers

future = workers.submit(my_long_calculation, project_copy)
future.add_done_callback(lambda f: print(f.result()))
```

Copy the project first (copy.deepcopy) when the calculation runs while the user keeps editing, and change widgets only from the main thread (use a Qt signal or QTimer.singleShot(0, ...)).

Labels, legends and display settings

Module	Functions
`flowchaos.core.layer_style`	`label_settings_for_key`, `store_settings`, `display_settings`, `store_display`, `layer_fields`
`flowchaos.core.legends`	`legend_settings`, `store_legend`, `class_breaks`
`flowchaos.core.map_views`	`NODE_VIEWS`, `LINK_VIEWS`, `node_values`, `link_values`

Change them inside document.apply so they are undoable and saved with the project.

33

Project model

The data a project holds: nodes, links, results and settings.

NetworkProject

flowchaos.core.project.NetworkProject

Field	Type	Meaning
`name`	str	project name
`crs`	str	coordinate system, for example `"EPSG:32736"`
`nodes`	list of `NetworkNode`	junctions, reservoirs and tanks
`links`	list of `NetworkLink`	pipes, pumps and valves
`results`	dict or None	the stored results (see below)
`settings`	dict	everything else saved with the project
`analysis_sections`	dict	analysis options, profiles, curves, controls and rules, as sections of rows

Method	Returns
`node(id)`, `link(id)`	the element with that id
`link_length_m(link)`	the length used by the analysis
`link_path_points(link)`	the drawn points of a link, from start to end
`next_node_name(type)`, `next_link_name(type)`	a free name with the project's prefixes
`to_dict()`, `NetworkProject.from_dict(data)`	the project as plain data and back

NetworkNode

Field	Meaning
`id`	unique id (use `flowchaos.core.project.new_id()` for new elements)
`name`	the name shown to users
`node_type`	`"JUNCTION"`, `"RESERVOIR"` or `"TANK"`
`x`, `y`	position in the project's coordinate system
`elevation_m`	hydraulic elevation (for a reservoir: its water level)
`demand_lps`	design withdrawal
`attributes`	a dict of everything else: ground level, starting concentration, imported fields, labels

Useful attribute keys: ground_elevation_m, initial_quality, fields (every column of the file the element came from), flowchaos_layer_id (its imported layer), hydrant.

NetworkLink

Field	Meaning
`id`, `name`	as for nodes
`link_type`	`"PIPE"`, `"PUMP"` or `"VALVE"`
`start_node`, `end_node`	node ids
`diameter_mm`	internal diameter
`roughness`	in the units of the head loss formula
`status`	`"OPEN"`, `"CLOSED"` or `"CV"` (check valve)
`attributes`	`vertices` (bend points), `material`, `pressure_class`, `bulk_coeff`, `wall_coeff`, `size_locked`, imported `fields`

Results

After a run, project.results holds:

```
{
    "method": "...",
    "nodes": [{"name": "J1", "head_m": 112.4, "pressure_m": 31.2, "demand_lps": 1.5, "quality":
    "links": [{"name": "P1", "flow_lps": 12.3, "velocity_mps": 0.81, "headloss_m": 0.42, "statu
    "time_steps": [{"time_seconds": 0, "nodes": [...], "links": [...]}, ...],   # extended per
}
```

Settings

project.settings keeps what is saved with the project, for example: layer_labels, layer_styles, layer_display, legends, contour_layers, layouts, metadata, project_info, elevation_source, predefined_scales. Plugins should use their own key: "plugin:<id>".

34

Plugin examples

Complete plugins to copy and adapt.

Pipe length by diameter

A table the user can copy into a spreadsheet.

```
from collections import defaultdict

from PySide6.QtWidgets import QDialog, QTableWidget, QTableWidgetItem, QVBoxLayout


def register(host):
    def run():
        project = host.document.project
        lengths = defaultdict(float)
        for link in project.links:
            if link.link_type == "PIPE":
                lengths[link.diameter_mm] += project.link_length_m(link)
        dialog = QDialog(host.main_window)
        dialog.setWindowTitle("Pipe length by diameter")
        table = QTableWidget(len(lengths), 2, dialog)
        table.setHorizontalHeaderLabels(["Diameter (mm)", "Length (m)"])
        for row, (diameter, length) in enumerate(sorted(lengths.items())):
            table.setItem(row, 0, QTableWidgetItem(f"{diameter:g}"))
            table.setItem(row, 1, QTableWidgetItem(f"{length:,.1f}"))
        QVBoxLayout(dialog).addWidget(table)
        dialog.resize(360, 400)
        dialog.exec()   # every table can be copied with right-click or Ctrl+C

    host.add_plugin_action("Pipe Length by Diameter", run, "Total length of pipe for each diame
```

Mark low-pressure junctions

Runs the analysis, then selects and zooms to the junctions below a pressure.

```
from PySide6.QtWidgets import QInputDialog

from flowchaos.core.solver_backend import run_with_best_backend


def register(host):
    def run():
        limit, ok = QInputDialog.getDouble(host.main_window, "Low pressure", "Minimum pressure
        if not ok:
            return
        project = host.document.project
        results = run_with_best_backend(project)
        low = {r.name for r in results.nodes if r.pressure_m < limit}
        ids = [("node", n.id) for n in project.nodes if n.name in low and n.node_type == "JUNC
        host.canvas.set_selection(ids)
        if ids:
            host.canvas.zoom_to_selection()
        host.statusBar().showMessage(f"{len(ids)} junction(s) below {limit:g} m.", 8000)

    host.add_plugin_action("Select Low-Pressure Junctions", run)
```

Set a material from the diameter (undoable)

```
def register(host):
    def assign(project):
        for link in project.links:
            if link.link_type == "PIPE":
                link.attributes["material"] = "HDPE PE100" if link.diameter_mm < 300 else "Duct

    host.add_plugin_action("Assign Materials by Size",
                           lambda: host.document.apply("Assign materials by size", assign))
```

A window of your own, with work in the background

```
import copy

from PySide6.QtCore import QObject, Signal
from PySide6.QtWidgets import QLabel, QPushButton, QVBoxLayout, QWidget

from flowchaos.core import workers
from flowchaos.core.solver_backend import run_extended_with_best_backend


class _Done(QObject):
    finished = Signal(object)


class PeakWindow(QWidget):
    def __init__(self, host):
        super().__init__()
        self.host = host
        self.setWindowTitle("Peak velocity over a day")
        self.label = QLabel("Press Run.")
        button = QPushButton("Run 24 hours")
        button.clicked.connect(self.run)
        layout = QVBoxLayout(self)
        layout.addWidget(self.label)
        layout.addWidget(button)
        self.done = _Done()
        self.done.finished.connect(self.show_result)

    def run(self):
        project = copy.deepcopy(self.host.document.project)   # the user may keep editing
        self.label.setText("Running...")
        workers.submit(lambda: self.done.finished.emit(run_extended_with_best_backend(project,

    def show_result(self, results):
        peak = max((abs(l.velocity_mps), l.name, f.time_seconds // 3600)
                   for f in results.frames for l in f.hydraulics.links)
        self.label.setText(f"Highest velocity {peak[0]:.2f} m/s in {peak[1]} at hour {peak[2]}


def register(host):
    host.add_plugin_action("Peak Velocity Over a Day",
                           lambda: host.show_tool_window("peak-velocity", lambda: PeakWindow(ho
```

Testing a plugin

Keep the calculation in a function and test it without a window:

```
from flowchaos.core.examples import sample_water_network

def test_total_length():
    project = sample_water_network()
    total = sum(project.link_length_m(l) for l in project.links if l.link_type == "PIPE")
    assert total > 0
```